

Revised Final Wastewater Preliminary Engineering Report

City of Alturas
200 W. North Street
Alturas, California



Prepared for:

City of Alturas



June 2021

518004.200

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CONSULTING ENGINEERS & GEOLOGISTS, INC.

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Reference: 518004.200

June 18, 2021

Kim Dodds, Interim Director of Public Works
City of Alturas
200 W. North Street
Alturas, CA 96101

**Subject: Final Wastewater Preliminary Engineering Report
Proposition 1 Wastewater Planning Grant**

Dear Mr. Dodds:

We are pleased to submit to you the enclosed copy of the Revised Final Wastewater Preliminary Engineering Report (PER). prepared under your Proposition 1 Wastewater Planning Grant. We have incorporated responses to comments on the Final PER from the CVRWQCB, and SWRCB-DFA. Please review and let us know if you have any further comments. Please submit to SWRCB-DFA via your FFAST account at your convenience.

If you have any questions, please feel free to contact me at 541-827-7855 or arasmussen@shn-engr.com.

Respectfully submitted,

SHN Engineers & Geologists

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Regional Principal/Senior Civil Engineer

AHR:ahr

Enclosures: Final PER

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Reference: 518004.200

Revised Final Wastewater Preliminary Engineering Report

Prepared for:
City of Alturas

Funding for this project has been provided by the California State Water Resources Board under a Proposition 1 Small Community Wastewater Planning Grant, Agreement No. D17-04002.

Prepared by:



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June 2021

QA/QC: AHR



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Abbreviations and Acronyms

BOD	Biochemical Oxygen Demand
CDO	Cease and Desist Order
CVRWQCB	Central Valley Regional Water Quality Control Board
DFA	Division of Financial Assistance
FAA	Federal Aviation Administration
FEMA	Federal Emergency Management Agency
I/I	Infiltration and Inflow
MG	Million Gallons
MGD	Million Gallons per Day
NPDES	National Pollution Discharge Elimination System
OMB	Office of Management of Budget
PER	Preliminary Engineering Report
PVC	Polyvinyl Chloride
SCADA	Supervisory Control and Data Acquisition
SHN	SHN Engineers & Geologists
SRF	State Revolving Fund
SSES	Sanitary Sewer Evaluation Study
SWRCB	State Water Resources Control Board
TSO	Time Schedule Order
TSS	Total Suspended Solids
USDA-RD	United States Department of Agriculture Rural Development
USEPA	United States Environmental Protection Agency
UV	Ultraviolet
VFD	Variable Frequency Drive
WDR	Waste Discharge Requirements
WWTP	Wastewater Treatment Plant
WWTF	Wastewater Treatment Facility

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Executive Summary

The City of Alturas' Wastewater Treatment Plant (WWTP) currently discharging to the Pit River has had difficulty meeting permitted effluent limits for various constituents, including zinc, copper, aluminum, biological oxygen demand (BOD), total suspended solids (TSS), total coliform, toxicity, and total suspended solids (TSS). The Central Valley Regional Water Quality Control Board has noted that land discharge is the more feasible long-term solution to remain in compliance with future discharge regulations. The WWTP, which having some upgrades in 2006, has components that are well beyond their useful life and need to be replaced.

Several options were evaluated in this Preliminary Engineering Report (PER). Three alternatives were investigated in detail:

- Alternative 1: Rehabilitate the existing WWTP and continue to discharge to the Pit River;
- Alternative 2: Rehabilitate the existing WWTP and use land discharge with infiltration/evaporation ponds offsite; and
- Alternative 3: Decommission the existing WWTP, pump the raw wastewater to an offsite new WWTP utilizing aeration ponds, and use land discharge with infiltration/evaporation ponds at the offsite location.

A planning level alternatives analysis was conducted, with estimated project costs as follows:

- Alternative 1: \$12.9 million
- Alternative 2: \$11.7 million
- Alternative 3: \$5.8 million



Perc. ponds per Bryan?

Alternative 3 was found to be the most cost-effective solution, both in terms of capital cost and long-term operational costs, through the Net Present Value analysis. Alternative 3 is the recommended alternative.

1.0 Introduction

1.1 Background

The City of Alturas (“City”) owns and operates a wastewater utility which consists of a collection system and wastewater treatment plant. The system serves the City’s population of approximately 2,600. The wastewater treatment plant (WWTP) is located along the north bank of the North Fork Pit River at the confluence with the South Fork Pit River. The WWTP provides primary and secondary treatment and had a major upgrade in 2006. At the time of project initiation, treated effluent was discharged to the Pit River under Waste Discharge Requirements (WDR) Order R5-2014-0033 and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0078921, issued by the Central Valley Regional Water Quality Control Board (CVRWQCB).

The City has had difficulty meeting permitted effluent limits for various constituents, including zinc, copper, aluminum, biological oxygen demand (BOD), total suspended solids (TSS), total coliform, and toxicity. A Time Schedule Order (TSO) R5-2014-0034-01 (as amended by Order No. R5-2015-0111) was issued specifying interim limits for zinc, copper, and total coliform. At the time of the draft Preliminary Engineering Report (PER), renewed WDRs were proposed that include new and recalculated effluent limits, which have been issued recently under Order R5-2020-0004. A corresponding Cease and Desist Order (CDO) R5-2020-0005 was also issued that includes interim effluent limits for copper, zinc, total coliform, and aluminum.

The City obtained a Proposition 1 Wastewater Planning Grant (Agreement No. D17-04002) from the State Water Resources Control Board (SWRCB) Division of Financial Assistance (DFA) to assist the City to develop a recommended path forward to achieve regulatory compliance.

1.2 Purpose

The purpose of this Preliminary Engineering Report (PER) is to evaluate alternatives and provide a recommended alternative to allow the City to obtain funding for final design and construction and achieve consistent regulatory compliance.

1.3 Format

The SWRCB does not require a specific format for wastewater treatment facility planning documents. The format of this PER generally follows the format guidelines provided by the United States Department of Agriculture Rural Development (USDA-RD), which will allow the City the flexibility to pursue USDA-RD funding in addition to SWRCB SRF funding.

2.0 Project Planning

2.1 Location

The City's existing WWTP is located just south of the City limits along the north bank of the North Fork Pit River (Figure 1). The WWTP is located on a portion of a property owned by the City, as shown in Figure 2. This property is also used for the City's dog pound and for storage of excess City equipment. Figure 1 also shows the location of the property where facilities are proposed under the recommended alternative as described later in this PER. The wastewater system service area is described in Section 3.

2.2 Environmental Resources Present

Environmental resources present in the planning area are described in the associated CEQA documents, which are being developed concurrently to this PER. Portions of the WWTP site are within the 100-year floodplain as defined by the Federal Emergency Management Agency (FEMA). The 100-year floodplain is shown on Figure 3. No significant changes to existing conditions are anticipated within the defined floodplain and, therefore, no significant downstream impacts are anticipated.

2.3 Population Trends

The current population of the City's wastewater service area is approximately 2,600. The City's population has been in decline at an average rate of 1.15% since the 2010 census, when the population was 2,833. Currently, the population appears relatively stable and is not expected to grow significantly within the planning horizon of this study, which is the next 20 years.

2.4 Community Engagement

The Public Works Department, through its Director, communicates with and provides updates to the City Council, which meets monthly. Meeting minutes are available at City Hall and online at the City's website for residents to view. If special announcements or communication is needed, the City can include written announcements with sewer and water bills, which are sent out monthly. The City also puts announcements on its website.

3.0 Existing Facilities

3.1 General

The City's WWTP is a Class II trickling filter facility with the following major components:

- Headworks, including grit removal
- Grinder
- Influent Pump Station
- Primary Clarifier
- Trickling Filter
- Secondary Clarifiers
- Digester
- Sludge Drying Beds
- Disinfection
- Outfall, including high water pump station

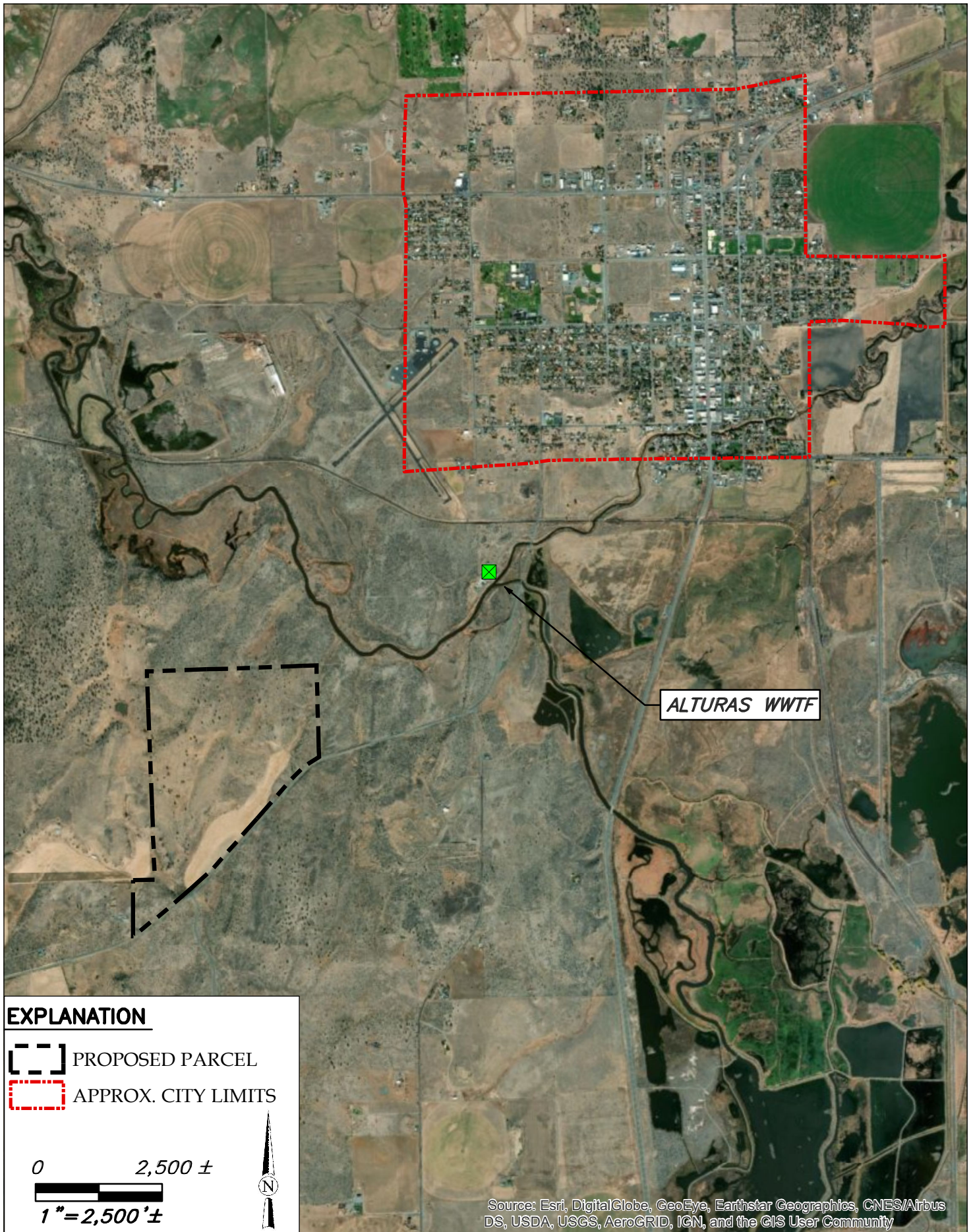
A site plan of the current WWTP is presented in Figure 4. The process flow diagram showing the existing treatment process is shown in Figure 5.

The wastewater collection system is not part of this study, but has been described and evaluated in a separate Sanitary Sewer Evaluation Survey (SSES), which was recently completed by SHN as part of the Proposition 1 Wastewater Planning Grant associated with this PER.

3.2 History

The exact year when the original WWTP was constructed is not known. Improvements were constructed in 1965, 1974, and 2006. Table 1 provides a list of the WWTP system components, when constructed (if known), when renovated, and some brief comments. Detailed discussion of the current status of each major component is found in Section 3.3.

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ALTURAS WWTF

EXPLANATION

-  PROPOSED PARCEL
-  APPROX. CITY LIMITS

0 2,500 ±
1" = 2,500' ±



Source: Esri, DigitalGlobe, GeoEye, Earthstar Geographics, CNES/Airbus DS, USDA, USGS, AeroGRID, IGN, and the GIS User Community



City of Alturas
Preliminary Engineering Report
Alturas, CA

Project Location

SHN 518004

December 2019

Figure_1

Figure 1

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City of Alturas
Wastewater Planning Study
Alturas, California

Wastewater Treatment Plant Boundaries
Plan View
SHN 518004

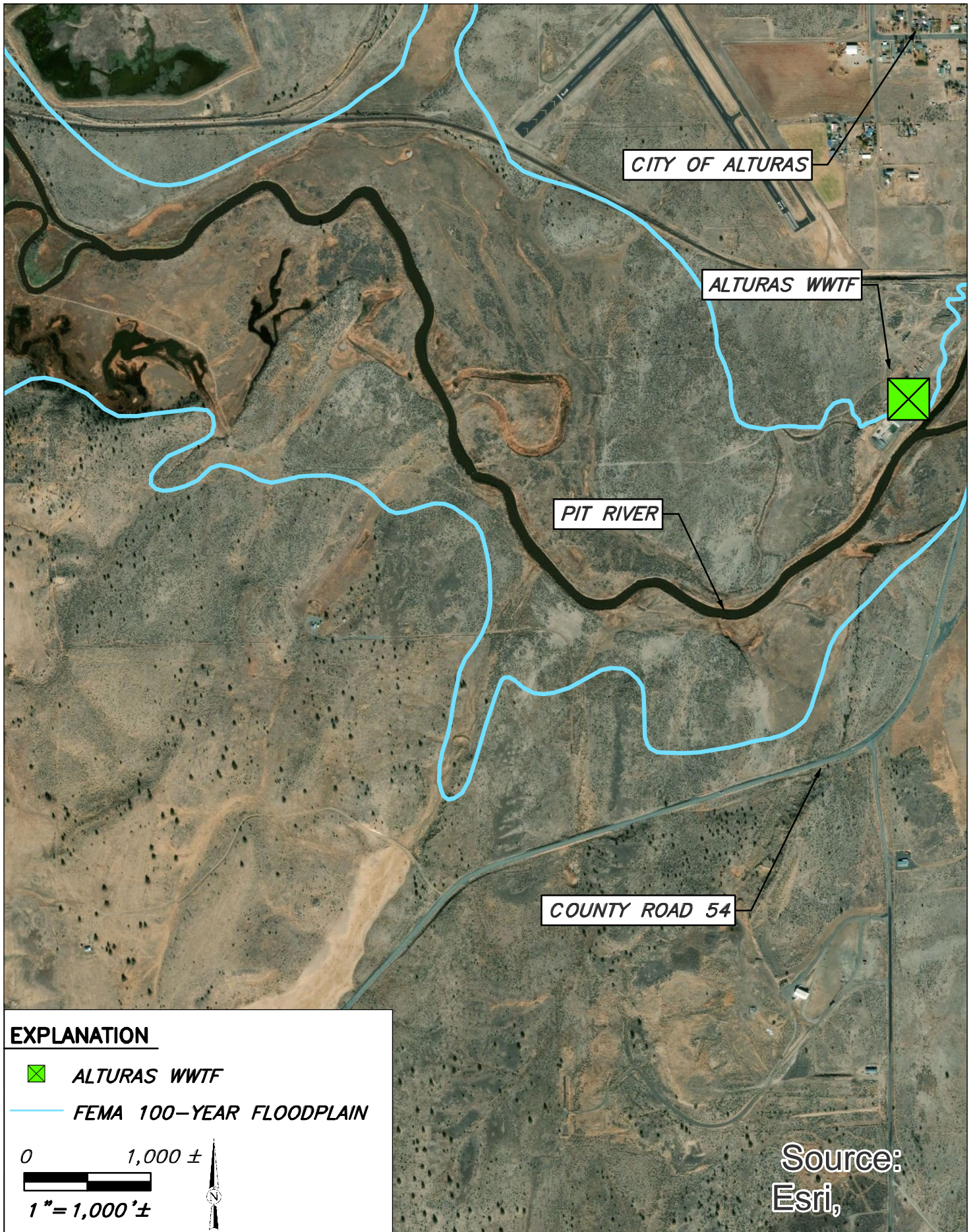
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518004-ALTURAS-WW

Figure 2

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EXPLANATION

 **ALTURAS WWTF**

 **FEMA 100-YEAR FLOODPLAIN**

0 1,000 ±

1" = 1,000' ±

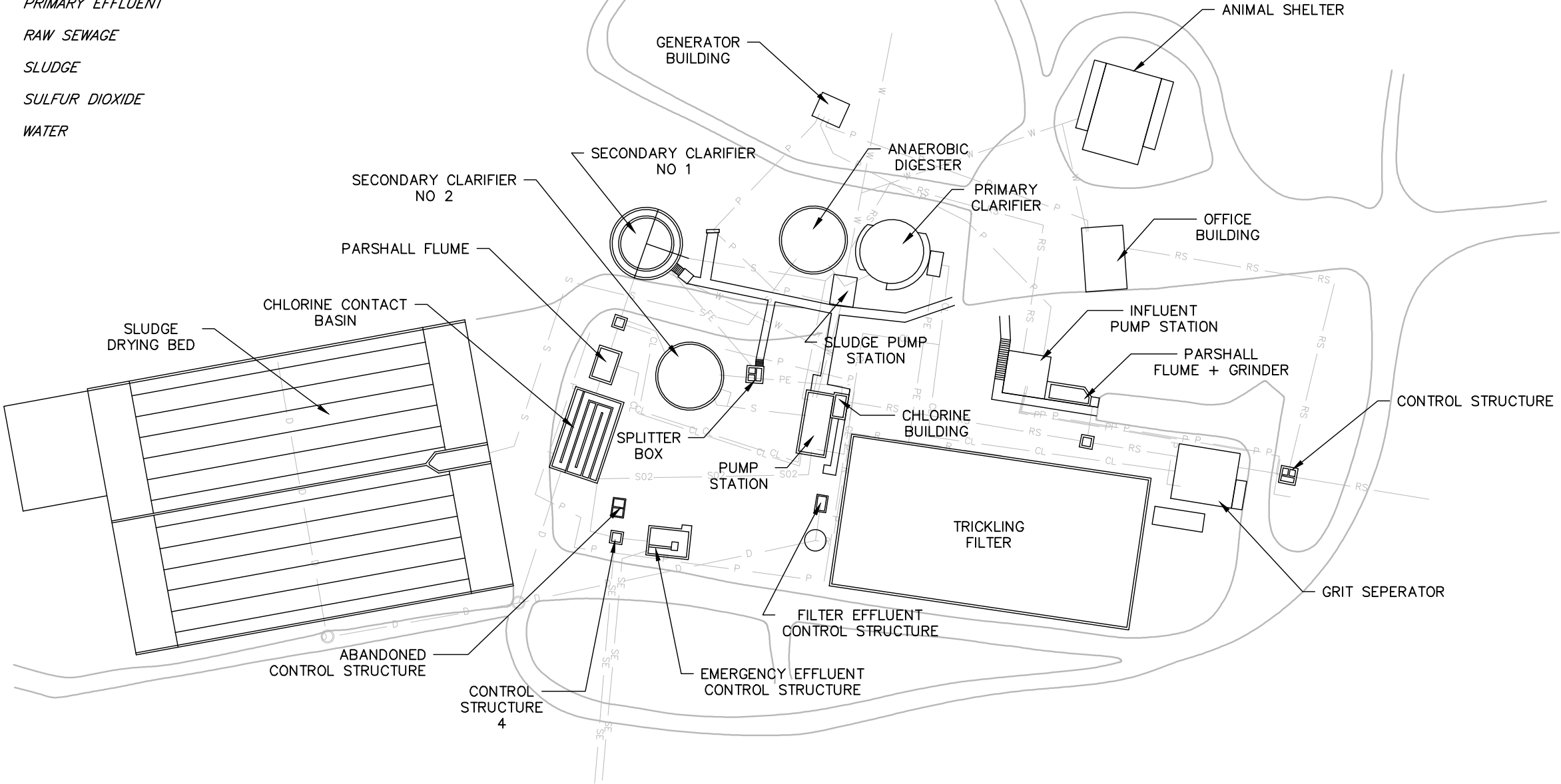


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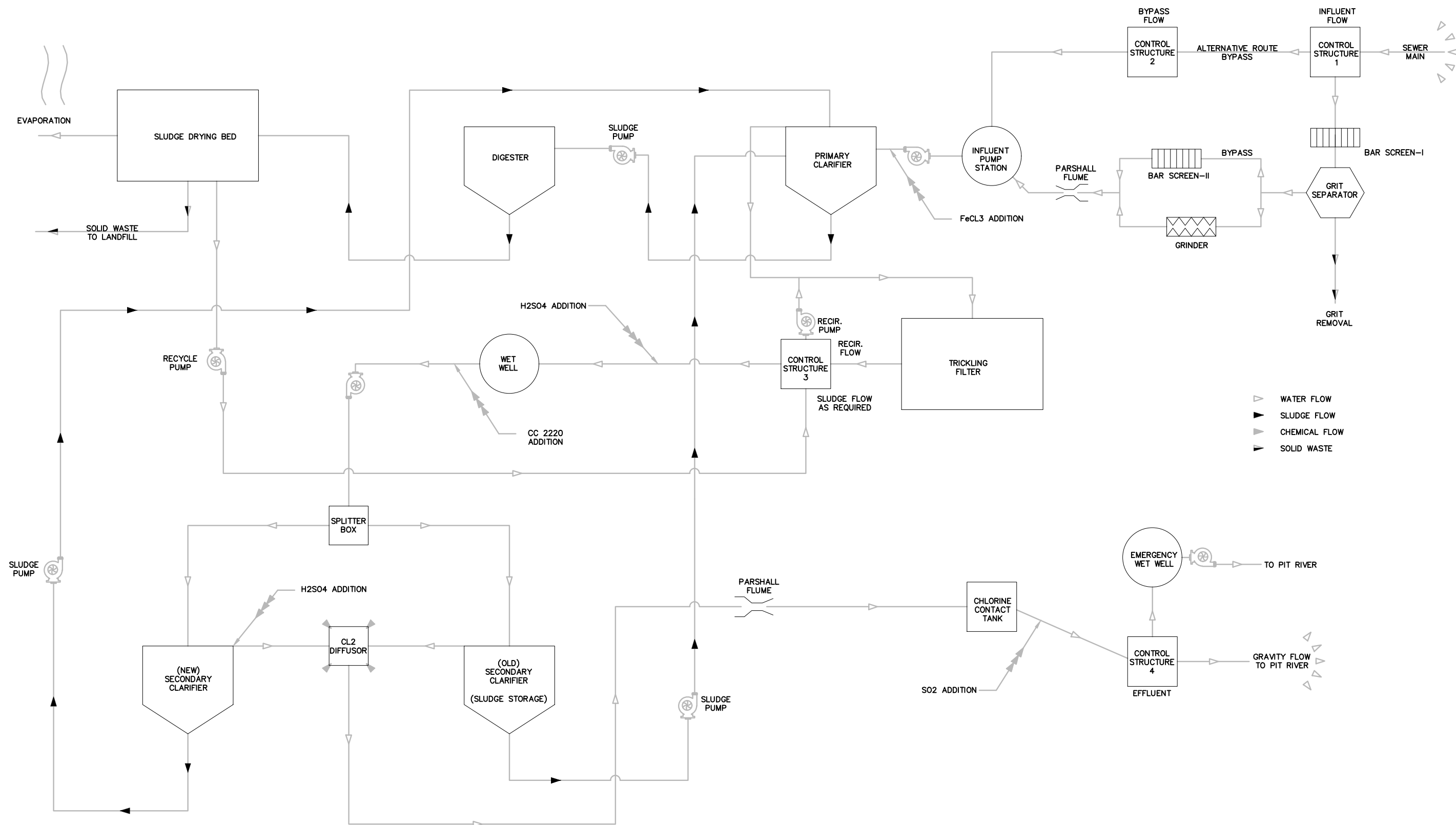
- CL CHLORINE
- D DRAIN
- P POWER
- PE PRIMARY EFFLUENT
- RS RAW SEWAGE
- S SLUDGE
- S02 SULFUR DIOXIDE
- W WATER



City of Alturas
Wastewater Planning Study
Alturas, California
December 2019

Existing Conditions
Plan View
SHN 518004
518004-ALTURAS-WW
Figure 4

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Table 1. Wastewater Treatment Plant System Components
Wastewater Preliminary Engineering Report
Alturas, California

System Component	Year Constructed	Years Renovated	Description of Renovation/Comments ¹
Grit Chamber	1978	-	
Grinder	1965	2006	Originally a comminutor; replaced with grinder in 2006
Influent Pump Station	1965	-	Replaced old pump station, which is now the recirculation pumps for the trickling filter
Primary Clarifier	Before 1965	1965, 2006	Only minor renovations in 2006
Trickling Filter	Before 1965	1965, 2006	Switched from gravel to plastic media
TF Recirculation Pumps	1978?	2006	
Secondary Clarifiers	1978 (#1) 2006 (#2)	-	Clarifier #1 is currently used for sludge storage
Sludge Pump Station	Before 1965	-	
Digester	Before 1965	1965	
Chlorine Contact Chamber	1978	2006	Additional parallel contact chamber
Sludge Drying Beds	1978	2006	Concrete basin with drain lines
Effluent Pump Station	1978	-	Used only when Pit River is high
1. See text for additional description and comments			

3.3 Condition of Existing Facilities

3.3.1 General

This section discusses the condition of the existing facilities, first with respect to regulatory compliance, then by individual major treatment component. Year of construction and renovation for each system component was provided in Table 1.

The existing WWTP was originally designed for an average dry weather flow (ADWF) of 0.5 million gallons per day (MGD), with a peak day flow of 1.0 MGD. Current ADWF is approximately 0.33 MGD, and peak daily flows have been as high as 0.91 MGD. Additional discussion about design flows can be found in Section 5.2.

3.3.2 Regulatory Compliance

The City's WWTP has been regulated by the Central Valley Regional Water Quality Control Board (CVRWQCB) under Waste Discharge Requirements (WDR) Order No. R5-2014-0033 (NPDES No. CA0078921). At the time the WDRs were issued, it was recognized that the City was not able to comply immediately with the effluent requirements for copper, zinc, and total coliform. The City has had compliance schedules to meet effluent limits for copper and zinc since 2006 in the permit or Time Schedule Orders (TSO). Therefore, Time Schedule Order (TSO) R5-2014-0034-01 (as amended by Order No. R5-2015-0111) was issued by the CVRWQCB. This TSO gave interim compliance limits for copper, zinc, and total coliform. The final compliance date for copper and zinc was May 18, 2020.

A review of effluent water quality data for 2015-2018 revealed that effluent water quality often did not meet the regulatory limits stipulated in the WDR and TSO for the following constituents: Aluminum, copper, zinc, BOD, TSS, and toxicity. The City was assessed \$15,000 in civil liability for effluent limitation violations of copper and zinc in 2015. The \$15,000 was treated as a permanently suspended administrative civil liability since the City completed a compliance project designed to correct the violations.

After the Draft PER was released and before this Final PER was completed, the CVRWQCB issued on April 1, 2020 a new permit under WDR Order No. R5-2020-0004, which provides interim limits through March 31, 2030 for the following constituents: Chronic Whole Effluent Toxicity, BOD, TSS, and ammonia. A CDO R5-2020-0005 was also issued and which carries the interim effluent limits for copper and zinc with a compliance date of May 18, 2020 and establishes new interim effluent limits for total coliform and aluminum with a final compliance date of March 31, 2025.

The City has had historical challenges meeting the chronic toxicity with nine out of 36 toxicity tests since 2014 having failed. The newly issued permit has the following compliance schedule for chronic toxicity:

- Until March 31, 2030, chronic whole effluent toxicity shall not exceed 16 toxicity units and a percent effect of 25% at 6.25% effluent, any endpoint as the median of up to three consecutive chronic toxicity tests within a six-week period.
- After March 31, 2030, chronic whole effluent toxicity shall not exceed 1 toxicity units and a percent effect of 25% at 100% effluent for any endpoint as the median of up to three consecutive chronic toxicity tests within a six-week period.

Suspected reasons for the toxicity violations have included ammonia and, most recently, surfactants. Other potential causes include chlorine, metals, non-polar organics, other treatment chemicals, and total dissolved solids. The most recent Toxicity Identification Evaluation by Pacific EcoRisk dated August 2019 from samples collected July 8 and 10, 2019 concluded the following:

"Toxicity was persistent with a 54.4% reduction in growth relative to the laboratory control...By weight of evidence, the combined visual observation of foam during air sparging and the overall improvements in cell growth from all treatments suggest that surface-active compounds (i.e., surfactants) may be causing or contributing to toxicity. Further analysis of all treatments for anionic and nonionic surfactants is recommended."

The CVRWQCB has indicated that if the City continues to discharge to the Pit River, the renewed WDR permit (R5-2020-0004) includes effluent limits for previously nonregulated constituents which include arsenic, Bis (2-ethylhexyl) phthalate, ammonia, and nitrate+nitrite. As discussed later in this PER, significant upgrades to the WWTP would be required to address these new effluent limits.

3.3.3 Headworks

The headworks consists of a manual bar screen, a grit removal chamber, and a grinder. The manual bar screen is located in the inflow channel to the grit chamber and is in good condition. The bar screen is cleaned manually daily as needed.

The grit removal chamber is a detritus tank, which is constant-level, short detention settling tank. Grit is removed from the bottom by a rotating rake arm that sweeps the grit to a collection sump where the grit is pumped to grade level into a wheelbarrow. The existing flow vanes, piping, grit pump, and rake are old and beyond their useful life. Much of the exposed metal shows significant signs of rust. While there is a bypass around the grit chamber, the grit chamber has never been taken offline for cleaning according to City staff.

The grinder grinds solids into smaller pieces that can be removed in the primary clarifier. This was originally a comminutor installed in 1965 but was replaced with a grinder in 2006 and appears to be in good condition. A bypass channel adjacent to the comminutor has a manual bar screen and allows for the grinder to be taken offline or for use as a high flow bypass if needed.

In general, the in-ground concrete structures housing the grit chamber and grinder are in good condition. This includes handrails and safety railing.

Figure 6 shows the bar screen and grit chamber. Figure 7 shows the grinder.



Figure 6. Existing bar screen and grit chamber



Figure 7. Grinder and bar screen bypass.

3.3.4 Influent Pump Station

The influent pump station consists of a concrete wet well and adjacent dry well. In the dry well are three centrifugal pumps with draw horizontally out of the wet well. One pump is rated at 350 gallons per minute (gpm) and is used for low flows. The other two pumps are rated at 750 gpm and are used for high flows and for redundancy. The drywell pumps and piping are shown in Figure 8.

It is not known whether the pumps have been replaced since original installation in 1965. The pumps appear to be past their useful life. In the last couple of years, the pumps have been frequently offline and in need of repair. The City is currently in the process of replacing all three pumps for proper redundancy. The pumps currently do not have variable frequency drives (VFD). When the pumps turn, the full flow occurs which leads to short circuiting in the primary clarifier as discussed in the following section.

The original piping from the pumps to the primary clarifier has been replaced with a new alignment due to a leak in the original piping. The old pipe was removed from the influent pump station and the pipe capped. The new piping consists of 12-inch-diameter polyvinyl chloride (PVC) much of which runs along the wall as shown in Figure 9. The interior PVC piping, which is restrained with chains and runs vertically from the dry well to grade level, was intended to be temporary but has been in place since 2006. In December 2006, the pipe failed and flooded the dry well. A shut-off float has been installed in the pump station to turn off the pumps if the pipe breaks inside the building. Failure of the pipe could allow raw wastewater to spill inside the lab and electrical rooms, which are located at the ground floor. Between the influent pump station and

the primary clarifier, the PVC pipe was laid at a shallow depth (16-20 inches), and a boulder barrier has been placed to keep vehicular traffic from driving over the pipe alignment.



Figure 8. Influent pump station.

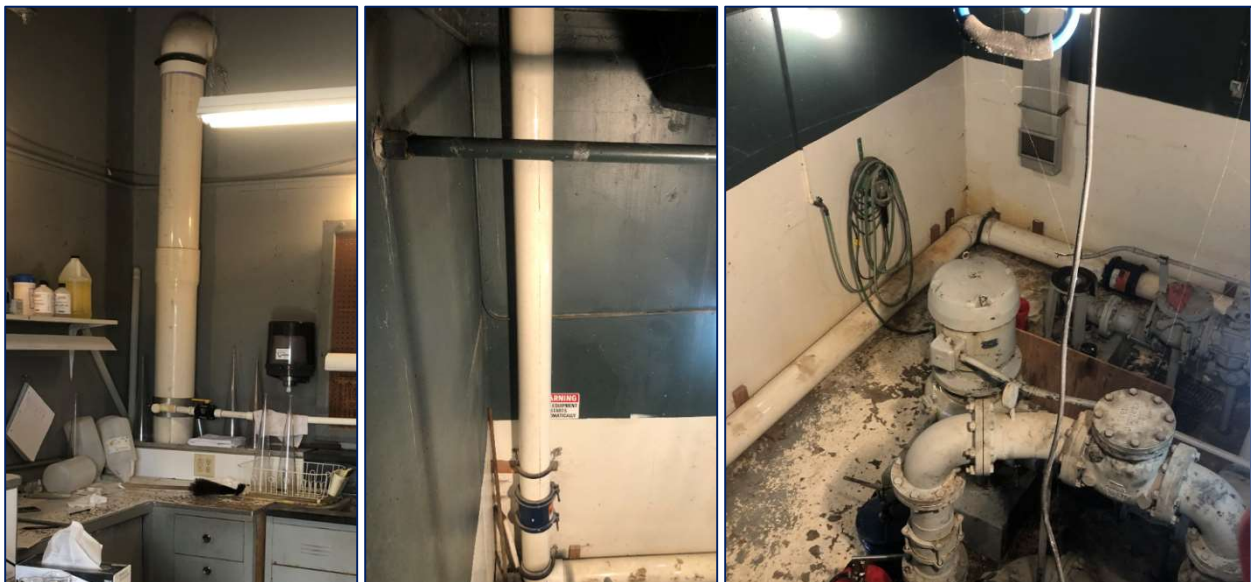


Figure 9. "Temporary" PVC force main from influent pump station.

Modifications to the influent pump station were needed to allow for installation of the new discharge piping. This retrofit has eliminated the ability of the operator to use a crane to remove pumps for servicing. Currently, the operator has to pull the pump and carry it up a narrow staircase with a ninety degree turn in it.

3.3.5 Primary Clarifier

The 30-ft-diameter primary clarifier has been in place since before the major WWTP upgrade in 1965, at which time it received significant upgrades, including a new skimming mechanism, new weir baffles, new scum box, and new drive housing. Minor upgrades were done in 2006. The internal raking and skimming mechanisms, while generally performing adequately, are well beyond their expected life, and many of the parts are difficult to find. The walkway associated with the drive housing is constructed of wood, which is currently in poor condition. Photos of the primary clarifier are shown in Figure 10.



Figure 10. Primary clarifier.

With only one primary clarifier, there is no redundancy. Currently, the clarifier experiences short circuiting which leads to inadequate solids removal. These solids end up in the trickling filter. The solids removal process utilizing a grinder and primary clarifier leads to significant amounts of plastics in the primary sludge. According to operations staff, the high amounts of plastics and other non-organic material creates problems in the poorly functioning anaerobic digester (described further in a later section).

3.3.6 Trickling Filter

The City's WWTP has utilized a trickling filter (TF) for biological treatment since before 1965. The existing trickling filter is rectangular in shape and uses fixed emitters separated into two zone. From at least 1965 to 2006, the trickling filter media consisted of gravel. In 2006, the gravel was replaced with random dump plastic honeycomb-shaped media (Bio-Pac SF#30), new underdrains were installed, and sidewalls were rehabilitated. The existing filter has a footprint of approximately 10,400 square feet and has a media depth of six feet. Figure 11 shows the photos of the trickling filter.

Effluent from the trickling filter is pumped either back to the TF for recirculation or to the secondary clarifier. Photos of the pump station are shown in Figure 12. In 2006, the pumps were upgraded with new motors and VFDs. While the pump station is old and may not meet current code requirements, it appears to be working adequately.



Figure 11. Trickling Filter (top) showing media (bottom left) and exterior wall where cracking has occurred in the CMU block wall (bottom right).

The issues with the trickling filter are as follows:

- Flow is not well distributed over the existing media.
- Wind affects the spray from the emitters, and the operator has created windrows to block wind.
- Freezing conditions can occur on the top of the media during winter, leading to reduced treatment efficiency.
- Operator access to sprayers is unsafe, since the operator must walk on the plastic media.
- Chemical addition is needed for pH adjustment.
- There is inadequate recirculation during wet weather months, leading to reduced treatment efficiency.
- The west wall of the trickling filter appears to have tilted, with cracking showing at the northwest and southwest corners.
- While the trickling filter has two zones and, therefore, technically has redundancy, there is no interior wall separating the two zones.
- The second zone does not have windrows in the media so it more susceptible to wind effects than the first (western) zone.
- There is inadequate nitrification occurring in the trickling filter.



Figure 12. Recirculation flow control structure (left) and pump station to secondary clarifier (right).

3.3.7 Secondary Clarifiers

Secondary Clarifier #1 was constructed in 1978. In 2006, a redundant new secondary clarifier (#2) was constructed. Since that time, clarifier #1 has generally been offline and used as sludge storage, which has removed the redundancy that was originally intended. Figure 13 shows clarifier #1, and Figure 14 shows clarifier #2.

Issues with clarifier #1 are as follows:

- The internal structure, including weirs, rake arms, and skimmer arms are showing significant signs of rust due to lack of use.
- Concrete surface of the clarifier structure is showing signs of pitting.
- The scum pipe valve is difficult to open and close.
- The skimming arm and scum baffle have been removed.

Clarifier #2 appears to be in good working order.



Figure 13. Secondary Clarifier #1 (currently offline).



Figure 14. Secondary Clarifier #2.

3.3.8 Sludge Pump Station

The sludge pump station has been in service since before the 1965 upgrade. Sludge from the trickling filter and secondary clarifiers is pumped from each respective unit through the sludge pump station to the primary clarifier. Sludge from the primary clarifier is pumped through this station to the digester. Sludge from the digester is then pumped to the sludge drying beds.

There is only one pump in the sludge pump station which does all the pumping in the various directions described above. The operation is manual and requires valves to be opened/closed to allow for the sludge to be moved in the proper direction. The piping is old, and the pump, while operational, appears to be well beyond its useful life. See photos in Figure 15.



Figure 15. Sludge pump station and piping gallery.

3.3.9 Digester

The anaerobic digester is in a significant state of disrepair and is not functioning as originally designed. It predates the 1965 upgrade, but upgrades performed in 1965 include a new sludge mixer and redwood roof. Figure 16 shows the current conditions of the digester.



Figure 16. Digester.

The digester is no longer functioning in an anaerobic state due to significant openings from missing plywood panels on the upper sides and roof. The shaft and blade of the mixer have broken away and fallen onto the digester floor. The supports for the mixer have failed, causing the mixer motor to drop a few inches. Operations staff has installed 4-inch by 4-inch wood boards to provide support to the mixer motor to ensure

it does not fall into the digester. Operator access to the roof is dangerous due to the missing or loose plywood panels.

3.3.10 Sludge Drying Beds

The sludge drying beds were constructed in 2006, replacing ones that were constructed in 1978. There are two separate beds located adjacent to each other. Each bed is concrete lined with a sloped bottom. Four-inch-wide slot drains run longitudinally to allow water to drain from the sludge. The slot drains contain a three-inch-diameter perforated pipe covered by 3/8-inch pea gravel to prevent solids from draining. Figure 17 shows the existing sludge drying beds.

The sludge drying beds appear to be prone to plugging in the slot drain gravel, causing slow drainage. At times, this has required storing sludge in secondary clarifier #1 due to lack of capacity. The gravel in the slot drains was recently replaced in at least one of the beds and has improved the drainage ability. The design of the slot drain appears to cause frequent plugging requiring more frequent gravel replacement.

The drain water pump station dates from 2006 and appears to be functioning properly. This pump station pumps drain water from the drying beds to the primary clarifier.



Figure 17. Sludge drying beds.

3.3.11 Disinfection

Since the effluent is currently discharged to the Pit River, the disinfection process includes chlorination and dechlorination. Chlorination is performed using chlorine gas, and dechlorination is performed using sulfur dioxide gas. In 2006, the chlorine contact chamber was expanded to include a second flow path for effluent which provides more capacity during wet weather flows and helps maintain proper contact time. The system is functioning properly. However, the cost to obtain gaseous chlorine and sulfur dioxide has increased significantly in recent years due to changes in the supplier and the hazardous nature of both transporting and storing the gas. Figure 18 shows the chlorination building. Figure 19 shows the chlorine contact chamber.



Figure 18. Disinfection building



Figure 19. Chlorine contact chamber. Chlorine injection occurs in the upper structure in the background and dechlorination (sulfur dioxide injection) occurs at the chamber outfall in the foreground.

3.3.12 Outfall

The outfall consists of two pipes that discharge to the Pit River. Under most flow conditions in the Pit River, the WWTP effluent is discharged by gravity directly to the Pit River. When water levels in the Pit River are very high, the effluent is directed to a pump station is pumped to a higher elevation discharge pipe. The effluent pump station is used very infrequently.

3.3.13 Backup Power System

A diesel generator installed in 1978 provide backup power to the WWTP in the event of a power outage. Operations staff has indicated that it is old and not currently tested and is, therefore, not reliable. The structure housing the generator appears to be in good condition.

3.3.14 SCADA System

The WWTP currently operates manually with no Supervisory Control and Data Acquisition (SCADA) system. There are alarms which alert the operator by phone in case of pump failures.

3.3.15 Plant Classification

The Alturas WWTP is currently classified as a Class II facility, requiring a Grade II operator. As discussed later in this report, the difficulty of finding licensed wastewater treatment operators with Grade III or higher factors significantly into the treatment methods being investigated.

3.4 Financial Status of Existing Facilities

The 2006 WWTP upgrade, which included the trickling filter and secondary clarifier, was funded through a loan from the Clean Water State Revolving Loan Fund (SRF). The current principal balance is approximately \$800,000, and annual payments are \$95,844. The SRF loan requires that the funded components be operated and maintained for 20 years. Under the current repayment schedule, the loan will be paid off in 2029.

3.5 Water/Energy/Waste Audits

No energy or waste audits have been performed at the WWTP. A Sanitary Sewer Evaluation Study (SSES) was recently performed as part of the grant funding for this PER. The results of the SSES showed that there is significant infiltration and inflow (I/I) into the system due to high groundwater and age of infrastructure. However, it was determined that it would be more cost effective to address the I/I flows at the WWTP rather than construct I/I reduction projects in the collection system. The SSES results along with an augmented analysis are summarized in Appendix 1.

4.0 Need for Project

4.1 Health, Sanitation, and Security

The main need for the project is to eliminate the effluent water quality violations that the City has experienced over the last several years. The CVRWQCB has indicated that the Pit River is a sensitive water and would prefer to see the City use land disposal for the effluent. Due to the frequency of violations of the current effluent limits, the CVRWQCB is concerned with the City's ability to meet the current and future effluent limits if they continue to discharge to the Pit River, as demonstrated by the new effluent limits for arsenic, Bis (2-ethylhexyl) phthalate, ammonia, nitrate+nitrite in the renewed WDR Order No. R5-2020-0004.

4.2 Aging Infrastructure

The secondary need for this project is due to aging infrastructure at the WWTP. As described in Section 3, a number of the unit processes are very old and not functioning properly. There is a need to rehabilitate or replace many of the processes. Operator safety issues are also of concern as discussed on Section 3.

4.3 Reasonable Growth

The City of Alturas has had a relatively stable population that has been slowly declining over the last couple of decades. Since there is not significant change in population expected within the planning horizon of this report, growth is not a factor.

5.0 Alternatives Considered

5.1 Initial Alternatives Development

The overall approach was to develop up to three main alternatives that meet the goals stated in Section 4. The main alternatives considered centered around continuing to discharge to the Pit River or utilize land disposal through a combination of evaporation, percolation, and irrigation for the treated effluent. Initially, the preferred direction by the City was to maintain the WWTP and its current secondary level of treatment for all alternatives; however, the WWTP could not be upgraded above its current classification in order to keep the same operator grade level. Later in the process, with staff changes at the City, an additional alternative of decommissioning the existing WWTP and building a simplified WWTP offsite was also suggested. A do nothing alternative was not considered because such alternative would not bring the City into compliance with the current WDR/NPDES permit.

Other alternatives that are typically reviewed include the following:

- Optimize current WWTP operations
 - This was not pursued since some of the treatment process units, e.g. the digester, are in such poor condition that only operational changes would not be sufficient.
- Interconnect with other existing systems
 - There are no other nearby systems.
- Build new centralized facilities for regional/joint management
 - There are no other nearby systems.
- Develop centrally managed decentralized systems, including small cluster or individual systems.
 - Since Alturas is a small community, a decentralized system was not considered given that the topography and infrastructure layout lend itself well to a single WWTP location and there is no advantage to a decentralized system.

After much discussion and research, three main alternatives were developed for further analysis:

- Alternative 1: Upgrade WWTP with Pit River discharge
- Alternative 2: Upgrade WWTP with land discharge
- Alternative 3: New WWTP with land discharge

The specific details of each alternative, along with analysis of subalternatives, are described in the following sections. The next section discusses the general design criteria which apply to all the alternatives and provided a specific framework for cost basis.

5.2 General Design Criteria and Constraints

Various design criteria are applicable to multiple or all alternatives and are discussed in this section. Design criteria that are specific to individual alternatives are discussed in the respective sections.

The existing WWTP is designed for an average dry weather flow of 0.5 million gallons per day (MGD) and a peak flow of 0.8 MGD. Based on influent flow data from 2008-2017, current average dry weather flow is approximately 0.33 MGD, and recent daily peak flows have been as high as 0.91 MGD. Instantaneous peaks have been up to 1.2 MGD. For this study, the alternatives were designed around 0.5 MGD for design dry weather flow and 1.0 MGD for peak day flow.

Based on the current permit and discussions with CVRWQCB regulators regarding likely future regulations under each alternative, the following treatment levels for BOD and TSS were used as a basis for design:

- BOD: 10 milligrams per liter (mg/L) for Alternative 1, 30 mg/L for Alternatives 2 and 3.
 - Under the current permit, BOD of 30 mg/L is allowed when the dilution ratio in the Pit River is greater than 20:1.
- TSS: 10 mg/L for Alternative 1, 30 mg/L for Alternatives 2 and 3.
 - Under the current permit, TSS of 30 mg/L is allowed when the dilution ratio in the Pit River is greater than 20:1.

Several other constituents with effluent limits in the new WDR Order No. R5-2020-0004 are not included in the general design but apply only to Alternative 1, where the City would continue to discharge to the Pit River. Those constituents are divided into two categories: (1) New constituents in the 2020 permit that were not in the 2014 permit; and (2) constituents with recalculated effluent limits in the 2014 permit that are also in the 2020 permit.

The new constituents in the 2020 permit include the following: ammonia, nitrate+nitrite, toxicity, bis (2-ethylhexyl) phthalate, and arsenic. These constituents are new in the 2020 permit which was issued after the draft PER was started. Therefore, they are not used as part of the design considerations, but there will discussion of the general approach and additional costs/work required to meet these new limits in Section 5.4.1.

The recalculated constituents from the 2014 permit that are also in the 2020 permit are the following: copper, zinc, and aluminum. Additional work may be required for Alternative 1 as discussed in Section 5.4.1 to meet final limits in the 2020 and future permits.

Another regulatory constraint is the Central Valley Salinity Alternatives for Long-Term Sustainability (CV-SALTS) program, regulated under individual basin plans and Resolution R5-2020-0057. CV-SALTS is divided into salt (salinity) and nitrate control programs.

The City's requirements under the salt control program within CV-SALTS is currently limited to participation in a regional Phase I Prioritization and Optimization (P&O) Study, which would continue regardless of which alternative is chosen. The purpose of the Phase I P&O Study is to determine whether the City would be subject to permit limits for salinity or continue operating with no additional salinity removal measures needed.

The City's requirements under the nitrate control program within CV-SALTS varies by alternative. Under Alternative 1, where the City will maintain its NPDES permit, the 2020 permit includes nitrate+nitrite limits as described above. WWTPs with land disposal of effluent, such as described in Alternatives 2 and 3, typically have not had nitrate limits unless that there is demonstrated or reason to suspect potential adverse impacts to groundwater quality, as required by the State's antidegradation policy. For Alternatives 2 and 3, we do not anticipate adverse impact to groundwater from nitrates, based on various factors, including depth to groundwater and experience with similar facilities. However, monitoring wells placed around the disposal ponds will allow for ongoing monitoring of groundwater quality and determination of impacts, if any. Additional detail regarding the nitrate issue for Alternatives 2 and 3 is provided in the respective sections below.

Redundancy and emergency bypass ability are generally required under USEPA guidelines for WWTPs. As such, redundant units and/or bypass piping/channels are incorporated in the design.

One of the key constraints affecting the design is that the existing WWTP must remain in operation at all times during construction of the new facilities. Another major constraint is the proximity of the WWTP to the Alturas airport. Current FAA regulations (AC 150/5200-33A) generally prohibit open water bodies within a 5,000-foot radius from the nearest operation area (that is, edge of runway) for an airport serving piston-powered aircraft due to concerns with bird strikes. Open water attracts various bird species, and the presence of these birds can increase collisions between aircraft and birds. The 5,000-foot buffer is shown on the figures for Alternatives 2 and 3 with respect to proposed pond locations.

5.3 Sub-Alternatives Analysis

5.3.1 General

Analysis of various options within each alternative was needed in order to provide definition to each alternative. For example, “upgrade WWTP sludge digestion” is inadequate, for there is more than one option for doing so. The following sections discuss the evaluation that was done for each of the major process items. Some of the options are applicable to one, two, or all three of the major alternatives as discussed in each section.

Many of the various components of the existing WWTP are well beyond their useful life. While some of them could be rehabilitated with minor costs, it is our opinion that this simply delays needed upgrades. Therefore, the approach was to recommend full replacement when it would be in the best long-term interests of the City for the respective alternatives.

5.3.2 Headworks and Primary Treatment

5.3.2.1 Alternatives 1 and 2

The existing headworks and primary treatment consist of bar screen, grit chamber, grinder, and primary clarification. The following options were considered:

- Upgrade or replace the various unit processes as needed (grit chamber, primary clarifier); and
- Replace grit chamber and primary clarification with screening.

Upgrade of the grit removal would entail either replacing the internal mechanisms or replacing the entire unit with a new package grit removal system. The primary clarifier currently has no redundancy, so a replacement would entail construction of two new clarifiers. Another option is to construct a new primary clarifier and rehabilitate the existing one. Rehabilitation of the existing primary clarifier would consist of replacement of the internal mechanism.

Replacing the grit chamber and primary clarification with a single screening unit process could be done either by (1) installing the screening unit within the existing grit chamber along the influent channel downstream of the bar screen, or (2) installing a new unit on the existing bypass line located along the south side of the grit chamber. There are several types of screening mechanisms on the market. After some research and discussion with vendors, a screw screening unit, such as the Parkson Helisieve (www.parkson.com) or the Franklin Miller Spiralift (www.franklinmiller.com), was determined to be the most appropriate type of screen (see Figure 20 for a typical screw screen). Only coarse (6 mm, or ¼ inch)

screening would be needed for effective grit and solids removal based on the downstream process of trickling filtration. The existing grinder would be decommissioned.



Figure 20. Typical helical screw screen.

The main advantage of replacing the grit and primary clarifier with a screw screen would be to reduce the inorganic solids loading (namely from plastics) to the digester and the sludge drying beds. According to operations staff, the presence of significant levels of plastics in the digester has caused upsets in the process. This process will increase the solids generation and handling needs at the front end of the WWTP.

After review of the options with respect to preliminary costs, constructability issues, and future operations and maintenance (O&M) costs, it was determined that replacing the existing headworks and primary treatment with a single screw screen with 6 mm opening size with a capacity of up to 1.0 MGD would be the best overall solution. For this study, it was assumed that the new screen would be placed on the bypass line in a new concrete chamber. The influent channel would have an emergency bypass. The screw screen includes an integral dewatering system conveys the dewatered solids to a hopper, which could be a dumpster.

5.3.2.2 Alternative 3

Under Alternative 3, the existing WWTP is decommissioned and all wastewater is conveyed to a new aeration pond treatment facility offsite. This will require a pump station located at the WWTP along the existing entrance road. Two options were considered: (1) Provide coarse screening to remove grit and solids prior to pumping or (2) pump all raw influent sewage to the new WWTP.

The first option, providing coarse screening prior to pumping, was determined to be the best option. Removal of grit and solids prior to pumping will reduce solids buildup in the forcemain and in the aeration ponds. The coarse screen system would be the same as for Alternatives 1 and 2 but placed in a different location as described under the Alternative 3 description.

5.3.3 Influent Pump Station

As discussed in Section 3, the current influent pump station (IPS) is generally in good shape other than the pumps and the forcemain to the primary clarifier. The needed upgrades include replacing all three pumps and the forcemain, redirecting the pumped flow to the new trickling filters. The pumps would have the same as the current flows, i.e. one pump at 350 gpm and two pumps at 750 gpm. All three pumps would include VFDs. This option is applicable only to Alternatives 1 and 2. Alternative 3 would require a package influent pump station as described later.

5.3.4 Trickling Filter

The main considerations with the existing trickling filter are twofold: (1) the media does not appear to provide the level of treatment originally anticipated and (2) due to the configuration with the bubbler sprayers, the surface area of the filter is not fully utilized. Further, one of the side walls is not in good condition and needs to be repaired.

Several options were evaluated, including the following:

- Replace the media with gravel as used previously;
- Replace the media with another engineered plastic media;
- Replace the emitters with a spray system that provides better flow distribution over the media;
- Repair the existing outer wall that is leaning slightly;
- Replace the existing trickling filter with new, more traditionally round shaped units;
- Replace the trickling filter with another treatment process, such as aerated ponds, activated sludge or membrane bioreactors.

Various factors were applied during evaluation of these alternatives, including operational issues, winter weather effects, redundancy, future regulations, and treatment efficiency. While rehabilitating the existing trickling filter is feasible, it was not favored based on the history of the existing unit and industry norms. Replacing the trickling filter with new round units is preferable because the round units will have the following advantages:

- No need for operator access to repair fixed emitters.
- Improved treatment efficiency from more efficient flow distribution.
- Improved nitrification.
- Improved redundancy.

Replacing the trickling filter with aeration ponds, which does not require a change in operator grade level, became Alternative 3. The aeration ponds needed to be moved offsite due to the proximity to the Alturas airport. Upgrading the trickling filter with another process such as activated sludge or membrane bioreactors would change the classification of the WWTP to a Class III Facility, thereby requiring a Grade III operator. This was not a preferred option for the City, since finding a Grade III operator is very uncertain as discussed earlier.

For Alternatives 1 and 2, we are recommending two parallel trickling filter treatment trains, each with two trickling filters. Each has a capacity of 0.5 MGD. Each filter is 30 feet in diameter and looks similar to what is shown in Figure 21. The media would be cross flow rigid media that can support a worker. The first filter of each train is for BOD removal, and the second is for nitrification. The new trickling filters would be placed just northwest of the existing digester (see Alternative 1) in order to allow the existing trickling filter to remain in operation during construction.

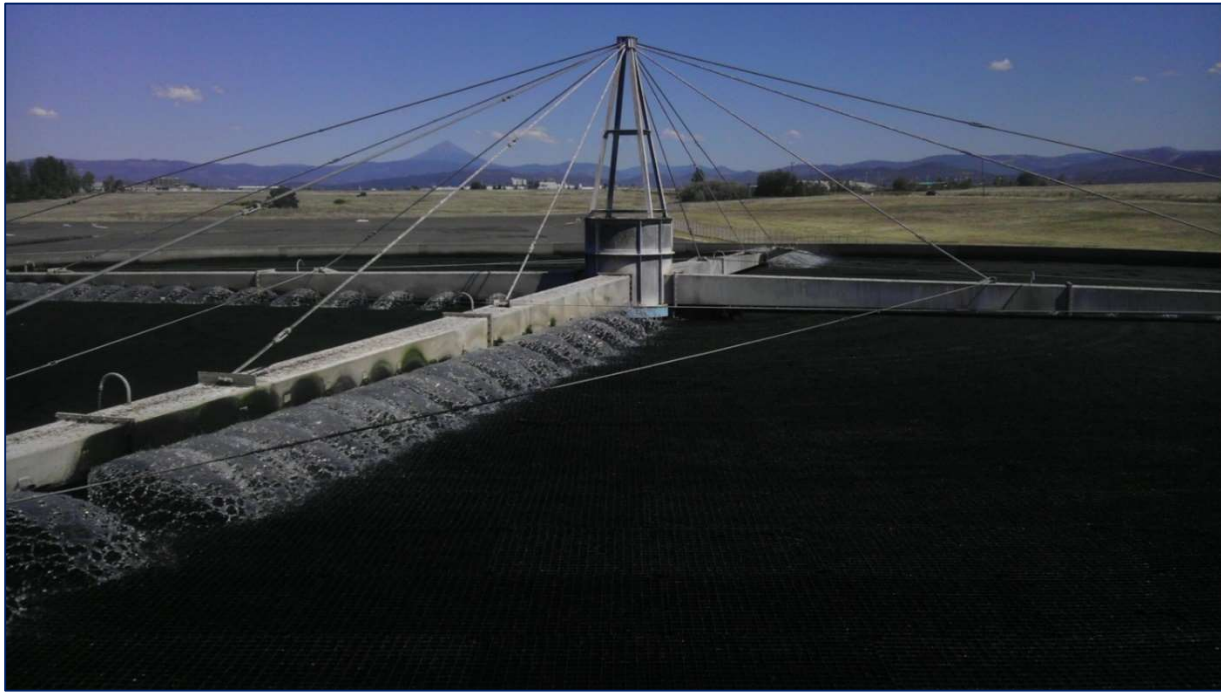


Figure 21. Typical trickling filter showing media as is recommended for Alturas.

5.3.5 Secondary Clarifier

Since clarifier #1 is not currently being used for clarification but for sludge storage, there is no redundancy. Two options were considered: (1) rehabilitate the clarifier #1 and (2) construct a new redundant clarifier. With the existing concrete tank and associated piping being in good condition, it was determined that replacement of the internal clarifier mechanisms was preferred. The internal mechanisms include the overflow weirs, rake arm, skimming arm, catwalk, motor and drives. The sludge pump is also assumed to require replacement. This is applicable to Alternatives 1 and 2.

5.3.6 Digester

Three options were considered for sludge digestion:

- Rehabilitate the existing anaerobic digester;
- Replace the digester with new anaerobic digesters;
- Replace the digester with new aerobic digesters.

Rehabilitation of the existing digester would include replacing the cover and upper side walls, replacing the mixer, and installing a gas flare. Anaerobic digestion produces methane gas, which must either be flared off

or otherwise contained. One method of methane use at WWTPs is to burn it as a fuel source for power generation. However, the quantities produced at the Alturas WWTP are too low to make beneficial use economical. Therefore, a gas flare with an associated air quality permit is needed. Rehabilitating the existing digester does not add redundancy without adding a second unit. Based on a cursory review of potential costs, it was determined that full replacement was a preferred option.

Sludge digestion can be performed either aerobically or anaerobically. There are advantages and disadvantages from an operational perspective for each method. The key deciding factor is that anaerobic digestion produces methane gas that must be flared off, requiring an air quality permit. It was therefore determined that replacement with two new aerobic digesters (for redundancy) was the selected option to use for Alternatives 1 and 2.

5.3.7 Disinfection

Alternative 1 is the only alternative where effluent disinfection is needed. The CVRWQCB confirmed that disinfection is not required for land disposal that utilizes evaporation and percolation as the effluent disposal method. Current and future permit requirements (for direct discharge to the Pit River) would include a limit on total residual chlorine, which necessitates dechlorination if chlorine is used as the disinfectant.

Currently, the City uses gaseous chlorine and sulfur dioxide for disinfection and dechlorination, respectively. As discussed in Section 3, cost and availability for remote deliveries of small quantities has become an issue. Other disinfection alternatives that were evaluated include the following:

- Liquid sodium hypochlorite (disinfection) and sodium bisulfite (dechlorination)
- Ultraviolet disinfection
- Onsite generation
- Ozone generation

Liquid sodium hypochlorite is commonly used as a disinfectant in order to avoid the regulatory safety requirements around chlorine gas. Sodium hypochlorite comes in a 12.5% solution, which does degrade with time, thus requiring deliveries at least quarterly. Sodium bisulfite for dechlorination also comes in a liquid solution.

Ultraviolet (UV) disinfection is also a popular disinfection method. It does require a fair amount of electrical energy, but no dechlorination is needed since UV does not introduce chlorine, which is harmful to aquatic life. UV units have historically been prone to fouling, requiring a fair amount of maintenance. One company, Enaqua, has introduced a non-contact UV unit, which reduces maintenance costs (www.enaqua.grundfos.com).

Onsite generation is a method of applying electricity to a solution of salt and water to produce sodium hypochlorite. This is one of the safest disinfection methods, as inert salt is stored on site. However, dechlorination is still required, which would entail using liquid sodium bisulfite.

Ozone is a powerful disinfectant, more powerful than chlorine, but is not a stable substance. This requires onsite generation of ozone, which is an energy intensive operation. No dechlorination is needed for this method.

After a cursory review of these methods, it is recommended that the disinfection system be changed to use liquid sodium hypochlorite and sodium bisulfite for Alternative 1. This is based on ease of operation, safety concerns, and cost. Conversion would entail replacing the chlorine and sodium hydroxide cylinders with liquid sodium hypochlorite and sodium bisulfite in the existing chlorination/dechlorination room and installing new chemical feed pumps. It is assumed the existing feed lines are in good condition. Frost protection of the lines near the injection points must be considered during final design.

It should be noted that for Alternatives 2 and 3, if disinfection were required in the future, the process could easily be added to the proposed configurations. Further, dechlorination is not needed if chlorination is used since the effluent is not being discharged to a water body.

5.3.8 Sludge Drying Beds

Operator feedback on the condition of the sludge drying beds has been mixed. The City has recently replaced the media located in the slot drains in half of the beds, and this improved the ability of the beds to drain more consistently. Replacement of the drying beds with a revised design to improve leachate drainage was considered but ultimately dismissed due to the recent media replacement. While not a superior design, the existing beds appear to be functioning well when properly operated.

5.3.9 Land Discharge

For Alternatives 2 and 3, land discharge of treated effluent is proposed. Land discharge can be in a variety of forms, including evaporation, percolation, irrigation, or a combination of these. Often, winter storage is needed, especially when irrigation is used, since irrigation is not needed during the winter months. A significant amount of land is needed for land discharge. The amount needed depends on various factors including wastewater flow (dry weather and wet weather), climate conditions (precipitation and evaporation), soil conditions (percolation), and crops (if irrigating). In most cases, the public must be prevented from having access to the disposal sites due to public health concerns. Groundwater quality must not be impacted, which typically restricts nitrogen levels. Compliance with this would be done through required groundwater quality monitoring. Also, no overflow from storage ponds or runoff from irrigated areas is allowed, as this would be considered a direct discharge to a water body which is a violation of land discharge permit.

Various factors were considered when evaluating each property. These factors included the following:

- Federal Aviation Administration (FAA) wildlife hazard mitigation regulations;
- Mapped floodplain;
- Proximity to the Pit River
- Depth to groundwater
- Soils
- Irrigation operations

The FAA discourages the creation of new water bodies within 5,000 feet of an airport, specifically 5,000 feet from the edge of runways. The water bodies act as bird attractants and, therefore, could increase hazardous bird strikes with aircraft. However, there are methods that are used to deter birds from coming to the water bodies and that are currently used at other pond locations near airports. These techniques include wires strung across the ponds, visual deterrents, audio deterrents, covers, and floating balls. A discussion of various techniques (except floating balls) is provided in Appendix 2. Wires strung across ponds

would require periodic, though infrequent, removal to provide maintenance access to the ponds. The floating balls are plastic balls partially filled with water and are used frequently to provide cover on ponds and open tanks to deter birds. The balls also reduce evaporation by approximately 90%. Mechanical evaporators are another solution that increases evaporation but also has a side benefit of deterring birds due to its noise when operating.

The Federal Emergency Management Agency (FEMA) has mapped the 100-year and 500-year floodplains along the Pit River (see Figure 3). All of the proposed parcels for land discharge are outside, or mostly outside, of the mapped floodplain. No pond structures would be proposed within the 100-year floodplain.

The CVRWQCB expressed concern based on recent litigation that if percolation ponds are placed too close to the Pit River then the risk of hydraulic connectivity with the river increases. Such hydraulic connectivity is difficult to ascertain without groundwater tracer studies. As such, it was determined on a qualitative basis to place any percolation areas as far away as possible from the Pit River.

Depth to groundwater is an important factor when especially percolation is used for disposal. No degradation of the groundwater quality is allowed. The greater the depth to groundwater, the better, since soils can provide additional treatment while water percolates through the soil matrix.

Soils, especially their permeability and infiltration capacities, are important factors in locating percolation and irrigation areas. The greater the infiltration ability of the soil, the smaller the infiltration area and pond volumes are required. Initial review of soils was conducted using the online web soil survey from the National Resources Conservation Service (NRCS), which provide maps of soil types and soil characteristics. Additional in field infiltration tests were performed at the preferred locations (discussed below), which provided data for preliminary sizing of percolation areas.

Irrigation is commonly used as an effluent disposal method. However, this can only be done during the growing season, which requires winter storage, with the exception of a forested irrigation, which is done in some locations year-round. Typically, irrigation is used for a fodder crop, such as grass or alfalfa hay. In general, wastewater utilities that utilize irrigation own the land where irrigation and winter storage take place, but it is possible to have long-term agreements with third parties to take the effluent for irrigation use. An irrigation disposal system often adds operational costs for the management and/or operation of the irrigation system by utility staff. Irrigation is used when recycled water has significant value and when more passive methods, such as evaporation and percolation, are not as feasible.

With these considerations in mind, several parcels were identified as potential land discharge locations. These are shown on Figure 22 and include the following:

- Adjacent Modoc Refuge property
- Mill Site (Duck Ponds)
- Irrigation Field along Highway 299
- Modoc County Landfill
- Manteca property along County Road 54

The Modoc National Wildlife Refuge property located immediately west of the City's WWTP site was initially the preferred alternative for land discharge. Percolation tests were conducted, and it was determined that

the site soils have decent permeability, indicating that approximately three to six acres of ponds would be needed to provide the area required for percolation and evaporation. Mechanical evaporators were also considered to increase evaporation.

The depth to groundwater at the Refuge property, especially during the winter months was not known. The distance from the proposed pond locations and the Pit River is under 1,000 ft. Without further studies, including installing monitoring wells and performing groundwater tracer studies, this location left an uncertain potential risk, as indicated by CVRWQCB staff, for hydraulic connectivity with the Pit River.

The Refuge property is within a 5,000-foot radius of the Alturas airport, which is within the FAA wildlife hazard mitigation zone. After informal discussion with the FAA by City staff, it was determined that getting FAA approval for siting the ponds would be very difficult. Modoc Refuge staff indicated that the Federal government, which owns the property, could not sell the land outright but could swap with equivalent land. The City does not own land that would likely be considered equivalent on a monetary basis to this property. A significant portion of this property is located within the 100-year mapped floodplain. For these reasons, it was concluded that pursuing this option was not favorable.

The Mill Site consists of a former mill site where portions of the parcel have been used to create duck hunting ponds which are currently filled with potable groundwater. After review of the site, this was not considered a feasible alternative because of the potential of public contact with the treated effluent would require higher levels of treatment and because the hunting club was not willing to take the effluent without compensation.

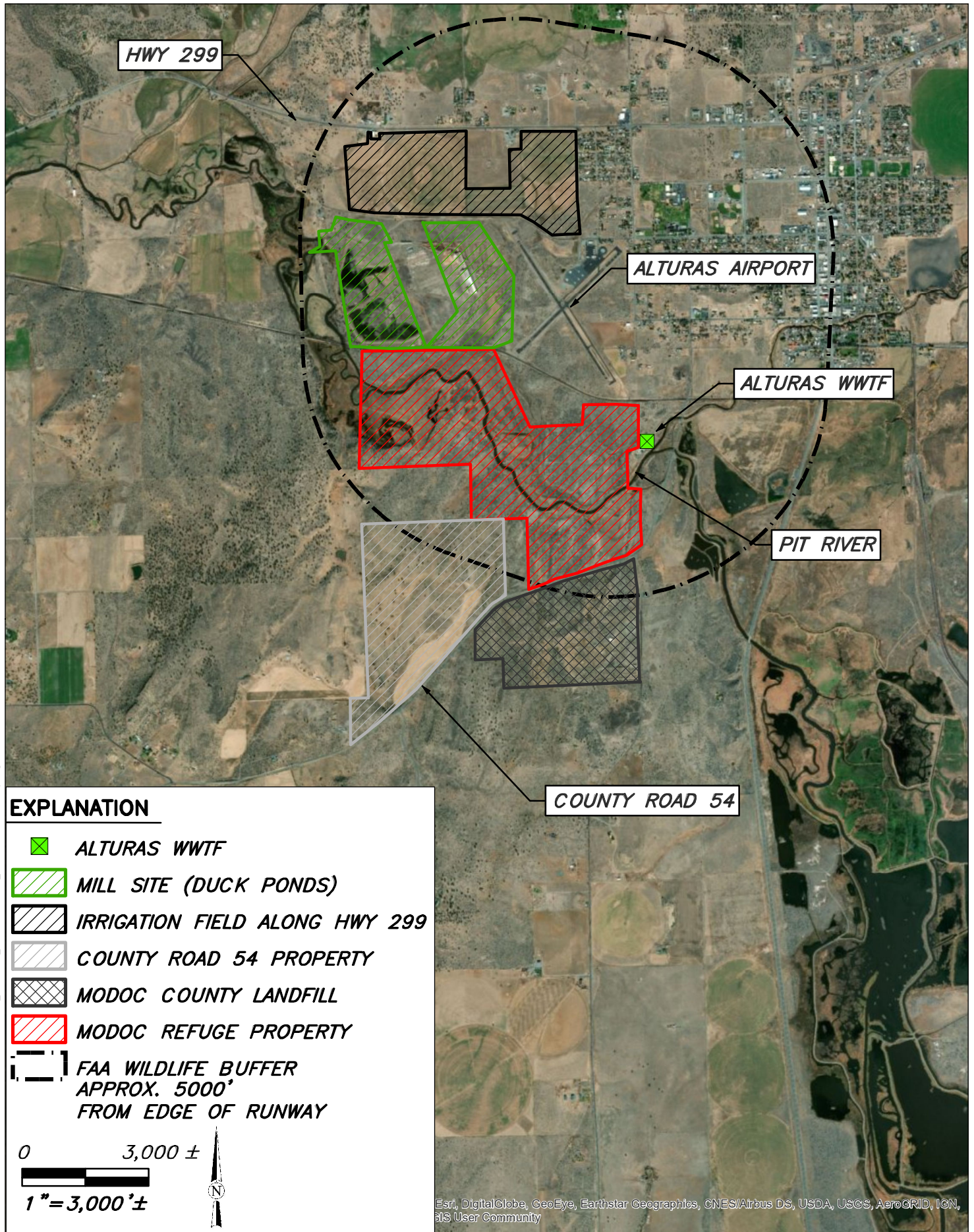
The irrigation field along Highway 299 was considered for irrigation disposal. However, winter effluent storage would still be needed, and additional land would be needed since this parcel does not have sufficient area for storage ponds. These ponds would need to be placed close by, which would be within the 5,000-foot wildlife mitigation zone around the airport. Therefore, this parcel was eliminated from further consideration.

According to the NRCS soil survey, well draining soils are found in large areas south of the Pit River along County Road 54. One site, owned by Modoc County, is where the County Landfill is located. Portions of the property are conducive to having percolation ponds, but are in too close proximity to the landfill, which would likely not be acceptable to the CVRWQCB. This site was therefore eliminated from further consideration.

A 278-acre parcel along County Road 54 (Manteca Property, see Figure 22) with well-draining soils was identified. This property is located within two miles of the WWTP. Based on a preliminary review of pumping costs, it was determined that pumping to a land disposal site within five miles of the WWTP was acceptable. Percolation tests were performed (see Appendix 3) which confirmed the ability of this site to provide good percolation opportunity, with a calculated minimum required infiltration area of approximately three acres, with two two-acre ponds being recommended.

The Manteca property is hilly with nestled shallow gulches which are conducive to siting percolation ponds and aeration treatment ponds. The depth to groundwater varies based on the overlying topography, which varies by more than 60 feet between the upper elevations to the lower elevations. At the lower elevations

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City of Alturas
Preliminary Engineering Report
Alturas, California
February 2020

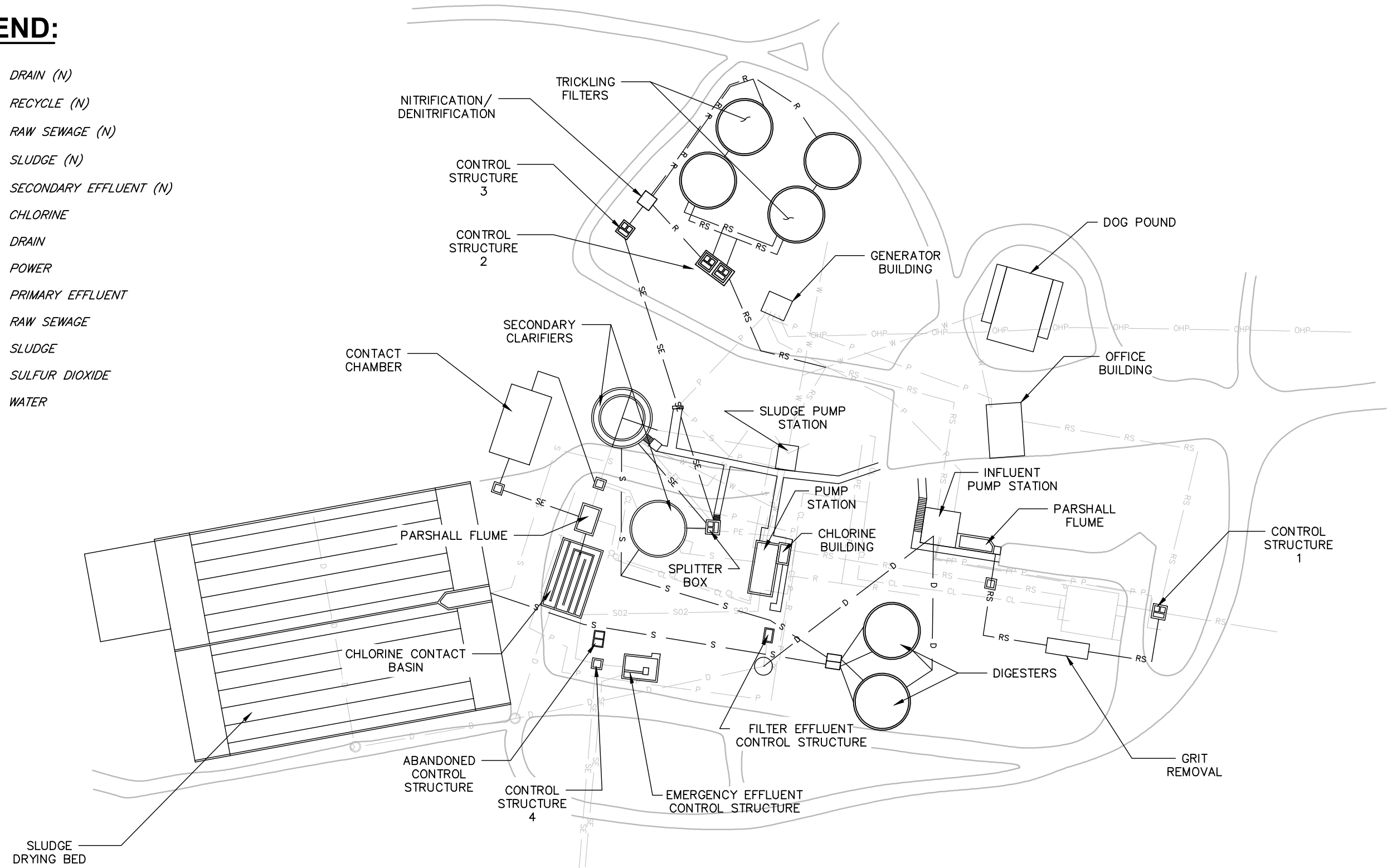
Proposed Disposal Sites
with FAA Perimeter
SHN 518004
FIGURE_22

Figure 22

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LEGEND:

— D —	DRAIN (N)
— R —	RECYCLE (N)
— RS —	RAW SEWAGE (N)
— S —	SLUDGE (N)
— SE —	SECONDARY EFFLUENT (N)
— CL —	CHLORINE
— D —	DRAIN
— P —	POWER
— PE —	PRIMARY EFFLUENT
— RS —	RAW SEWAGE
— S —	SLUDGE
— S02 —	SULFUR DIOXIDE
— W —	WATER



PLAN
0 FEET 60



City of Alturas
Wastewater Planning Study
Alturas, California

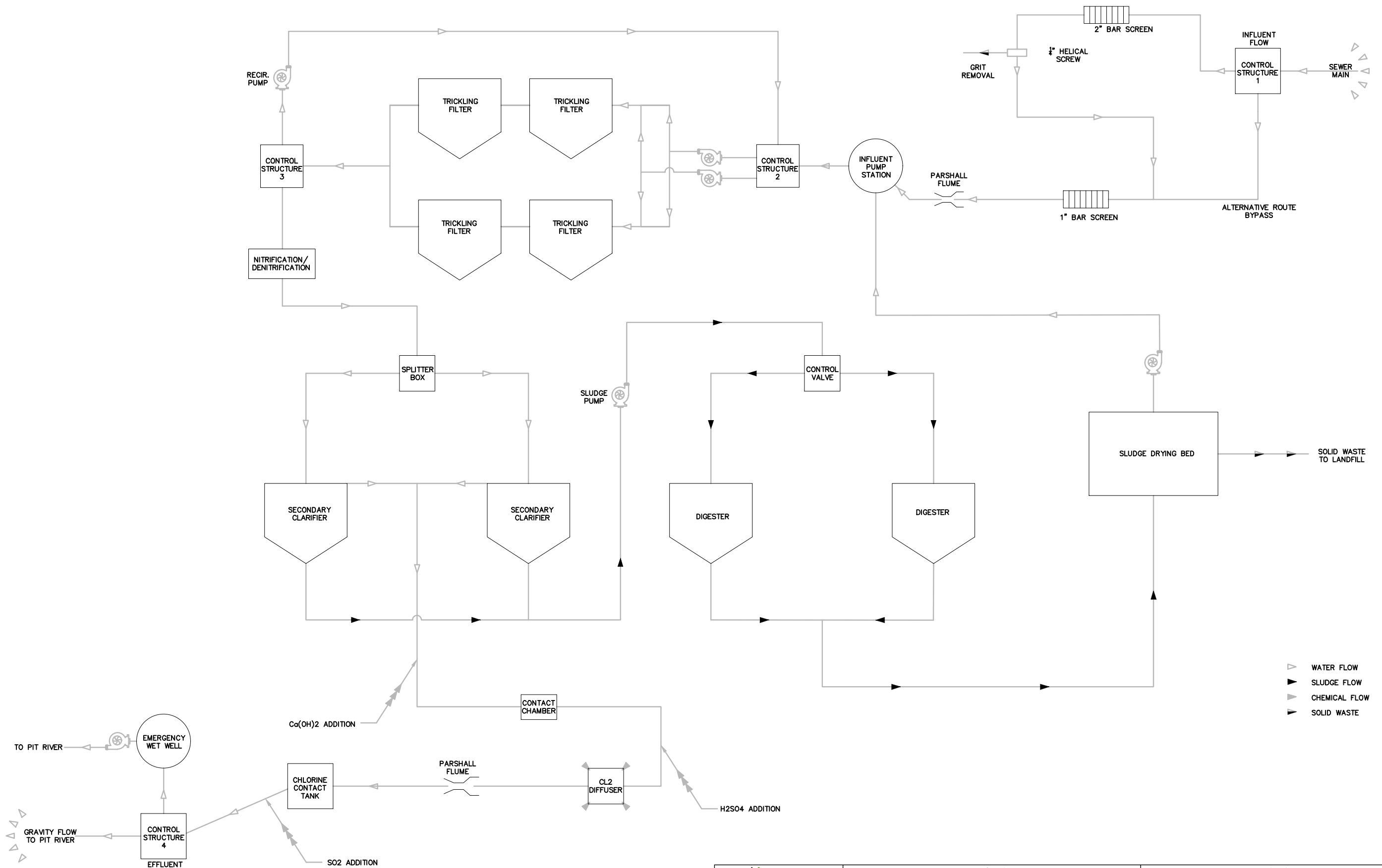
December 2019

518004-ALTURAS-WW

Alternative 1
Plan View
SHN 518004

Figure 23

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City of Alturas
Wastewater Planning Study
Alturas, California

December 2019

518004-ALTURAS-WW

Proposed Flow Diagram
Alternative 1
SHN 518004

Figure 24

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along County Road 54, groundwater was encountered at a depth of 9.5 feet in Exploratory Pit (1), while no groundwater was encountered up to 11 feet in Exploratory Pit (2). The ground elevation where the percolation ponds would be located is approximately 60 feet higher than the ground elevation around Exploratory Pit (1). The straight distance to the Pit River is approximately 2,300 feet. Based on the results of these initial investigations, the Manteca property was considered to be the most suitable location for effluent disposal and was therefore considered for Alternatives 2 and 3. The City has contacted the owner and an agreement for an option to purchase is currently being negotiated. Monitoring wells, which would be installed during final design in the vicinity of the percolation ponds, would be used to determine depths to groundwater at the location of the percolation ponds.

For Alternatives 2 and 3, two evaporation/percolation ponds will be constructed for disposal of the treated effluent, each designed to infiltration at least 0.5 MGD. Each pond will have a bottom area of two acres with a depth of six feet. Embankment side slopes will be 3H:1V on the inside and 2H:1V on the outside. It is anticipated that soils will be suitable and a balanced cut and fill construction will be feasible to minimize construction costs. Preliminary sizing calculations with respect to infiltration are presented in Appendix 3.

5.3.10 Other options evaluated

Other options that were considered but dismissed as either infeasible, too complicated, or too uncertain include the following:

- Polishing treatment surface wetland;
- Polishing treatment subsurface wetland;
- Equalization ponds
- Effluent disposal into geothermal reinjection wells.

Polishing treatment wetlands, both surface and subsurface, were considered for Alternative 1. They were determined not to be needed for Alternatives 2 and 3, since land discharge has fewer effluent limitations than river discharge. These wetlands provide additional treatment to remove BOD, TSS, metals, and nitrogen. The wetlands are not the ultimate disposal locations, as the effluent pass through them and must be sent to a discharge location at the downstream end. For Alternative 1, the discharge would be to the Pit River, and for Alternatives 2 and 3, the discharge would be percolation basins. It was determined that polishing wetlands would not be needed for Alternatives 2 or 3, since the proposed process would provide sufficient treatment and since land discharge has fewer regulatory requirements than a river discharge.

A surface wetland would have open water, which attracts birds. Since the surface wetland would need to be located near the WWTP, it would be within the FAA wildlife hazard mitigation zone, and would, therefore, be difficult to permit. Research into subsurface wetlands found that, while these can be effective for treatment, available removal data were difficult to find. Without performing pilot tests, which were not part of this study, it would be difficult to predict the specific treatment efficiency with certainty. Further, the ability of subsurface wetlands to treat constituents that are likely to be regulated in the future was unknown due to lack of data. Subsurface wetlands are also prone to plugging, which increases maintenance costs. Thus, treatment wetlands were not pursued further.

Equalization ponds were considered as an option to reduce flow spikes in the treatment train during high winter flows after a significant storm or snowmelt event. When flows are equalized, a relatively constant treatment flow can be maintained which can reduce the size of various unit processes. Based on flow data

from 2017, which has exceptionally high wet weather flows during the first half of the year, it was determined that equalization volumes of approximately 500,000 gallons would be needed. This would require large tanks or ponds, which would be a new open water source within the FAA wildlife mitigation zone. It was, therefore, determined that equalization ponds were not an economical option for peak flow reduction.

Another option considered was the disposal of treated effluent into an unused geothermal well. This has been used effectively in other locations. After discussion with City staff, it was determined that no geothermal wells for this use were available, as there are very few geothermal wells in Alturas and the surrounding area.

5.4 Alternative 1: Rehabilitate WWTP with River Discharge

5.4.1 Description

Alternative 1 consists of improvements to the existing WWTP and continued discharge to the Pit River. Based on an evaluation of the existing treatment processes (see Section 3) and the sub-alternative analyses discussed in Section 5.3, the improvements consist of the items listed in Table 2. A plan view showing the configuration of the upgraded facility is shown in Figure 23. A process flow diagram is shown in Figure 24.

In general, decommissioning of the unit processes is assumed to consist of removing internal moving mechanisms but keeping the existing concrete structures in place. The only exceptions are the trickling filter, which needs to be removed to make room for the new digesters, and the existing digester, which has a failing superstructure that needs to be removed and no exterior railing to prevent someone from falling into the below grade structure.

The new screw press screen, as discussed earlier, will consist of a screening unit such as the Franklin Miller Spiralift with integrated weather protection (-40 degrees Fahrenheit), integral screen washing, and dewatering unit. A below grade concrete structure will be constructed over the existing bypass line with an open concrete channel in which the screen is placed. The washed screenings will be conveyed to grade level where they will be dewatered and placed in a dumpster, which will be emptied on a weekly basis. A bypass channel will be incorporated to allow for the screen to be taken offline if needed.

The new trickling filters will be round concrete structures partly below grade using structured sheet, plastic crossflow media (by e.g. Brentwood). As discussed earlier, the trickling filters will consist of two parallel treatment trains of two units each, for a total of four 30-foot-diameter trickling filters.

Secondary Clarifier #1 will get new internal parts including rake arms, skimmers, weir plates, and motor due to the deterioration that has occurred from lack of use. The new weir plates would be of fiberglass material, rather than steel, to prevent future rusting.

The disinfection system would consist of two chemical feed pumps, one for chlorination and one for dechlorination. Full redundancy is not needed, since extra chemical feed pumps can be kept on hand for replacement in the event of a pump failure.

The new aerobic digesters will consist of two units for redundancy and will be located where the trickling filter is currently. A unit such as manufactured by WesTech is recommended.

Table 2. Alternative 1 Component Description
City of Alturas Wastewater Planning Study

Item	Additional Description
Decommission the grit chamber, grinder, and primary clarifier; replace with new screw press screening	The existing structures will remain while all mechanical equipment is removed. The new screen will be placed on the existing bypass line.
Improve influent pump station	Replace existing pumps with new pumps with VFDs ¹ ; replace PVC ² force main piping and redirect to the new trickling filter location; and install new crane hoist for pump removal.
Decommission existing trickling filter and replace with new round trickling filters	New filters to be located in different location to allow the existing filter to remain in service. New filters will be 30-foot diameter round units, two parallel treatment trains for redundancy and high flows, each train with two filters in series. Includes flow control box and recirculating pumps.
Install new nitrification/denitrification process	Assume packaged engineered unit process.
Rehabilitate Secondary Clarifier #1	Replace all internal moving mechanisms and weir plate.
Install new metals removal process	Use chemical precipitation process with pH adjustment.
Replace disinfection system	Switch from gaseous to liquid chlorination and dechlorination
Replace anaerobic digester with aerobic digester	Two units for redundancy are recommended. Placed at the location of the former trickling filters.
Replace associated yard piping and site electrical as needed to connect the relocated treatment processes	
1. VFD: Variable Frequency Drive 2. PVC: Polyvinyl Chloride	

For Alternative 1, limits on zinc, copper, aluminum, toxicity, bis (2-ethylhexyl) phthalate, ammonia, nitrate+nitrite, and arsenic must be considered in accordance with the 2020 permit. As discussed in Section 5.2, design considerations for these constituents apply only to this alternative. The following paragraphs describe the general approaches to each of these constituents.

Zinc and copper removal can be accomplished in a number of ways, including chemical precipitation or ion exchange. For this alternative, a chemical precipitation process is assumed since it is commonly used in industrial wastewater processes. Hydrated lime will be added to the secondary effluent coming from the secondary clarifier in order to bring the pH up to above 10.3. Zinc and copper will precipitate out as the effluent flows through a contact clarifier. The pH will be lowered using sulfuric acid prior to disinfection. Sludge will be sent to the sludge drying beds.

Aluminum, which is found in the polymer currently used to assist the existing trickling filter process, is assumed to be removed from the wastewater since the improved trickling filters will not require the coagulant.

Toxicity is not a particular constituent but can result from a number of causes, including chlorine, ammonia, non-polar organics, metals, other treatment chemical additives, surfactants, and total dissolved solids. It is anticipated that many of these potential causes will be reduced or eliminated as a result of the improved treatment process. If toxicity is persistent, additional study would be required.

Removal of bis (2-ethylhexyl) phthalate in municipal wastewaters has been researched recently, and studies have indicated that this phthalate can be removed by sorption to primary and secondary sludges. Additional removal may occur during the denitrification process. It would be anticipated that the recommended unit processes would provide sufficient treatment for this phthalate.

A new nitrification/denitrification process would be added to remove nitrogen to meet the final ammonia and nitrite+nitrate limits. Ammonia conversion to nitrate will take place partly in the trickling filter process, with the remaining conversion as well as the denitrification process (removal of nitrate+nitrite) will take place in a separate engineered nitrification/denitrification process such as the NitrOx+D process by Triplepoint Environmental.

Arsenic can be removed from wastewater through a number of means, and the state of the science is still emerging. The most prevalent method of arsenic removal is by chemical precipitation through the addition of ferric chloride followed by filtration. This process would likely be in series with the zinc and copper removal process described earlier.

5.4.2 Design Criteria

General design criteria were described in Section 5.2. Specific design criteria for individual components are discussed below.

The pumps for the influent pump station will match the existing, i.e. one at 350 gpm and two at 750 gpm. The head requirements will be determined during final design since the pumped flow would be directed to the new trickling filters.

5.4.3 Environmental Impacts

This alternative would disturb previously disturbed areas within and around the footprint of the existing WWTP.. Additional discussion of environmental impacts are discussed in the CEQA documentation.

5.4.4 Land Requirements

No new land is required for Alternative 1.

5.4.5 Potential Construction Problems

Potential construction problems include rock excavation for buried yard piping and site electrical and construction sequencing that does not allow for continued use of the existing WWTP during construction.

5.4.6 Sustainability Considerations

This section discusses any design consideration with respect to water conservation, water reuse, energy efficient design, operational simplicity, appropriate technology, and other considerations.

Since this alternative is an upgrade of the existing system with continued discharge to the river, there were no significant sustainability considerations that affected the design, other than the recommendation that variable frequency drives be added to the influent pump station pumps. Solar panels could be installed in the future on unused portions of this property to offset electrical costs.

5.4.7 Costs

Capital costs for Alternative 1 are shown in Table 3. Anticipated operational costs are presented in Table 4. Additional detail is provided in Appendix 4. It should be noted that the operational costs listed below exclude the current SRF Loan repayment of \$95,844 annually through 2028 since this cost is the same for all alternatives.

Table 3. Alternative 1 Opinion of Probable Project Costs^{1,2,3}
City of Alturas Wastewater Planning Study

Item	Description	Item Cost
1	Decommission/Demolition	\$25,000
2	Demolish existing trickling filter	\$41,000
3	Helical Screw Screen	\$309,000
4	Trickling Filters (4)	\$1,744,000
5	Nitrification/Denitrification	\$900,000
6	Aerobic Digesters (2)	\$772,000
7	Metals Removal	\$1,500,000
8	Influent Pump Station Improvements (pumps, VFD, force main)	\$200,000
9	Convert disinfection system	\$50,000
10	Refurbish Secondary Clarifier #1	\$300,000
11	Replace Generator	\$57,000
	Mobilization (12%)	\$708,000
	Subtotal:	\$6,606,000
	Contingency (30%)	\$1,982,000
	Subtotal Construction:	\$8,588,000
	Engineering/Construction Management (18%)	\$1,546,000
	Environmental/Permitting (5%)	\$430,000
	Administration/Legal (4%)	\$344,000
	Total Project:	\$12,890,000
1. See Appendix 4 for additional detail 2. Costs do not include upgrades to meet proposed effluent limits as described in the text. 3. Items not explicitly listed are incorporated with the listed items.		

Table 4. Alternative 1 Opinion of Annual Operational Costs^{1,2}
City of Alturas Wastewater Planning Study

Description	Item Cost
Labor	\$85,000
Chemicals	\$95,000
Testing/Reporting	\$35,000
Sludge/solids Hauling	\$2,000
Electrical Costs	\$28,500
Permitting	\$5,000
Annual O&M	\$250,500
1. See Appendix 4 for additional detail 2. Some minor operational costs have not been included because they are assumed to be relatively equal for all alternatives. Detailed operational costs will be shown in a Rate Study being performed as part of this study.	

5.5 Alternative 2: Rehabilitate WWTP with Land Discharge

5.5.1 Description

Alternative 2 consists of improvements to the existing WWTP (same as Alternative 1) but with land discharge. Based on an evaluation of the existing treatment processes (see Section 3) and the sub-alternative analyses discussed in Section 5.3, the improvements consist of the items listed in Table 5. A plan view showing the configuration of the upgraded facility is shown in Figure 25. Figure 26 shows the effluent force main alignment and disposal pond locations. A process flow diagram is shown in Figure 27.

The discussion under Alternative 1 provides additional detail regarding the new unit processes common between both Alternatives 1 and 2.

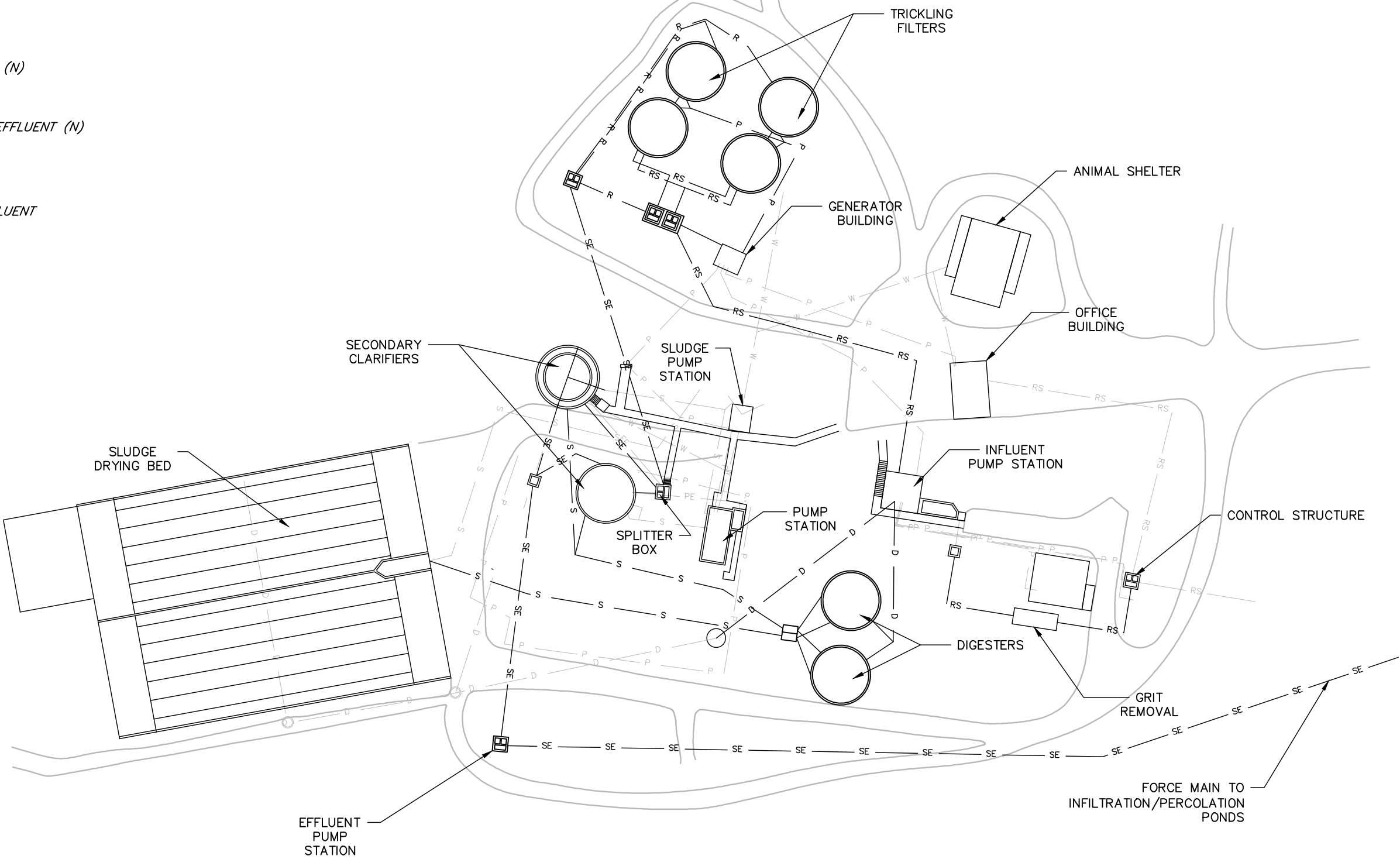
The new effluent pump station would be a package lift station located between the existing chlorine contact chamber and the Pit River outfall. A force main would run from the pump station along the WWTP access road to County Road 54, where it would run along the road shoulder, crossing the Pit River on the road bridge, to the entrance of the Manteca property. The force main would continue along the internal access road to the location of the evaporation/percolation ponds.

Effluent disposal at the site is recommended to be via **infiltration/evaporation ponds**. Average annual precipitation in the area is 13 inches, and average annual pan evaporation is 51 inches. This indicates that there is a net loss to evaporation from the surface. As discussed in the next section, percolation ability is significant and is estimated, based on field tests, to be in excess of 400 feet per year. The ponds will be located in a small, perched gulch near the upper elevations of the property where well-draining soils were encountered.

Two evaporation/percolation ponds will be constructed for disposal of the treated effluent, each designed to infiltration at least 0.5 MGD. Each pond will have a bottom area of two acres with a depth of six feet. Embankment side slopes will be 3H:1V on the inside and 2H:1V on the outside. It is anticipated that soils will be suitable and a balanced cut and fill construction will be feasible to minimize construction costs.

LEGEND:

— D —	DRAIN (N)
— R —	RECYCLE (N)
— RS —	RAW SEWAGE (N)
— S —	SLUDGE (N)
— SE —	SECONDARY EFFLUENT (N)
— D —	DRAIN
— P —	POWER
— PE —	PRIMARY EFFLUENT
— RS —	RAW SEWAGE
— S —	SLUDGE
— W —	WATER



PLAN

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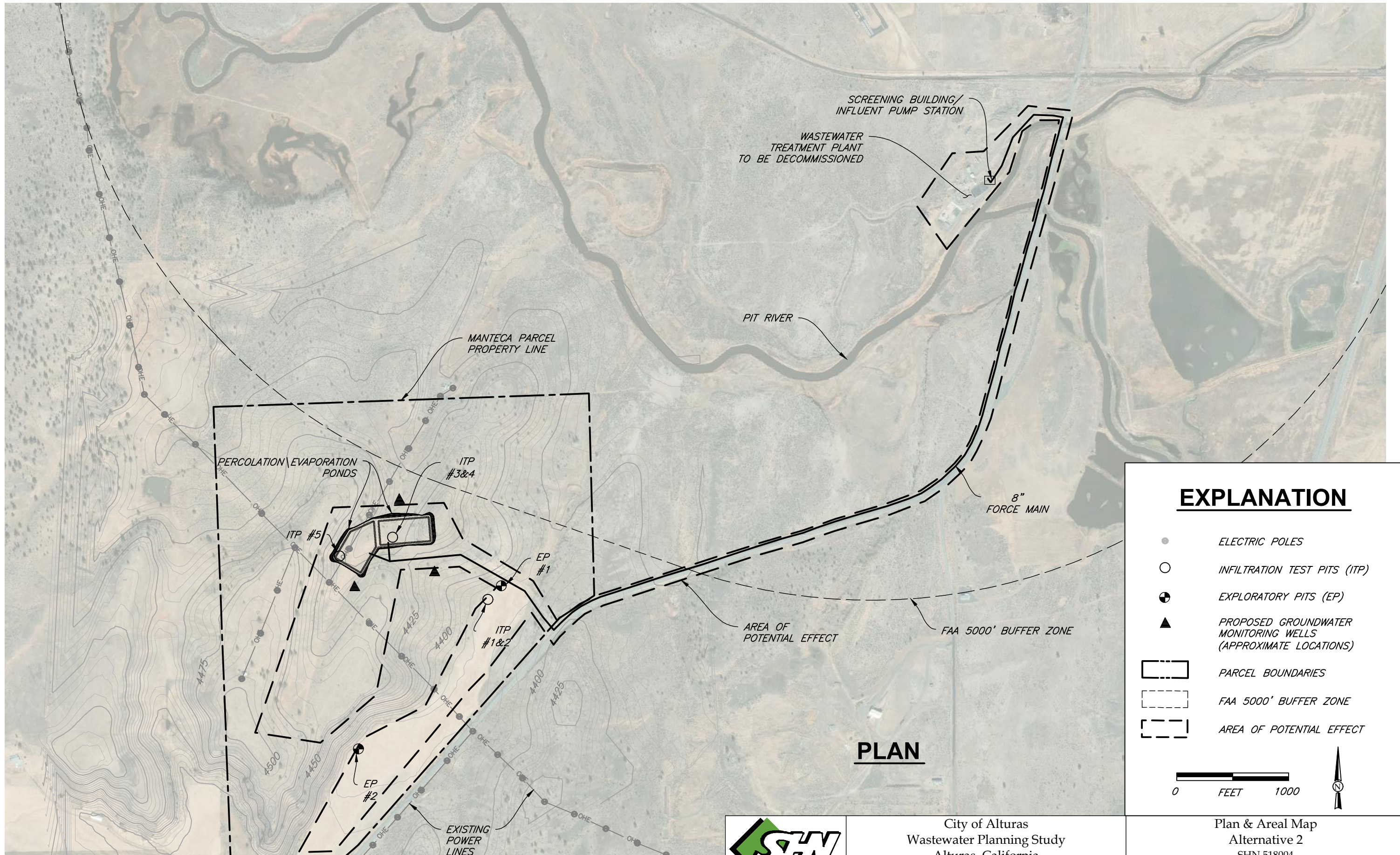
City of Alturas
Wastewater Planning Study
Alturas, California

December 2019

Alternative 2
Plan View
SHN 518004

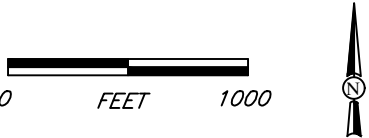
Figure 25

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EXPLANATION

- ELECTRIC POLES
- INFILTRATION TEST PITS (ITP)
- ⊙ EXPLORATORY PITS (EP)
- ▲ PROPOSED GROUNDWATER MONITORING WELLS (APPROXIMATE LOCATIONS)
- ▭ PARCEL BOUNDARIES
- - - FAA 5000' BUFFER ZONE
- - - AREA OF POTENTIAL EFFECT



PLAN



City of Alturas
Wastewater Planning Study
Alturas, California

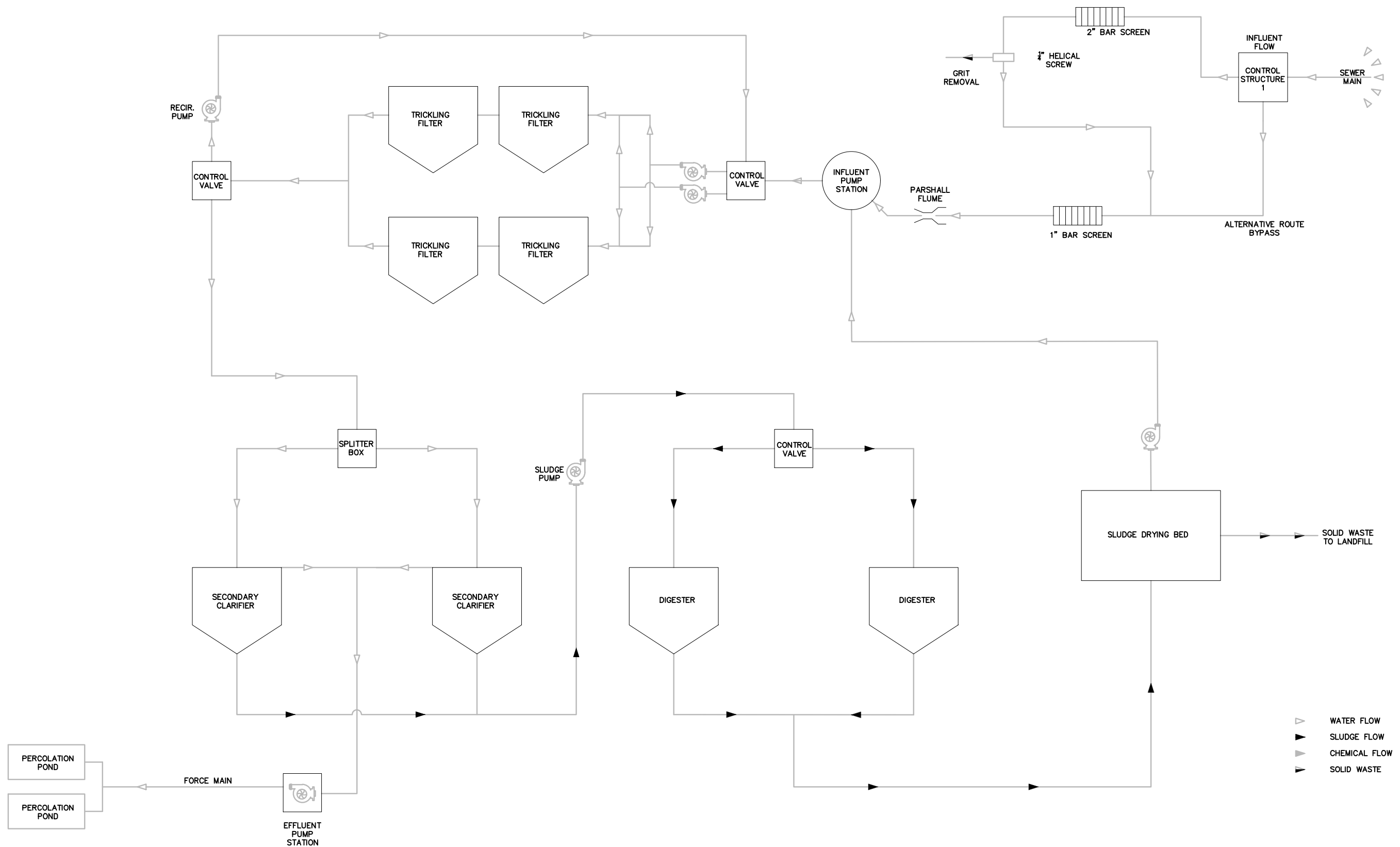
November 2020

518004-ALTURAS-WW2

Plan & Areal Map
Alternative 2
SHN 518004

Figure 26

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Table 5. Alternative 2 Component Description
City of Alturas Wastewater Planning Study

Item	Additional Description
Decommission the grit chamber, grinder, and primary clarifier; replace with new screw press screening	The existing structures will remain while all mechanical equipment is removed. The new screen will be placed on the existing bypass line.
Improve influent pump station	Replace existing pumps with new pumps with VFDs ¹ ; replace PVC ² force main piping and redirect to the new trickling filter location; and install new crane hoist for pump removal.
Decommission existing trickling filter and replace with new round trickling filters	New filters to be located in different location to allow the existing filter to remain in service. New filters will be 30-foot diameter round units, two parallel treatment trains for redundancy and high flows, each train with two filters in series. Includes flow control box and recirculating pumps.
Rehabilitate Secondary Clarifier #1	Replace all internal moving mechanisms and weir plate.
Decommission disinfection system	No disinfection will be required for this alternative.
Replace anaerobic digester with aerobic digester	Two units for redundancy are recommended. Placed at the location of the former trickling filters.
Replace associated yard piping and site electrical as needed to connect the relocated treatment processes	
New Effluent Pump Station	To be located at the current location of the effluent pipe.
New Force main	Eight-inch diameter pipe to the effluent disposal site
New Evaporation/Percolation Ponds	Two ponds able to infiltrate 0.5 MGD, approximately 2-acre bottom area for each pond.
1. VFD: Variable Frequency Drive 2. PVC: Polyvinyl Chloride	

Monitoring wells will be required to monitor groundwater quality and verify that the percolated effluent does not degrade groundwater quality. Existing groundwater quality will affect design criteria for final design. Typically, three monitoring wells are placed around the infiltration areas that will provide adequate coverage since exact groundwater flow direction is not known. Three wells would be placed around the pond locations. The depths of the monitoring wells would extend to the highest groundwater level in order to be able to take groundwater samples. Until further investigation is performed, groundwater depth is not known in this specific location where the wells will be located.

No electricity is needed at the infiltration/evaporation pond site. Therefore, no new electrical service is anticipated or included in the cost estimate. If site lighting were desired, Surprise Valley Electric has indicated that they would provide a service drop at no up-front cost.

5.5.2 Design Criteria

General design criteria were described in Section 5.2. Specific design criteria for individual components were discussed previously (those common with Alternative 1) or are discussed below.

The package effluent pump station would have three pumps, one at 350 gpm and two at 750 gpm, which matches the pump capacity configuration of the existing influent pump station. Total design head is estimated at approximately 200 ft. It is expected that the pump station will be a wetwell and drywell configuration using centrifugal pumps, but an option to use submersible pumps is left up to final design.

The force main would be eight inches in diameter to minimize headlosses and travel time and consist of high-density polyethylene (HDPE), except for the two Pit River crossings locations, where it will be ductile iron pipe. The trench for the force main will be approximately 18 inches wide and five feet deep.

The disposal ponds are based on a percolation value of one foot per day, which was estimated from field testing and applying a factor of safety of four. Additional discussion of the percolation test results is provided in Appendix 3. Groundwater monitoring wells would be required around the percolation ponds and installed prior to construction. This will help determine baseline groundwater quality, which would refine the design criteria for the final design.

With respect to the nitrate control program under CV-SALTS (see Section 5.2), the City would monitor nitrate at the monitoring wells around the disposal ponds. As aforementioned, nitrate impacts are not anticipated due to various factors, including depth to groundwater and experience with similar facilities. However, if the monitoring data indicate that additional nitrogen (specifically nitrate) removal is needed, this alternative allows for future upgrade to remove nitrogen, such as an engineered nitrification/denitrification unit process or seasonal irrigation. The specific solution, if needed, would be determined after review of the monitoring data results after at least two years of operation of the new facility or as directed by the CVRWQCB.

5.5.3 Environmental Impacts

Ground disturbing activities for this alternative are twofold: (1) Within the existing WWTP property, which has previously been disturbed, and (2) along County Road 54 and at the effluent disposal property. Additional discussion of environmental impacts are discussed in the CEQA documentation.

5.5.4 Land Requirements

A new parcel of land, located outside the Alturas city limits, will be required to site the disposal ponds. A parcel has been identified (County Road 54 property) as discussed earlier. Also, a utility easement or similar permission will be needed from Modoc County to place the force main along County Road 54.

5.5.5 Potential Construction Problems

As with Alternative 1, potential construction problems include rock excavation for buried yard piping and site electrical and construction sequencing that does not allow for continued use of the existing WWTP during construction.

Additional potential construction problems could be conflicts with other utilities along County Road 54.

5.5.6 Sustainability Considerations

This section discusses any design consideration with respect to water conservation, water reuse, energy efficient design, operational simplicity, appropriate technology, and other considerations.

Since this alternative is an upgrade of the existing system, there were no significant sustainability considerations that affected the design, other than the recommendation that variable frequency drives be added to the influent pump station pumps. Solar panels could be installed in the future on unused portions of this property and the property where the disposal ponds are located to offset electrical costs.

The effluent disposal ponds are simple in design and operation and therefore provides operational simplicity through the use of an appropriate disposal technology.

5.5.7 Costs

Capital costs for Alternative 2 are shown in Table 6. Anticipated operational costs are presented in Table 7. Additional detail is provided in Appendix 4. It should be noted that the operational costs listed below exclude the current SRF Loan repayment of \$95,844 annually through 2028 since this cost is the same for all alternatives.

Table 6. Alternative 2 Opinion of Probable Project Costs^{1,2}
City of Alturas Wastewater Planning Study

Item	Description	Item Cost
1	Decommission/Demolition	\$25,000
2	Demolish existing trickling filter	\$41,000
3	Helical Screw Screen	\$309,000
4	Trickling Filters (4)	\$1,744,000
5	Aerobic Digesters	\$772,000
6	Effluent Pump Station	\$900,000
7	Effluent Force Main	\$582,000
8	Infiltration Ponds	\$157,000
9	Replace Generator	\$56,000
10	Influent Pump Station Improvements (pumps, VFD, force main)	\$200,000
11	Rehabilitate Secondary Clarifier #1	\$300,000
12	Effluent Flow Meter	\$15,000
13	Groundwater Monitoring Wells	\$60,000
	Mobilization (12%)	\$620,000
	Subtotal:	\$5,781,000
	Contingency (30%)	\$1,735,000
	Subtotal Construction:	\$7,516,000
	Land Acquisition ³	\$361,000
	Engineering/Construction Management (18%)	\$1,353,000
	Environmental/Permitting (5%)	\$376,000
	Administration/Legal (4%)	\$301,000
	Total Project:	\$11,642,000
1. See Appendix 4 for additional detail		
2. Items not explicitly listed are incorporated with the listed items.		
3. Land costs include basic sale at \$1,000/acre plus \$300/acres for closing costs, legal fees, survey, etc.		

Table 7. Alternative 2 Opinion of Annual Operational Costs^{1,2}
City of Alturas Wastewater Planning Study

Description	Item Cost
Labor	\$85,000
Chemicals	\$39,500
Testing/Reporting	\$24,500
Sludge/solids Hauling	\$2,000
Electrical Costs	\$30,900
Permitting	\$5,000
Annual O&M	\$186,900
1. See Appendix 4 for additional detail 2. Some minor operational costs have not been included because they are assumed to be relatively equal for all alternatives. Detailed operational costs will be shown in a Rate Study being performed as part of this study.	

5.6 Alternative 3: Convert to Aeration Ponds with Land Discharge

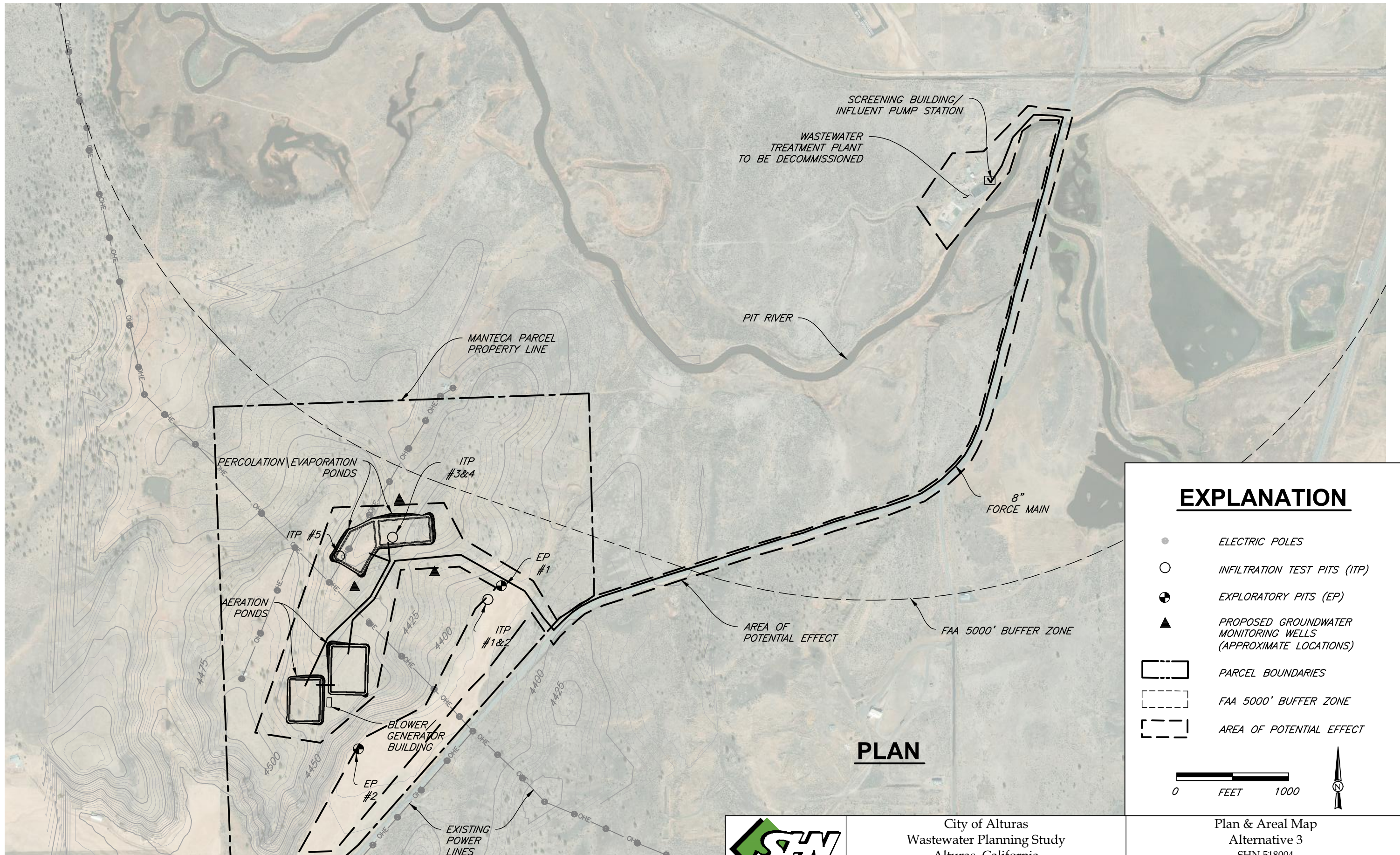
5.6.1 Description

Alternative 3 consists of decommissioning the existing WWTP and moving treatment to a new offsite location using aeration ponds with land discharge. The improvements consist of the items listed in Table 8. A plan view showing the configuration of the alternative is shown in Figure 28. A schematic layout of the new headworks and pump station is shown in Figure 29. A process flow diagram is shown in Figure 30.

Table 8. Alternative 3 Component Description
City of Alturas Wastewater Planning Study

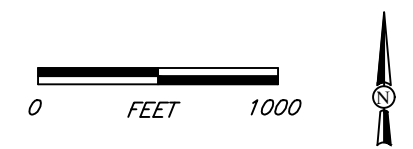
Item	Additional Description
Decommission the existing WWTP ¹	The existing structures will remain while all mechanical equipment is removed.
New Headworks with new screw press screening	The new headworks will be the same as Alternatives 1 and 2.
New Influent Pump Station	To be located adjacent to the new headworks.
New Force main	Eight-inch diameter pipe to the treatment pond site
New Aeration Ponds	Two ponds with coarse and fine bubble aeration
New Evaporation/Percolation Ponds	Two ponds able to infiltrate 0.5 MGD, approximately 2-acre bottom area for each pond.
1. WWTP: Wastewater Treatment Plant	

Raw wastewater would be diverted from near the entrance to the existing WWTP and pumped to the new treatment ponds located approximately two miles away. Two options for this process were evaluated: (1) Pump raw unscreened wastewater to the treatment ponds and (2) Remove solids prior to pumping to the treatment ponds. In either case, solids should be removed prior to the aeration ponds to prevent solids buildup in the ponds. Removing solids prior to pumping would reduce the potential for solids buildup in the force main. For this reason, a headworks prior to the influent pump station is recommended.



EXPLANATION

- ELECTRIC POLES
- INFILTRATION TEST PITS (ITP)
- ⊕ EXPLORATORY PITS (EP)
- ▲ PROPOSED GROUNDWATER MONITORING WELLS (APPROXIMATE LOCATIONS)
- ▭ PARCEL BOUNDARIES
- - - FAA 5000' BUFFER ZONE
- - - AREA OF POTENTIAL EFFECT



City of Alturas
Wastewater Planning Study
Alturas, California

December 2019

518004-ALTURAS-WW2

Plan & Areal Map
Alternative 3
SHN 518004

Figure 28

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City of Alturas
Wastewater Treatment Facility Project
Alturas, California

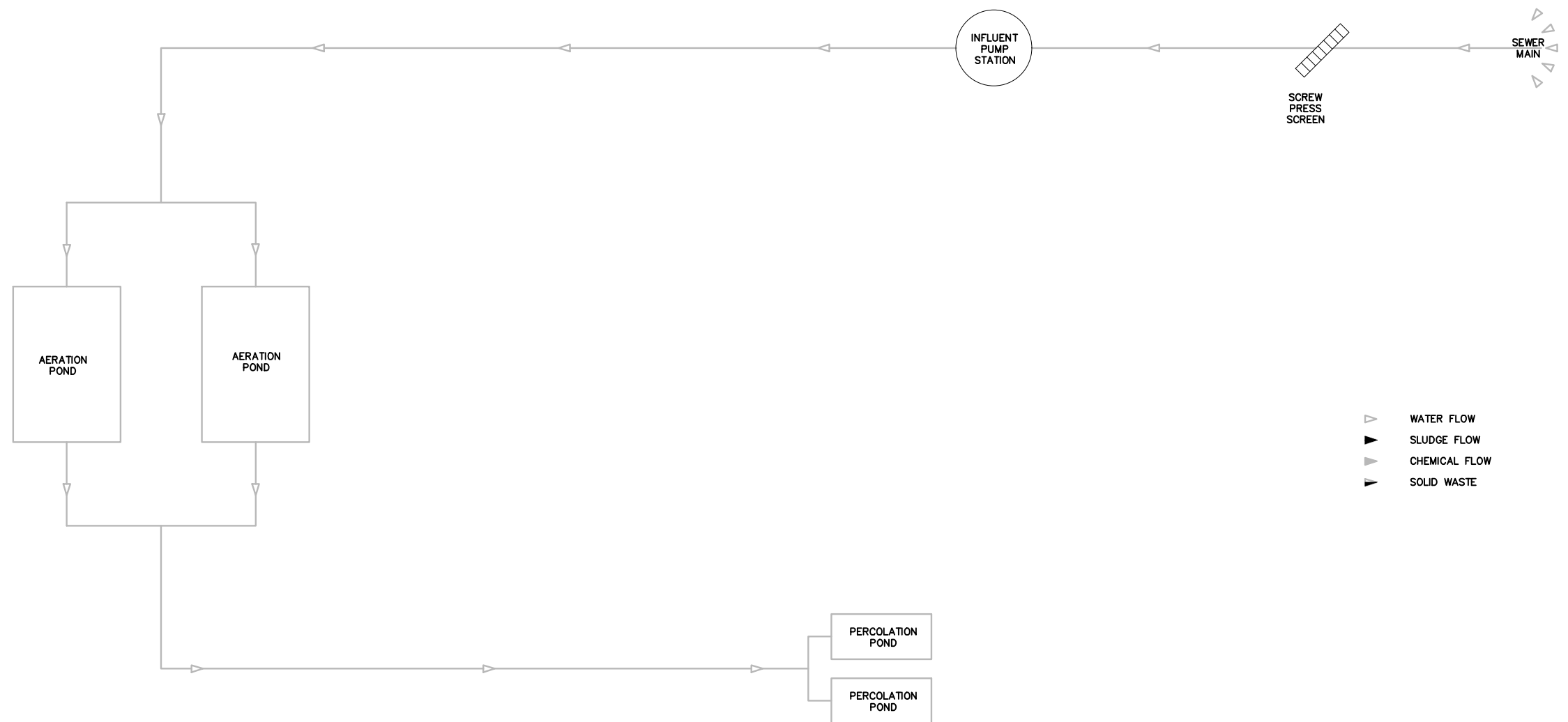
New Headworks & Pump Station
Alternative 3
SHN 518004

December 2019

518004-ALTURAS-WW2

Figure 29

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The new headworks and pump station would be located near the existing control structure and utilize the existing bypass line (see Figure 29). Another location has potential but is not preferred. This other location is along the access road to the WWTP and next to an existing manhole, where a diversion could be constructed. The final location can be determined during final design. A new standby generator would be required for the new pump station.

The influent pump station would have the same design criteria as the effluent pump station for Alternative 2 but located as described in the previous paragraph.

A force main would run from the pump station along the WWTP access road to County Road 54, where it would run along the road shoulder, crossing the Pit River on the road bridge, to the entrance of the Manteca property. The force main would continue along the internal access road to the location of the aeration ponds.

The aeration ponds will be lined ponds with coarse and fine bubble aerators. The preliminary design has been made around the MARS lagoon aerator manufactured by Triplepoint Water Technologies (www.triplepointwater.com). There would be a two-cell pond system. Both ponds measure 400 ft by 240 ft at the surface with 3:1 side slopes and a 12-ft water depth. Three blowers (two duty and one standby) at 30 horsepower (hp) each would be housed in a 20-ft by 20-ft blower building and would provide air to the aerators. Additional design data from this manufacturer is provided in Appendix 5.

As in Alternative 2, two evaporation/percolation ponds will be constructed for disposal of the treated effluent, each designed to infiltration at least 0.5 MGD. Each pond will have a bottom area of two acres with a depth of six feet. Embankment side slopes will be 3H:1V on the inside and 2H:1V on the outside. It is anticipated that soils will be suitable and a balanced cut and fill construction will be feasible to minimize construction costs.

The existing office and lab will remain at the current location and will be used for staff. No office facilities will be constructed at the new WWTP location. However, a small storage building will be included to store tools and spare equipment.

Monitoring wells will be required to monitor groundwater quality and verify that the percolated effluent does not degrade groundwater quality. Existing groundwater quality will affect design criteria for final design. Typically, three monitoring wells are placed around the infiltration areas that will provide adequate coverage since exact groundwater flow direction is not known. Three wells would be placed around the pond locations. The depths of the monitoring wells would extend to the highest groundwater level in order to be able to take groundwater samples. Until further investigation is performed, groundwater depth is not known in this specific location where the wells will be located.

New 480-volt, three phase, electrical service at the aeration ponds will be required. Based on estimate electrical usage, the WWTP would require less than 200 amps. A standby generator will be required and will be located adjacent to the blower building.

The existing WWTP will be decommissioned. However, doing so will require approval from the State since there is an outstanding loan through 2028 on the most recent upgrades to the WWTP. Once approval is

obtained from the State, the City will still be required to pay the outstanding balance of the previous loan in accordance with the loan covenants.

5.6.2 Design Criteria

General design criteria were described in **Section 5.2**. This section provides additional design criteria specific to this alternative.

The package influent pump station would have three pumps, one at 350 gpm and two at 750 gpm, which matches the pump capacity configuration of the existing influent pump station. Total design head is estimated at approximately 200 ft. It is expected that the pump station will be a wetwell and drywell configuration using centrifugal pumps, but an option to use submersible pumps is left up to final design.

The force main would be eight inches in diameter to minimize headlosses and travel time and consist of high-density polyethylene (HDPE), except for the two Pit River crossings locations, where it will be ductile iron pipe. The trench for the force main will be approximately 18 inches wide and five feet deep.

As mentioned previously, the aeration system and associated ponds were designed around a particular manufacturer (Triplepoint Water Technologies). Slight modifications would be expected if the specific aeration units are different.

The disposal ponds are based on a percolation value of one foot per day, which was estimated from field testing and applying a factor of safety of four. Additional discussion of the percolation test results is provided in Appendix 3. Groundwater monitoring wells would be required around the percolation ponds and installed prior to construction. This will help determine baseline groundwater quality, which would be the design criteria for the final design.

With respect to the nitrate control program under CV-SALTS (see Section 5.2), the City would monitor nitrate at the monitoring wells around the disposal ponds. As aforementioned, nitrate impacts are not anticipated due to various factors, including depth to groundwater and experience with similar facilities. However, if the monitoring data indicate that additional nitrogen (specifically nitrate) removal is needed, this alternative allows for future upgrade to remove nitrogen, such as an engineered nitrification/denitrification unit process or seasonal irrigation. The specific solution, if needed, would be determined after review of the monitoring data results after at least two years of operation of the new facility or as directed by the CVRWQCB.

5.6.3 Environmental Impacts

Ground disturbing activities for this alternative are twofold: (1) Within the existing WWTP property, which has previously been disturbed, and (2) along County Road 54 and at the effluent disposal property. Estimated extents of these disturbances (area and depth) are provided in Table 8a. Additional detail of environmental impacts for this alternative are discussed in the CEQA documentation.

5.6.4 Land Requirements

A new parcel of land, located outside the Alturas city limits, will be required to site the disposal ponds. A parcel has been identified (County Road 54 property) as discussed earlier. Also, a utility easement or similar permission will be needed from Modoc County to place the force main along County Road 54.

5.6.5 Potential Construction Problems

Potential construction problems could be conflicts with other utilities along County Road 54.

5.6.6 Sustainability Considerations

This section discusses any design consideration with respect to water conservation, water reuse, energy efficient design, operational simplicity, appropriate technology, and other considerations.

The aeration ponds and effluent disposal ponds are simple in design and operation and therefore provides operational simplicity through the use of an appropriate disposal technology. The aeration component would require a new service connection/backup power component for the site and is energy intensive, the City could install solar panels in the future on unused portions of the current WWTP property and where the new WWTP and disposal ponds are located to offset electrical costs.

From a plant classification standpoint, based on current State guidelines, this alternative does not require any change in current plant classification (Class II) or operator grade level (Grade II) which provides significant sustainability benefits in operations since higher-grade operators are difficult to hire in remote WWTPs.

5.6.7 Costs

Capital costs for Alternative 3 are shown in Table 9. Anticipated operational costs are presented in Table 10. Additional detail is provided in Appendix 4. It should be noted that the operational costs listed below exclude the current SRF Loan repayment of \$95,844 annually through 2028 since this cost is the same for all alternatives. As stated previously, the 2006 SRF loan must be repaid regardless of whether the funded facilities are still in use or have been decommissioned.

Table 9. Alternative 3 Opinion of Probable Project Costs^{1,2}
City of Alturas Wastewater Planning Study

Item	Description	Item Cost
1	Decommission/Demolition	\$25,000
2	Generators	\$124,000
3	Flow meter	\$15,000
4	Helical Screw Screen	\$309,000
5	Influent Pump Station	\$900,000
6	Influent Force Main	\$582,000
7	Aeration Basins	\$1,037,000
8	Infiltration Ponds	\$156,000
9	Groundwater Monitoring Wells	\$60,000
10	Blower Building	\$80,000
	Mobilization (12%)	\$395,000
	Subtotal:	\$3,263,000
	Contingency (30%)	\$979,000
	Subtotal Construction:	\$4,242,000
	Land Acquisition ³	\$361,000
	Engineering/Construction Management (18%)	\$764,000
	Environmental/Permitting (5%)	\$213,000
	Administration/Legal (4%)	\$170,000
	Total Project:	\$5,750,000
1. See Appendix 4 for additional detail 2. Items not explicitly listed are incorporated with the listed items. 3. Land costs include basic sale at \$1,000/acre plus \$300/acres for closing costs, legal fees, survey, etc.		

Table 10. Alternative 3 Opinion of Annual Operational Costs^{1,2}
City of Alturas Wastewater Planning Study

Description	Item Cost
Labor	\$85,000
Chemicals	\$0
Testing/Reporting	\$24,500
Sludge/solids Hauling	\$1,000
Electrical Costs	\$48,800
Permitting	\$5,000
Annual O&M	\$164,300
1. See Appendix 4 for additional detail 2. Some minor operational costs have not been included because they are assumed to be relatively equal for all alternatives. Detailed operational costs will be shown in a Rate Study being performed as part of this study.	

6.0 Selection of an Alternative

6.1 Life Cycle Cost Analysis

A life cycle net present value cost analysis was used to compare the alternatives. The present value cost analysis is a basic evaluation of alternative costs utilizing the present value factors. The interest rate used is the real 20-year Federal discount rate from Appendix C of the Office of Management and Budget (OMB) Circular A-94. The most recent publication of Circular A-94 Appendix C is from November 2019, and the real discount rate is 0.3%. Table 11 shows the present value analysis results.

Salvage values can be difficult to determine, as various portions of each alternative have different useful lives. For this analysis, the average useful life for each alternative was assumed to be 30 years, which means that after 20 years, the salvage value is 33% using a straight-line depreciation.

Table 11. Present Value Analysis¹
City of Alturas Wastewater Planning Study

Item	Alternative 1	Alternative 2	Alternative 3
Capital Cost ²	\$12,890,000	\$11,642,000	\$5,750,000
Annual O&M ³	\$250,500	\$186,900	\$164,300
Present Value O&M	\$4,855,598	\$3,622,799	\$3,184,729
Salvage Value ⁴	\$4,253,7006	\$3,841,860	\$1,897,500
Present Value Salvage	\$4,006,344	\$3,618,452	\$1,787,159
Net Present Value⁵	\$13,739,254	\$11,646,347	\$7,147,571
1. For 20-year period 2. Total project costs from tables in Section 5 in 2019 dollars. 3. O&M: Operations and Maintenance 4. Assumes straight line depreciation, 30-year life 5. Net Present Value = Capital Cost + Present Value O&M – Present Value Salvage.			

The analysis reveals that Alternative 3 has the lowest net present value.

6.2 Non-Monetary Factors

The CVRWQCB has expressed preference for the City to use land discharge rather than continue to discharge to the Pit River. This favors Alternatives 2 and 3 significantly over Alternative 1.

7.0 Proposed Project

7.1 Description

The proposed project is Alternative 3 as described earlier, which includes the following key features:

- Decommissioning of the existing WWTP;
- New offsite aerated pond treatment system;
- Land discharge of treated effluent utilizing infiltration and evaporation; and
- No change in operator grade level over current WWTP.

7.2 Project Schedule

The project schedule is dependent on acquiring funding for final design and construction. The current Proposition 1 wastewater planning study will be completed by late 2020 with the CEQA documentation and construction funding application being the final tasks of the study to be completed.

Funding for final design and construction will likely not be available until the latter half of 2021. Once notice to proceed is issued in late 2021, the schedule shown in Table 12 would be anticipated.

Table 12. Anticipated Project Schedule (Recommended Alternative)
City of Alturas Wastewater Planning Study

Activity	Estimated Start Date	Estimated Finish Date
Complete Proposition 1 Wastewater Planning Study, including funding application	N/A	October 31, 2021
SRWCB ¹ Funding Application Review	November 2021	July 2022
Monitoring Well Installation ²	August 2021	October 2021
Funding Awarded and Notice to Proceed	August 2022	August 2022
Preliminary investigations: Survey and Geotechnical	September 2022	December 2022
Final Design	September 2022	March 2023
Bidding ³	April 2023	May 2023
Construction	July 2023	June 2024
1. SWRCB: State Water Resources Control Board 2. It is anticipated that the City will self-fund the installation of the monitoring wells. 3. The City has expressed interest in issuing multiple bid packages to allow local contractors to bid; this may affect the construction schedule.		

7.3 Permit Requirements

Environmental clearances through the CEQA process is being completed under the current Proposition 1 Planning Grant. A Report of Waste Discharge will also be completed under the current grant. Other anticipated permit requirements to be obtained later include the following:

- Utility Easements from Modoc County for the force main along County Road 54;
- CVRWQCB Construction Stormwater permit;

- Building permits from Modoc County;
- SWRCB Division of Water Rights Change Permit; and
- RWQCB Groundwater Permit.

7.4 Sustainability Considerations

Sustainability considerations were discussed in Section 5.

7.5 Project Cost Estimate

The project cost estimate for the recommended alternative was presented in Table 9.

7.6 Annual Operating Budget

A separate Rate Study is being prepared as part of the Proposition 1 Wastewater Planning Grant. The Rate Study will include detailed information about operating budgets, proposed rate structure, income, and debt payments.

8.0 Conclusions and Recommendations

It is recommended that the City pursue Alternative 3 by securing the property and obtaining funding for final design and construction. This alternative provides the best long-term solution for the least cost and is preferred by the CVRWQCB.

Sanitary Sewer Evaluation Study Summary 1

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Technical Memorandum

Reference: 518004.200
Date: November 4, 2020
To: Jason Diven, City of Alturas
From: Phil McGovern, Anders Rasmussen
Subject: SSES Report & I/I Analysis Summary

Purpose

This document provides supplemental information for the Preliminary Engineering Report (PER) and augments the 2020 Final Sanitary Sewer Evaluation Survey (SSES) prepared by SHN. The purpose of this document is to provide a comparison of infiltration and inflow (I/I) reduction cost compared to addressing I/I flows at the wastewater treatment plant (WWTP).

SSES Summary

The 2020 Sanitary Sewer Evaluation Survey for the City of Alturas examined ten years (2007-2017) of influent flow data at the WWTP and determined that a significant amount of I/I is occurring in the collection system due in large part to high seasonal groundwater in various portions of the service area. The age and condition of the existing system is also a contributing factor to infiltration.

I/I is a common problem and cannot be completely avoided. Mitigation measures include I/I reduction within the collection system, or addressing increased flows at the WWTP, or a combination of both based on costs. The four alternatives for I/I reduction presented in the SSES are summarized in Table 1. The reader is referred to the full SSES report for additional information.





Table 1. I/I Reduction Alternatives and Total Project Cost Estimates
SSES Summary
Alturas, CA

Alternative	Description	I/I ¹ Reduction Rate ²		Estimated Costs Project Total ³
		Low	High	
1	Address smoke test findings	5%	10%	Minimal
2	Address smoke test findings, plus mainline rehabilitation/replacement, including manhole rehabilitation/replacement in seasonally high groundwater areas only.	20%	40%	\$ 22,943,640
3	Address smoke test findings, plus mainline and manhole rehabilitation/replacement, plus service lateral rehabilitation/replacement from main to structure in seasonally high groundwater areas only.	60%	80%	\$ 24,511,280
4	Do Nothing	0%	0%	Additional costs at WWTP –see discussion
1. I/I: Infiltration and Inflow 2. I/I reduction rates based on industry typical values and SHN experience but can vary based on specific system conditions and rehabilitation and/or construction methods. 3. Estimated Cost breakdown can be found in the original SSES document.				

Analysis

Flow data at the WWTP for the years 2008-2017 show that annual average flow through the WWTP is 125.77 million gallons (MG). For this period of record, the average annual Base Sanitary Flow (BSF) is estimated as 104.17 MG and average annual I/I volume is estimated as 21.60 MG.

Percentage reductions shown in Table 1 were applied to the estimated average I/I volume to determine the volume of I/I reduced by each alternative. The estimated cost for each alternative was then divided by the volume of I/I reduction to determine a capital cost per unit volume to be used as a comparison with addressing the additional flow at the WWTP. As a conservative measure, the average percentage reduction of the listed range was used for this calculation. Only Alternatives 2 and 3 are analyzed here, since there is no capital cost for Alternative 1.

Implementation of Alternative 2 is anticipated to reduce I/I volumes by 30%, that is, by approximately 6.48 MG. At an estimated cost of \$22.9 million, this is a capital cost of \$3.53 per gallon removed. Implementation of Alternative 3 is anticipated to reduce I/I volumes by 70%, that is, by approximately





15.12 MG. At an estimated cost of \$24.5 million, this is a capital cost of \$1.62 per gallon removed.

The design average dry weather flow (ADWF) for the WWTP is 0.5 million gallons per day (MGD) with a design peak flow of 1.0 MGD. Additional discussion related to this can be found in the PER. The current flows at the WWTP, including current I/I flows, are generally within those design limits. Therefore, no additional treatment or effluent storage capital costs are anticipated as a result of not reducing I/I flows. It should be noted that the SSES presented that additional effluent storage costs would be the only capital costs for I/I management at the WWTP; however, after additional review during the PER preparation, this additional cost does not appear to be needed given the design flows described above. This means that there is no additional capital cost at the WWTP for the additional I/I flow.

Additional operating costs at the WWTP to address I/I flows would generally only result from additional pumping from the influent pump station to the new WWTP under the preferred alternative. Annual pumping costs for the I/I volume reduced under Alternative 3, which is 15.12 MG, would be approximately \$1,500 per year based on a power cost of \$0.16 per kilowatt-hour (kWh) and energy consumption of 600 kWh per MG.

Conclusions

From a capital cost perspective, no more than Alternative 1 for I/I reduction is recommended. From an operational cost perspective, the additional pumping expense is minor and does not warrant the large capital cost of significant I/I reduction.



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Bird Mitigation Strategies 2

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Technical Memorandum

Reference: 518004.500
Date: November 12, 2018
To: Anders Rasmussen
From: Gretchen O'Brien
Subject: **Alturas Wastewater Pond - Bird Deterrent Methods Research**

Background

Bird strikes by airplanes is a common occurrence and most often happens when the aircraft is less than 500 feet off the ground during take-off and landing (DeVault et al., 2017). The Alturas Municipal Airport is just north of the Alturas wastewater treatment plant. Creating a treated wastewater retention pond with in such close proximity to the airport will require wildlife management mitigation measures to reduce the chance of bird strikes by aircraft. An internet research effort was conducted to create recommendations for implementing wildlife deterrents from the proposed pond(s).

Specific measures to deter birds from the proposed pond(s) near the Alturas Municipal Airport should take into consideration the types of species that occur in the region, the juxtaposition of other habitat, existing wildlife management techniques being utilized at the Alturas Municipal Airport, and necessary design features of the pond(s) to achieve project goals. Based on the location of the proposed pond(s) in the proximity of the Modoc National Wildlife Refuge and the waterfowl that breed, overwinter, or migrate through the area, the primary concern for bird strikes may be waterfowl and other larger water birds such as Canada Goose (*Branta canadensis*), although the Alturas Public Works director has reported that, to his knowledge, there has never been a bird strike by an aircraft at the Alturas Municipal Airport in the past ten years or so (Pers. Comm., Picotte, 2018).

Research Results Summary

Several studies have been conducted to test the efficacy of bird deterrents from airports, water sources, and agricultural crops. The common conclusions among the research papers investigated were that an integrative approach to bird deterrent methods, in combination with pond design and management, is the most effective.

Pond Design

Pond design recommendations include linear or circular shapes to minimize the perimeter and geographic isolation from other water sources (Blackwell et al., 2008). Geographic isolation of created ponds, away from other water sources, may decrease the probability of use by waterfowl (Blackwell et al., 2008). Bottom-lined ponds help to reduce vegetation growth which can provide a food source for waterfowl (National Academy of Sciences, 2011). Keeping the pond surface free of floating vegetation and the pond edges free of vegetative cover and woody debris can reduce the attractiveness to birds (Blackwell et al., 2008).

**Civil • Environmental • Geotechnical • Surveying
Construction Monitoring • Materials Testing
Economic Development • Planning & Permitting**

Exclusion Methods

Covering and dewatering ponds are considered the most effective bird deterrent from open water ponds, although this is not an option for the Alturas wastewater retention pond(s). Exclusion methods may need to be incorporated into the pond design. Overhead wires are the most researched and effective method of bird exclusion (National Academy of Sciences, 2011). Gridwire™ and Stealthnet™ products at BirdBarrier.com may be useful for waterfowl exclusion from ponds, installed in a grid pattern directly over the surface of the water. This will discourage birds from landing on the water while allowing for evaporation.

Deterrents

The National Academy of Sciences conducted a literature review to evaluate the relative effectiveness of bird repellent techniques. The research results determined that human effigies, or models, that moved frequently and were dressed in bright colors were the most effective as bird deterrents. This conclusion was also made by a study that specifically tested scarecrows and predator models for scaring birds away (Marsh et al., 1992). Studies also show that lifelike human effigies in combination with broadcasting a waterfowl distress call periodically was effective in deterring birds from a specific area (National Academy of Sciences, 2011; Marsh et al., 1992; DeVault et al., 2017). Birds often habituate to visual and auditory deterrents unless they are moved frequently, and by changing the look of an effigy (e.g. different clothing) and the distress call emitted, can also increase the effectiveness of the deterrents (DeVault et al., 2017, Marsh et al., 1992).

Conclusion

The combination of pond design, pond placement, and an integrative suite of avian deterrent methods will be the best approach to preventing the attraction of birds to a wastewater retention pond at the Alturas wastewater treatment plant. The following recommendations summarize a suggested plan for bird deterrents to help prevent bird strikes at the Alturas Municipal Airport.

Recommendations

- Design pond(s) in a circular or linear shape to reduce perimeter.
- Place pond(s) isolated, as far away as possible from other water sources.
- Manage vegetation and woody debris in and around the pond to reduce food and cover resources for birds.
- Use a combination of bird deterrent methods:
 - Exclusion: Grid wires placed over the surface of the pond(s), may also need anti-perch spikes on the wires to deter smaller birds from using the wires.
 - Visual deterrent: Lifelike human effigy dressed in brightly colored, loose-fitting clothing (to blow in the wind) placed on a floating island just big enough for the effigy (disallow space for birds to land). Most effective if the effigy moves and clothing changed periodically.
 - Audio deterrent: Broadcast a recording of various waterfowl distress calls, either periodically or motion detected. Most effective if the type of calls and location change periodically. According to the USDA, birds react most to sounds from 1 to 3 kHz.
 - Place all deterrents concurrently with pond creation before birds begin to investigate the resource.
- Coordinate with the Alturas Municipal Airport managers regarding wildlife management methods and monitoring bird strikes.

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Infiltration Testing 3

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**CONSULTING ENGINEERS & GEOLOGISTS, INC.**803 Main Street, Suite 401
Klamath Falls, OR 97601

Tel. 541/827-7855

JOB	518004		
SHEET NO.	1	OF	1
CALC'ED BY	AHR	DATE	11/2/2020
CHECKED BY	DATE		

Determine infiltration pond area required on Manteca (Hunter) property.

Approach for simplified calculation

1. Set pond area based on being able to infiltrate design flows.

$$\text{Area} = \text{daily flow} / \text{infiltration rate}$$

2. Neglect precipitation and evaporation, which are small in comparison to infiltration.
3. This location has a net annual evaporation, so these calculations are conservative.
4. Infiltration rates from field testing results, adjusting for a factor of safety.
5. Assume infiltration occurs only on bottom of pond and not along embankments.

Data:	ADWF =	0.5	MGD	See PER for discussion on design flow
	PWWF =	1	MGD	See PER for discussion on design flow
	Infiltration rate =	30		minutes/inch, based on field data from test pits 3, 4, and 5
	Factor of safety =	4		
	Design Infiltration Rate =	120		minutes/inch (field infiltration * factor of safety)
		=	12	inches/day
	Required Area for ADWF =	1.53		acres; this is pond bottom area.

Set one cell at 1.53 acres to handle the design flow with a second cell of equal size to handle peak wet weather flows. Add third pond for redundancy. Or have two ponds @ 2 acres each.

Since little storage is needed, set depth of pond at 6 ft.

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Memorandum

Reference: 518004.200
Date: December 11, 2019
To: Anders Rasmussen
From: Phil McGovern
Subject: **Infiltration Testing**

Introduction:

Infiltration is the permeation of a liquid into something by filtration; in this case we are attempting to ascertain an empirical rate at which water infiltrates a soil matrix. Several tests can be performed to determine this empirical value; an Open Pit Falling Head Test was used on the Hunter Current SCP Manteca (HCSPCM) Parcel on November 8, 2019.

Prior to the field tests, SHN obtained a copy of the National Resource Conservation Service (NRCS) web soil survey report for the HCSPCM parcel (See Attached). The test pits according to the NRCS report are located in two different soil matrices, with two different referenced hydrologic soil types (See Figure 1). Site 1 was identified as a Ladd Sandy Loam, with a hydrologic soil classification of C. Site 1 is location is approximately 400 feet from County Road 54. Site 2 was identified as a Tuff Outcrop-Cause eroded Complex, with a hydrologic soil classification of D. Site 2 is significantly further from the public and is currently the preferred location for effluent disposal because it is further away from the public, limiting public's view and possible interaction with the effluent disposal site.

Scope:

The scope of our field investigation was to determine percolation rate for the preliminary sizing of the proposed subsurface Rapid Infiltration Basins (RIB) located on the HCSPCM parcel. Secondly, SHN staff wanted to visit the parcel to determine if it is a feasible location for effluent disposal. Site 1 is where test pits one and two were dug and Site 2 was where test pits three (3), four (4), and five (5) were excavated (See Figure 1).



Groundwater:

Two exploratory pits were dug around the proposed locations; one exploratory pit was dug south of Site 1 & 2, and the other pit was dug to the northeast of the two Sites (See Figure 1). The exploratory pits were dug to a depth of depth of 10 ft. Groundwater infiltrated into the pit at an approximate depth of 9.5 ft.

Procedure:

The five test pits were dug with a backhoe loader; the pits were dug approximately 2' x 4' and a depth of approximately either two feet or four feet. The pits were filled with water prior to testing by the City of Alturas Staff. According to the testing procedure, the pits need to be soaked overnight prior to the test being performed in clay-based soils. For sandy soils if the water percolates through the soil matrix in less than 10 minutes (performed 2 times), you are free to begin testing. This was the case for pits one (1) and two (2). Pits three (3), four (4), and five (5) were soaked overnight.

The holes were refilled on the day of the test. The pits were filled with at least 12 inches of clean water, and the depth of the water in the pit was recorded at the beginning of the test. The height of the water was then recorded in 20 minute intervals. The test was performed for a total of two hours or until the pit was empty.

Results:

The estimated rate of percolation for the test pit/ RIB sites are as follows;

SITE 1

Pit one (1) percolation rate was recorded to be 80 minutes/inch. This value may be misrepresenting the true percolation rate for this location. It was determined that Pit one (1) needed to have a certain amount of water depth to drive the water into the soil matrix. Once the water height reached about 5 inches from the bottom of the pit, the water infiltration rate diminished significantly. This pit was refilled after an hour and the infiltration rate significantly increased, but followed the same trend and diminished once a depth of five inches of water was again reached.

Pit two (2) final percolation rate was determined to be 42 minutes/inch. This test pit was dug to 2 ft in depth and filled with 1 ft of water.

The final percolation rate for Site 1 was determined to be 42 minutes/inch or 34.3 in/day.



SITE 2

Pit three (3) final percolation rate was found to be 24 minutes/inch. The depth of this pit was approximately 2 ft.

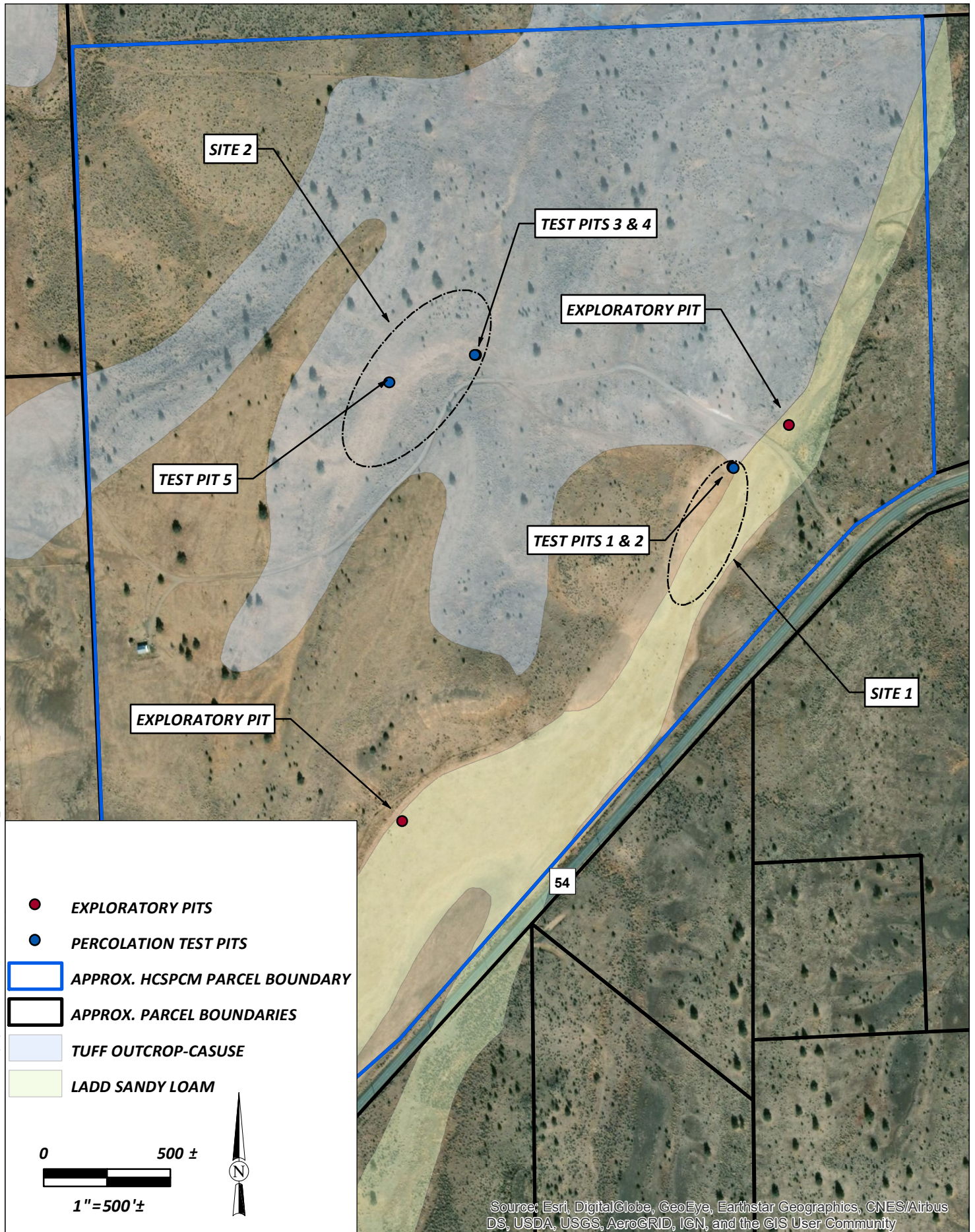
Pit four (4) final percolation rate was recorded to be 34 minutes/inch. The depth of this pit was approximately 4 ft.

Pit five (5) final percolation rate was recorded to be 31 minutes/inch. The depth of this pit was approximately 4 ft.

The overall estimated percolation rate used for the preliminary analysis for the sizing of infiltration ponds for Site 2, was 30 min/inch or 48 in/day.

One infiltration test was done for each of the pits, a more comprehensive infiltration test and soil analysis is recommended prior to final design. The procedure calls for multiple infiltration tests be done in the same pit to verify that the rates do not vary significantly from test to test.

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City of Alturas
Preliminary Engineering Report
Alturas, CA

Infiltration Testing

SHN 518004

December 2019

Infiltration_Testing

Figure 1



CONSULTING ENGINEERS & GEOLOGISTS, INC.

803 Main Street Suite 401, Klamath Falls, OR 97601 541-827-7855

SOILS PERCOLATION TEST DATA SHEET 1 of 2

CLIENT CITY OF ALTURAS DATE 11/8/2019
JOB REF. 218018 APN 315-211-03
TEST PIT No. PP-1 (4') TESTED BY PM
DEPTH TESTED 4 ft DTW >9.5'
PRE-SOAK 24 Hour NRCS CLAY/SANDY
LOAM

NOTE: PIT LOCATIONS FOR PIT ONE (1) AND TWO (2) WERE OFFSET BY 5' TESTING DIFFERENT SOIL LAYERS.

Reading No.	Start Time	Stop Time	Interval (Minutes)	Water Level Drop (Inches)	Percolation Rate (Minutes per Inch)
1	10:59	11:19	20	7 3/4	2 4/7
2	11:19	11:39	15	1 1/4	12
3	11:39	11:59	20	1/4	80
4	11:59	12:19	20	3/8	53 1/3
Tested again with Driving Head					
1	12:42	12:54	12	11 1/2	1
2	12:54	1:07	13	3	4 1/3
3	1:07	1:47	40	1/2	80

STABILIZED PERCOLATION RATE = 80

TEST PIT No. PP 2 (2FT) TESTED BY PM
DEPTH TESTED 24" DTW >9.5'
PRE-SOAK 24 Hour NRCS CLAY/SANDY
LOAM

NOTE: PIT LOCATION PIT TWO (2) OFFSET FROM PIT ONE (1) TESTING DIFFERENT SOIL LAYERS.

Reading No.	Start Time	Stop Time	Interval (Minutes)	Water Level Drop (Inches)	Percolation Rate (Minutes per Inch)
1	11:03	11:23	20	2 5/8	7 5/8
2	11:23	11:43	20	1 1/8	17 7/9
3	11:43	12:03	20	3/4	26 2/3
4	12:03	12:23	20	5/8	32
5	12:23	13:08	45	1 1/8	40
6	13:08	13:45	37	7/8	42 2/7

STABILIZED PERCOLATION RATE = 42



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TEST PIT No. PP-3 TESTED BY PM
DEPTH TESTED 24" DTW >9.5'
PRE-SOAK 24 Hour NRCS TUFF CASUSE
NOTE: PIT LOCATIONS FOR PIT THREE (3) AND FOUR (4) WERE OFFSET BY 5' TESTING

Reading No.	Start Time	Stop Time	Interval (Minutes)	Water Level Drop (Inches)	Percolation Rate (Minutes per Inch)
1	11:48	12:08	20	3 1/2	6
2	12:08	12:28	20	1 1/2	13
3	12:28	12:48	20	1	20
4	12:48	13:08	20	7/8	23
5	13:08	13:28	20	3/4	27
6	13:28	13:58	30	1 1/4	24

STABILIZED PERCOLATION RATE = 24

TEST PIT No. PP-4 TESTED BY PM
DEPTH TESTED 48" DTW >9.5'
PRE-SOAK 24 Hour NRCS TUFF CASUSE
SOIL LAYERS (TOP OF HILL BY TRAILER)

Reading No.	Start Time	Stop Time	Interval (Minutes)	Water Level Drop (Inches)	Percolation Rate (Minutes per Inch)
1	11:45	12:05	20	2 1/8	9
2	12:05	12:27	22	1 3/8	16
3	12:27	12:47	20	3/4	27
4	12:47	13:09	22	1	22
5	13:09	13:27	18	1	18
6	13:27	13:57	30	7/8	34

STABILIZED PERCOLATION RATE = 34



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TEST PIT No. PP-5 TESTED BY PM
DEPTH TESTED 24" DTW >10'
PRE-SOAK 24 Hour NRCS TUFF CASUSE

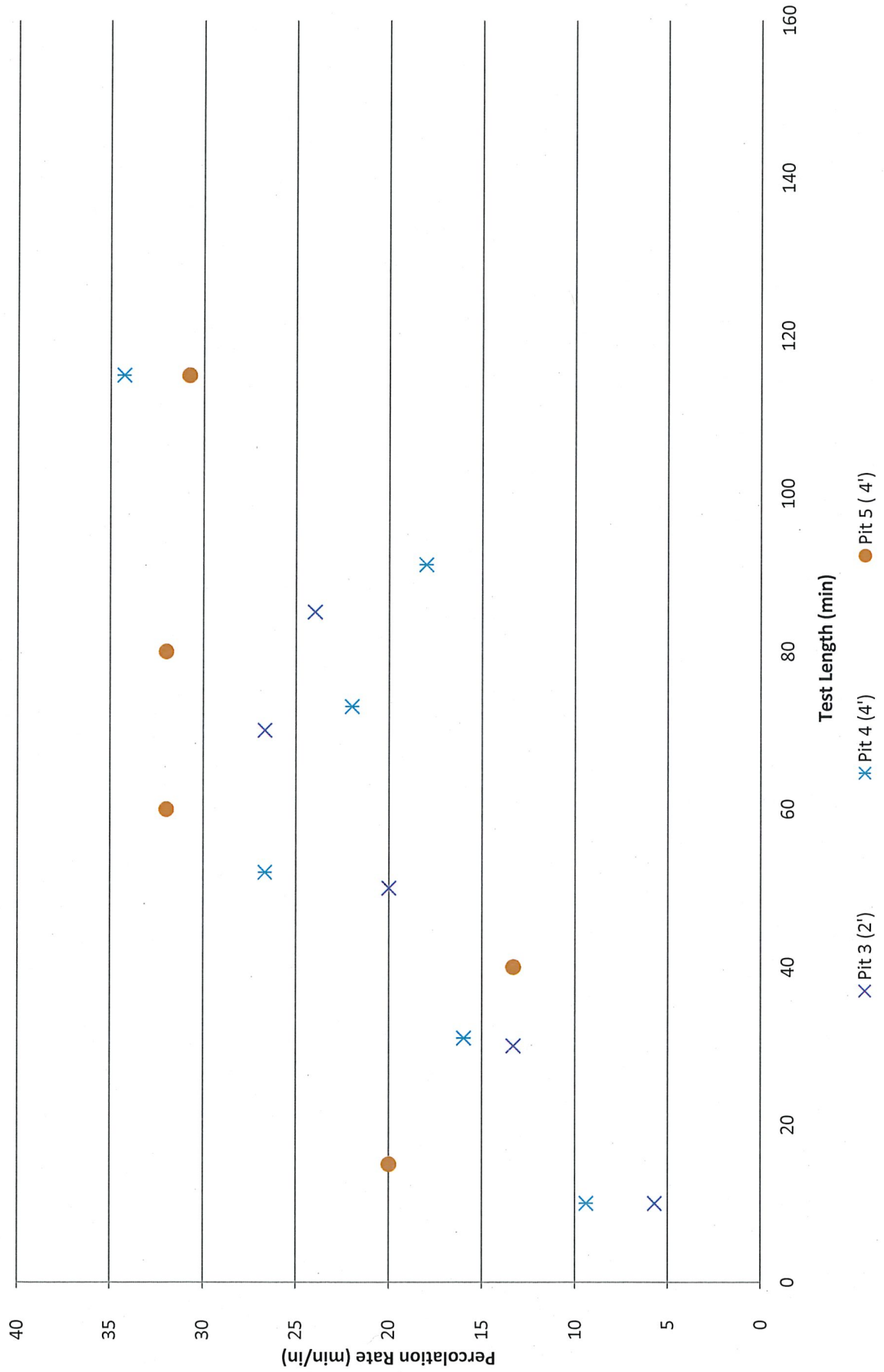
EXISTING POWERLINES (ABOVE TRAILER).

Reading No.	Start Time	Stop Time	Interval (Minutes)	Water Level Drop (Inches)	Percolation Rate (Minutes per Inch)
1	11:55	12:25	30	1 1/2	20
2	12:25	12:45	20	1 1/2	13
3	12:45	13:05	20	5/8	32
4	13:05	13:25	20	5/8	32
5	13:25	14:15	50	1 5/8	31

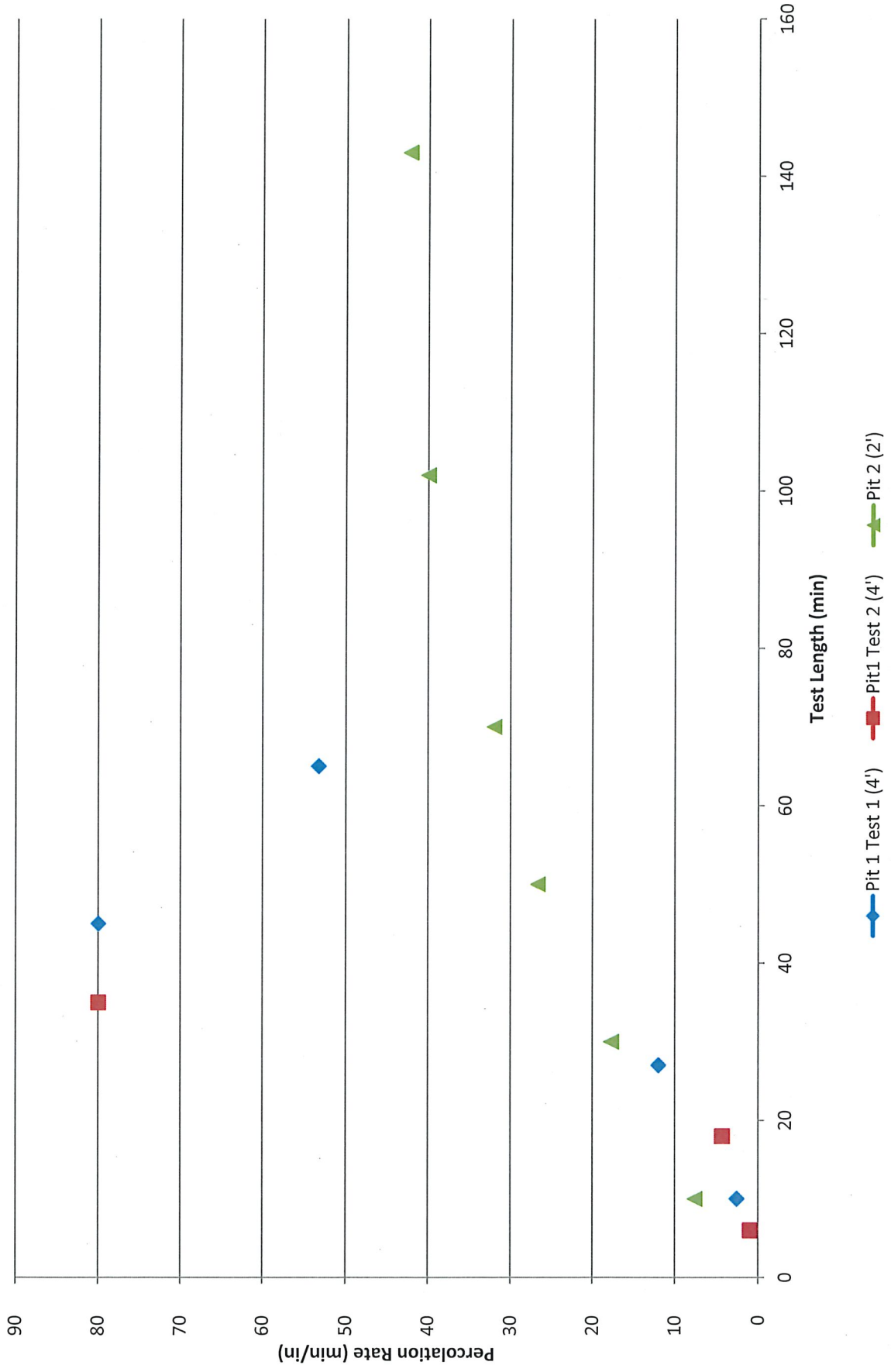
STABILIZED PERCOLATION RATE = 31

Hunter Properties Percolation Rates

Site Two (2) Alturas, CA



Hunter Properties Percolation Rate Site One (1) Alturas, CA





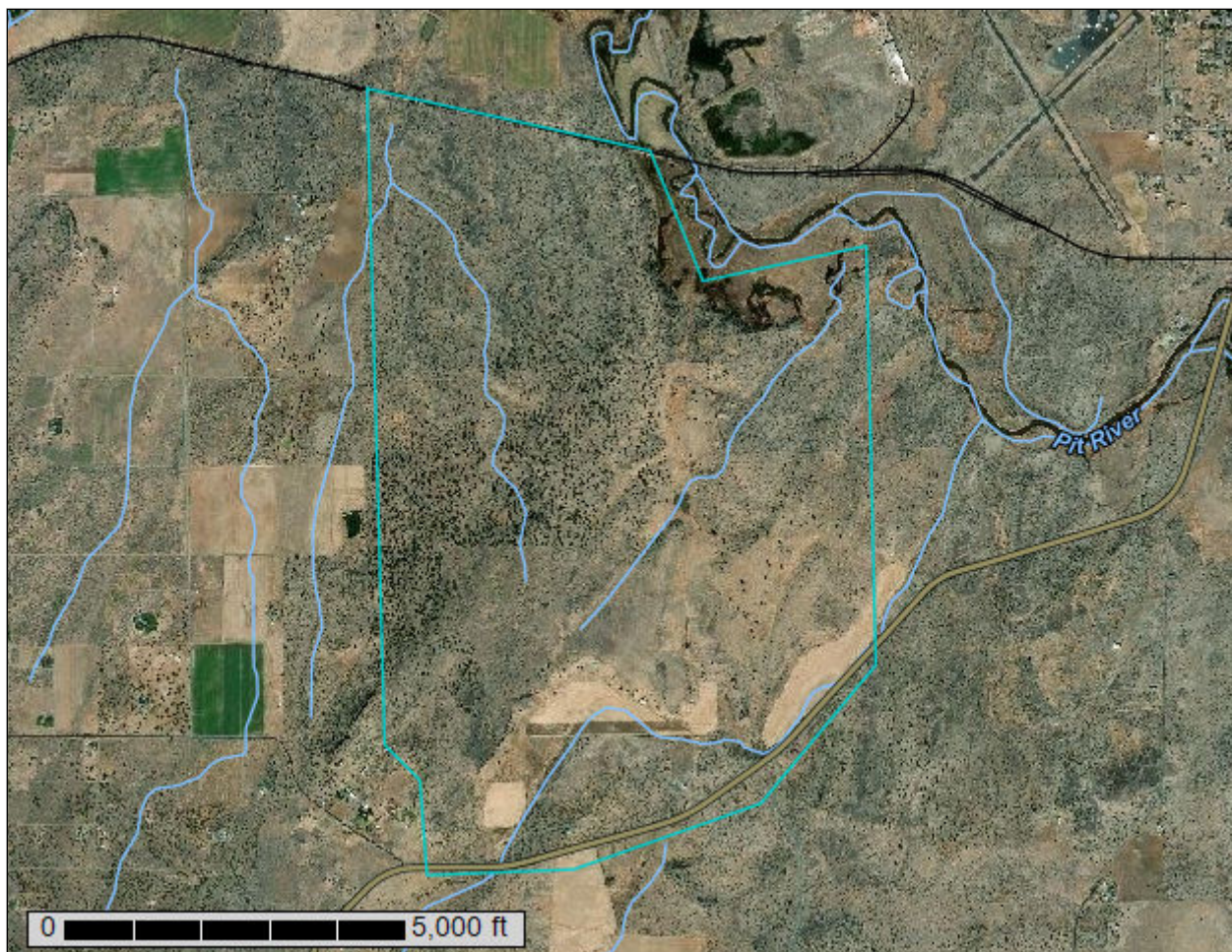
United States
Department of
Agriculture

NRCS

Natural
Resources
Conservation
Service

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Cooperative Soil Survey,
a joint effort of the United
States Department of
Agriculture and other
Federal agencies, State
agencies including the
Agricultural Experiment
Stations, and local
participants

Custom Soil Resource Report for **Modoc County, California, Alturas Area**



Preface

Soil surveys contain information that affects land use planning in survey areas. They highlight soil limitations that affect various land uses and provide information about the properties of the soils in the survey areas. Soil surveys are designed for many different users, including farmers, ranchers, foresters, agronomists, urban planners, community officials, engineers, developers, builders, and home buyers. Also, conservationists, teachers, students, and specialists in recreation, waste disposal, and pollution control can use the surveys to help them understand, protect, or enhance the environment.

Various land use regulations of Federal, State, and local governments may impose special restrictions on land use or land treatment. Soil surveys identify soil properties that are used in making various land use or land treatment decisions. The information is intended to help the land users identify and reduce the effects of soil limitations on various land uses. The landowner or user is responsible for identifying and complying with existing laws and regulations.

Although soil survey information can be used for general farm, local, and wider area planning, onsite investigation is needed to supplement this information in some cases. Examples include soil quality assessments (<http://www.nrcs.usda.gov/wps/portal/nrcs/main/soils/health/>) and certain conservation and engineering applications. For more detailed information, contact your local USDA Service Center (<https://offices.sc.egov.usda.gov/locator/app?agency=nrcs>) or your NRCS State Soil Scientist (http://www.nrcs.usda.gov/wps/portal/nrcs/detail/soils/contactus/?cid=nrcs142p2_053951).

Great differences in soil properties can occur within short distances. Some soils are seasonally wet or subject to flooding. Some are too unstable to be used as a foundation for buildings or roads. Clayey or wet soils are poorly suited to use as septic tank absorption fields. A high water table makes a soil poorly suited to basements or underground installations.

The National Cooperative Soil Survey is a joint effort of the United States Department of Agriculture and other Federal agencies, State agencies including the Agricultural Experiment Stations, and local agencies. The Natural Resources Conservation Service (NRCS) has leadership for the Federal part of the National Cooperative Soil Survey.

Information about soils is updated periodically. Updated information is available through the NRCS Web Soil Survey, the site for official soil survey information.

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How Soil Surveys Are Made

Soil surveys are made to provide information about the soils and miscellaneous areas in a specific area. They include a description of the soils and miscellaneous areas and their location on the landscape and tables that show soil properties and limitations affecting various uses. Soil scientists observed the steepness, length, and shape of the slopes; the general pattern of drainage; the kinds of crops and native plants; and the kinds of bedrock. They observed and described many soil profiles. A soil profile is the sequence of natural layers, or horizons, in a soil. The profile extends from the surface down into the unconsolidated material in which the soil formed or from the surface down to bedrock. The unconsolidated material is devoid of roots and other living organisms and has not been changed by other biological activity.

Currently, soils are mapped according to the boundaries of major land resource areas (MLRAs). MLRAs are geographically associated land resource units that share common characteristics related to physiography, geology, climate, water resources, soils, biological resources, and land uses (USDA, 2006). Soil survey areas typically consist of parts of one or more MLRA.

The soils and miscellaneous areas in a survey area occur in an orderly pattern that is related to the geology, landforms, relief, climate, and natural vegetation of the area. Each kind of soil and miscellaneous area is associated with a particular kind of landform or with a segment of the landform. By observing the soils and miscellaneous areas in the survey area and relating their position to specific segments of the landform, a soil scientist develops a concept, or model, of how they were formed. Thus, during mapping, this model enables the soil scientist to predict with a considerable degree of accuracy the kind of soil or miscellaneous area at a specific location on the landscape.

Commonly, individual soils on the landscape merge into one another as their characteristics gradually change. To construct an accurate soil map, however, soil scientists must determine the boundaries between the soils. They can observe only a limited number of soil profiles. Nevertheless, these observations, supplemented by an understanding of the soil-vegetation-landscape relationship, are sufficient to verify predictions of the kinds of soil in an area and to determine the boundaries.

Soil scientists recorded the characteristics of the soil profiles that they studied. They noted soil color, texture, size and shape of soil aggregates, kind and amount of rock fragments, distribution of plant roots, reaction, and other features that enable them to identify soils. After describing the soils in the survey area and determining their properties, the soil scientists assigned the soils to taxonomic classes (units). Taxonomic classes are concepts. Each taxonomic class has a set of soil characteristics with precisely defined limits. The classes are used as a basis for comparison to classify soils systematically. Soil taxonomy, the system of taxonomic classification used in the United States, is based mainly on the kind and character of soil properties and the arrangement of horizons within the profile. After the soil

scientists classified and named the soils in the survey area, they compared the individual soils with similar soils in the same taxonomic class in other areas so that they could confirm data and assemble additional data based on experience and research.

The objective of soil mapping is not to delineate pure map unit components; the objective is to separate the landscape into landforms or landform segments that have similar use and management requirements. Each map unit is defined by a unique combination of soil components and/or miscellaneous areas in predictable proportions. Some components may be highly contrasting to the other components of the map unit. The presence of minor components in a map unit in no way diminishes the usefulness or accuracy of the data. The delineation of such landforms and landform segments on the map provides sufficient information for the development of resource plans. If intensive use of small areas is planned, onsite investigation is needed to define and locate the soils and miscellaneous areas.

Soil scientists make many field observations in the process of producing a soil map. The frequency of observation is dependent upon several factors, including scale of mapping, intensity of mapping, design of map units, complexity of the landscape, and experience of the soil scientist. Observations are made to test and refine the soil-landscape model and predictions and to verify the classification of the soils at specific locations. Once the soil-landscape model is refined, a significantly smaller number of measurements of individual soil properties are made and recorded. These measurements may include field measurements, such as those for color, depth to bedrock, and texture, and laboratory measurements, such as those for content of sand, silt, clay, salt, and other components. Properties of each soil typically vary from one point to another across the landscape.

Observations for map unit components are aggregated to develop ranges of characteristics for the components. The aggregated values are presented. Direct measurements do not exist for every property presented for every map unit component. Values for some properties are estimated from combinations of other properties.

While a soil survey is in progress, samples of some of the soils in the area generally are collected for laboratory analyses and for engineering tests. Soil scientists interpret the data from these analyses and tests as well as the field-observed characteristics and the soil properties to determine the expected behavior of the soils under different uses. Interpretations for all of the soils are field tested through observation of the soils in different uses and under different levels of management. Some interpretations are modified to fit local conditions, and some new interpretations are developed to meet local needs. Data are assembled from other sources, such as research information, production records, and field experience of specialists. For example, data on crop yields under defined levels of management are assembled from farm records and from field or plot experiments on the same kinds of soil.

Predictions about soil behavior are based not only on soil properties but also on such variables as climate and biological activity. Soil conditions are predictable over long periods of time, but they are not predictable from year to year. For example, soil scientists can predict with a fairly high degree of accuracy that a given soil will have a high water table within certain depths in most years, but they cannot predict that a high water table will always be at a specific level in the soil on a specific date.

After soil scientists located and identified the significant natural bodies of soil in the survey area, they drew the boundaries of these bodies on aerial photographs and

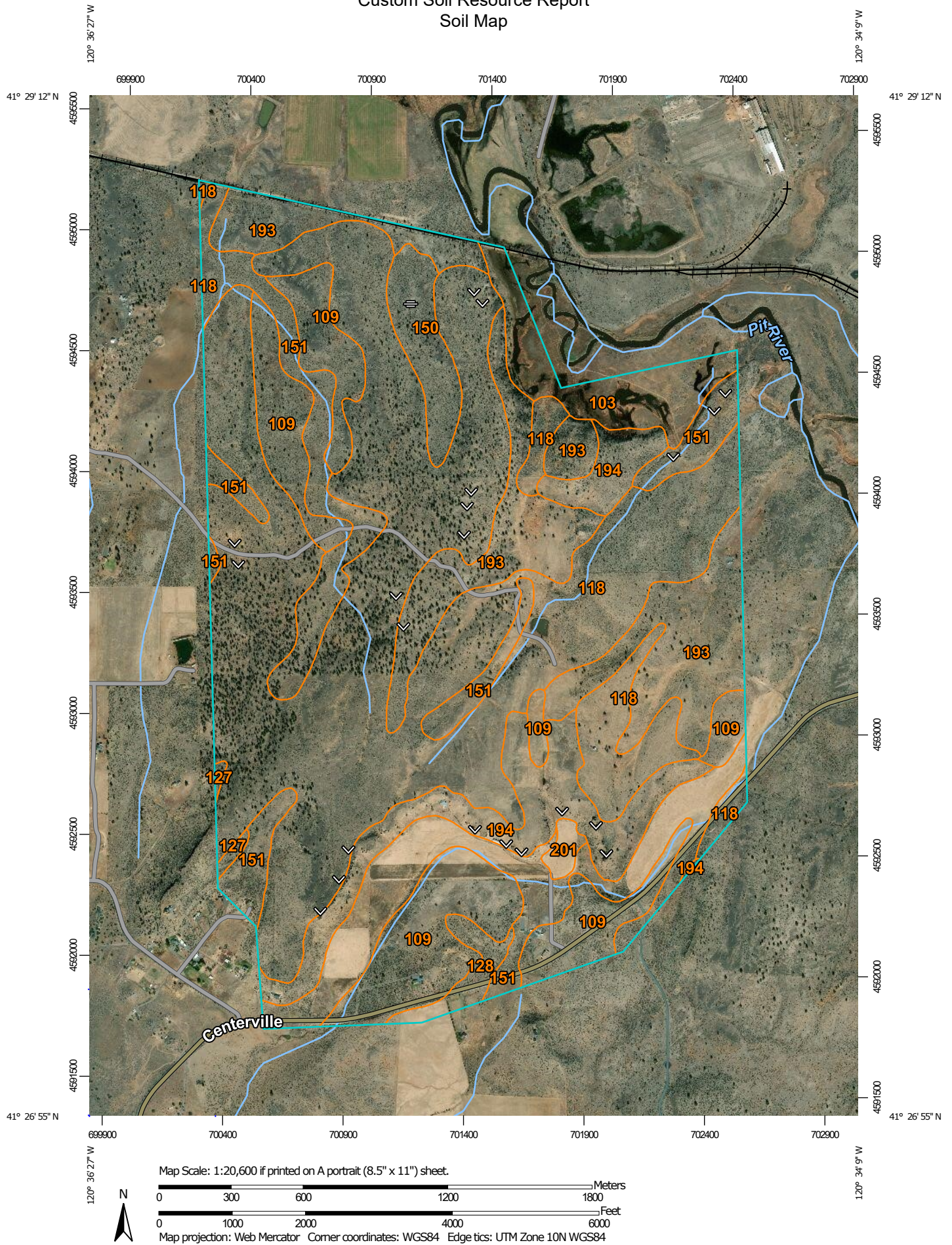
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identified each as a specific map unit. Aerial photographs show trees, buildings, fields, roads, and rivers, all of which help in locating boundaries accurately.

Soil Map

The soil map section includes the soil map for the defined area of interest, a list of soil map units on the map and extent of each map unit, and cartographic symbols displayed on the map. Also presented are various metadata about data used to produce the map, and a description of each soil map unit.

Custom Soil Resource Report Soil Map



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MAP LEGEND

Area of Interest (AOI)

 Area of Interest (AOI)

Soils

 Soil Map Unit Polygons

 Soil Map Unit Lines

 Soil Map Unit Points

Special Point Features

 Blowout

 Borrow Pit

 Clay Spot

 Closed Depression

 Gravel Pit

 Gravelly Spot

 Landfill

 Lava Flow

 Marsh or swamp

 Mine or Quarry

 Miscellaneous Water

 Perennial Water

 Rock Outcrop

 Saline Spot

 Sandy Spot

 Severely Eroded Spot

 Sinkhole

 Slide or Slip

 Sodic Spot

 Spoil Area

 Stony Spot

 Very Stony Spot

 Wet Spot

 Other

 Special Line Features

Water Features

 Streams and Canals

Transportation

 Rails

 Interstate Highways

 US Routes

 Major Roads

 Local Roads

Background

 Aerial Photography

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:24,000.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service

Web Soil Survey URL:

Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Modoc County, California, Alturas Area

Survey Area Data: Version 10, Sep 12, 2018

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: Aug 1, 2016—Oct 25, 2017

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend

Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
103	Alturas loam	57.8	3.7%
109	Bieber gravelly loam, 0 to 9 percent slopes	187.3	12.0%
118	Casuse sandy loam, 2 to 9 percent slopes	707.5	45.2%
127	Delma loam, 30 to 50 percent slopes	3.4	0.2%
128	Delma cobbly loam, 0 to 9 percent slopes	16.2	1.0%
150	Ladd sandy loam, 0 to 2 percent slopes	35.3	2.3%
151	Ladd sandy loam, 2 to 9 percent slopes	231.1	14.8%
193	Tuff outcrop-Casuse, eroded complex, 2 to 15 percent slopes	203.4	13.0%
194	Tuff outcrop-Casuse, eroded complex, 30 to 50 percent slopes	117.4	7.5%
201	Water	6.4	0.4%
Totals for Area of Interest		1,565.8	100.0%

Map Unit Descriptions

The map units delineated on the detailed soil maps in a soil survey represent the soils or miscellaneous areas in the survey area. The map unit descriptions, along with the maps, can be used to determine the composition and properties of a unit.

A map unit delineation on a soil map represents an area dominated by one or more major kinds of soil or miscellaneous areas. A map unit is identified and named according to the taxonomic classification of the dominant soils. Within a taxonomic class there are precisely defined limits for the properties of the soils. On the landscape, however, the soils are natural phenomena, and they have the characteristic variability of all natural phenomena. Thus, the range of some observed properties may extend beyond the limits defined for a taxonomic class. Areas of soils of a single taxonomic class rarely, if ever, can be mapped without including areas of other taxonomic classes. Consequently, every map unit is made up of the soils or miscellaneous areas for which it is named and some minor components that belong to taxonomic classes other than those of the major soils.

Most minor soils have properties similar to those of the dominant soil or soils in the map unit, and thus they do not affect use and management. These are called noncontrasting, or similar, components. They may or may not be mentioned in a particular map unit description. Other minor components, however, have properties

and behavioral characteristics divergent enough to affect use or to require different management. These are called contrasting, or dissimilar, components. They generally are in small areas and could not be mapped separately because of the scale used. Some small areas of strongly contrasting soils or miscellaneous areas are identified by a special symbol on the maps. If included in the database for a given area, the contrasting minor components are identified in the map unit descriptions along with some characteristics of each. A few areas of minor components may not have been observed, and consequently they are not mentioned in the descriptions, especially where the pattern was so complex that it was impractical to make enough observations to identify all the soils and miscellaneous areas on the landscape.

The presence of minor components in a map unit in no way diminishes the usefulness or accuracy of the data. The objective of mapping is not to delineate pure taxonomic classes but rather to separate the landscape into landforms or landform segments that have similar use and management requirements. The delineation of such segments on the map provides sufficient information for the development of resource plans. If intensive use of small areas is planned, however, onsite investigation is needed to define and locate the soils and miscellaneous areas.

An identifying symbol precedes the map unit name in the map unit descriptions. Each description includes general facts about the unit and gives important soil properties and qualities.

Soils that have profiles that are almost alike make up a *soil series*. Except for differences in texture of the surface layer, all the soils of a series have major horizons that are similar in composition, thickness, and arrangement.

Soils of one series can differ in texture of the surface layer, slope, stoniness, salinity, degree of erosion, and other characteristics that affect their use. On the basis of such differences, a soil series is divided into *soil phases*. Most of the areas shown on the detailed soil maps are phases of soil series. The name of a soil phase commonly indicates a feature that affects use or management. For example, Alpha silt loam, 0 to 2 percent slopes, is a phase of the Alpha series.

Some map units are made up of two or more major soils or miscellaneous areas. These map units are complexes, associations, or undifferentiated groups.

A *complex* consists of two or more soils or miscellaneous areas in such an intricate pattern or in such small areas that they cannot be shown separately on the maps. The pattern and proportion of the soils or miscellaneous areas are somewhat similar in all areas. Alpha-Beta complex, 0 to 6 percent slopes, is an example.

An *association* is made up of two or more geographically associated soils or miscellaneous areas that are shown as one unit on the maps. Because of present or anticipated uses of the map units in the survey area, it was not considered practical or necessary to map the soils or miscellaneous areas separately. The pattern and relative proportion of the soils or miscellaneous areas are somewhat similar. Alpha-Beta association, 0 to 2 percent slopes, is an example.

An *undifferentiated group* is made up of two or more soils or miscellaneous areas that could be mapped individually but are mapped as one unit because similar interpretations can be made for use and management. The pattern and proportion of the soils or miscellaneous areas in a mapped area are not uniform. An area can be made up of only one of the major soils or miscellaneous areas, or it can be made up of all of them. Alpha and Beta soils, 0 to 2 percent slopes, is an example.

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Some surveys include *miscellaneous areas*. Such areas have little or no soil material and support little or no vegetation. Rock outcrop is an example.

Modoc County, California, Alturas Area

103—Alturas loam

Map Unit Setting

National map unit symbol: jb3m

Elevation: 4,100 to 4,800 feet

Mean annual precipitation: 8 to 15 inches

Mean annual air temperature: 46 to 48 degrees F

Frost-free period: 80 to 130 days

Farmland classification: Farmland of statewide importance

Map Unit Composition

Alturas and similar soils: 85 percent

Minor components: 15 percent

Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Alturas

Setting

Landform: Terraces

Landform position (two-dimensional): Footslope

Landform position (three-dimensional): Talf

Down-slope shape: Linear

Across-slope shape: Linear

Parent material: Alluvium derived from basic igneous rock

Typical profile

H1 - 0 to 17 inches: loam

H2 - 17 to 42 inches: clay loam

H3 - 42 to 60 inches: stratified sandy loam to gravelly loam

Properties and qualities

Slope: 0 to 2 percent

Depth to restrictive feature: 40 to 50 inches to duripan

Natural drainage class: Moderately well drained

Runoff class: High

Capacity of the most limiting layer to transmit water (Ksat): Moderately low to moderately high (0.06 to 0.20 in/hr)

Depth to water table: About 30 to 60 inches

Frequency of flooding: Rare

Frequency of ponding: None

Salinity, maximum in profile: Very slightly saline to moderately saline (2.0 to 8.0 mmhos/cm)

Sodium adsorption ratio, maximum in profile: 15.0

Available water storage in profile: Moderate (about 6.8 inches)

Interpretive groups

Land capability classification (irrigated): 3s

Land capability classification (nonirrigated): 3s

Hydrologic Soil Group: C

Ecological site: ALKALI TERRACE (R021XE136CA)

Hydric soil rating: No

Minor Components

Rumbo

Percent of map unit: 5 percent
Hydric soil rating: No

Unnamed

Percent of map unit: 5 percent
Hydric soil rating: No

Buntingville

Percent of map unit: 3 percent
Hydric soil rating: No

Barnard, gravelly loam, 0-9%

Percent of map unit: 1 percent
Hydric soil rating: No

Salisbury, gravelly loam, 0-9%

Percent of map unit: 1 percent
Hydric soil rating: No

109—Bieber gravelly loam, 0 to 9 percent slopes

Map Unit Setting

National map unit symbol: jb3t
Elevation: 4,300 to 5,000 feet
Mean annual precipitation: 10 to 14 inches
Mean annual air temperature: 46 to 48 degrees F
Frost-free period: 70 to 80 days
Farmland classification: Not prime farmland

Map Unit Composition

Bieber and similar soils: 85 percent
Minor components: 15 percent
Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Bieber

Setting

Landform: Terraces
Landform position (two-dimensional): Footslope
Landform position (three-dimensional): Tread
Down-slope shape: Linear
Across-slope shape: Linear
Parent material: Alluvium derived from basic igneous rock

Typical profile

H1 - 0 to 6 inches: gravelly loam
H2 - 6 to 13 inches: gravelly clay loam
H3 - 13 to 18 inches: gravelly clay

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H4 - 18 to 22 inches: indurated

H5 - 22 to 60 inches: indurated

Properties and qualities

Slope: 0 to 9 percent

Depth to restrictive feature: 8 to 22 inches to duripan

Natural drainage class: Well drained

Runoff class: Very high

Capacity of the most limiting layer to transmit water (Ksat): Very low (0.00 to 0.00 in/hr)

Depth to water table: More than 80 inches

Frequency of flooding: None

Frequency of ponding: None

Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)

Available water storage in profile: Very low (about 2.4 inches)

Interpretive groups

Land capability classification (irrigated): 4e

Land capability classification (nonirrigated): 6e

Hydrologic Soil Group: D

Ecological site: HARDPAN TERRACE (R021XE137CA)

Hydric soil rating: No

Minor Components

Barnard

Percent of map unit: 3 percent

Hydric soil rating: No

Lovejoy

Percent of map unit: 3 percent

Hydric soil rating: No

Daphnedale

Percent of map unit: 2 percent

Hydric soil rating: No

Unnamed

Percent of map unit: 2 percent

Hydric soil rating: No

Casuse

Percent of map unit: 2 percent

Hydric soil rating: No

Pineal

Percent of map unit: 2 percent

Hydric soil rating: No

Delma

Percent of map unit: 1 percent

Hydric soil rating: No

118—Casuse sandy loam, 2 to 9 percent slopes

Map Unit Setting

National map unit symbol: jb43
Elevation: 4,300 to 4,700 feet
Mean annual precipitation: 8 to 14 inches
Mean annual air temperature: 46 to 50 degrees F
Frost-free period: 80 to 90 days
Farmland classification: Not prime farmland

Map Unit Composition

Casuse and similar soils: 85 percent
Minor components: 15 percent
Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Casuse

Setting

Landform: Terraces, escarpments
Landform position (two-dimensional): Shoulder
Landform position (three-dimensional): Riser
Down-slope shape: Linear
Across-slope shape: Linear
Parent material: Weakly cemented residuum weathered from tuff

Typical profile

H1 - 0 to 2 inches: sandy loam
H2 - 2 to 12 inches: clay loam
H3 - 12 to 60 inches: weathered bedrock

Properties and qualities

Slope: 2 to 9 percent
Depth to restrictive feature: 8 to 20 inches to paralithic bedrock
Natural drainage class: Well drained
Runoff class: Very high
Capacity of the most limiting layer to transmit water (Ksat): Very low (0.00 in/hr)
Depth to water table: More than 80 inches
Frequency of flooding: None
Frequency of ponding: None
Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)
Available water storage in profile: Very low (about 2.0 inches)

Interpretive groups

Land capability classification (irrigated): None specified
Land capability classification (nonirrigated): 6e
Hydrologic Soil Group: D
Ecological site: SHALLOW LOAMY (R021XE133CA)
Hydric soil rating: No

Minor Components

Unnamed

Percent of map unit: 8 percent

Hydric soil rating: No

Unnamed, moderately deep

Percent of map unit: 7 percent

Hydric soil rating: No

127—Delma loam, 30 to 50 percent slopes

Map Unit Setting

National map unit symbol: jb4d

Elevation: 4,300 to 5,300 feet

Mean annual precipitation: 10 to 16 inches

Mean annual air temperature: 46 to 48 degrees F

Frost-free period: 80 to 100 days

Farmland classification: Not prime farmland

Map Unit Composition

Delma and similar soils: 85 percent

Minor components: 15 percent

Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Delma

Setting

Landform: Escarpments, lake terraces

Landform position (two-dimensional): Backslope

Landform position (three-dimensional): Riser

Down-slope shape: Concave, linear

Across-slope shape: Convex, linear

Parent material: Lacustrine deposits derived from basic igneous rock

Typical profile

H1 - 0 to 13 inches: loam

H2 - 13 to 18 inches: clay

H3 - 18 to 60 inches: weathered bedrock

Properties and qualities

Slope: 30 to 50 percent

Depth to restrictive feature: 8 to 20 inches to paralithic bedrock

Natural drainage class: Well drained

Runoff class: Very high

Capacity of the most limiting layer to transmit water (Ksat): Very low (0.00 in/hr)

Depth to water table: More than 80 inches

Frequency of flooding: None

Frequency of ponding: None

Custom Soil Resource Report

Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)

Available water storage in profile: Very low (about 2.8 inches)

Interpretive groups

Land capability classification (irrigated): None specified

Land capability classification (nonirrigated): 7e

Hydrologic Soil Group: D

Ecological site: SHALLOW LOAMY (R021XE133CA)

Hydric soil rating: No

Minor Components

Daphnedale

Percent of map unit: 5 percent

Hydric soil rating: No

Casuse

Percent of map unit: 4 percent

Hydric soil rating: No

Ager

Percent of map unit: 3 percent

Hydric soil rating: No

Unnamed, eroded

Percent of map unit: 2 percent

Tuff outcrop

Percent of map unit: 1 percent

128—Delma cobbly loam, 0 to 9 percent slopes

Map Unit Setting

National map unit symbol: jb4f

Elevation: 4,300 to 5,300 feet

Mean annual precipitation: 10 to 16 inches

Mean annual air temperature: 46 to 48 degrees F

Frost-free period: 80 to 100 days

Farmland classification: Not prime farmland

Map Unit Composition

Delma and similar soils: 85 percent

Minor components: 15 percent

Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Delma

Setting

Landform: Lake terraces, escarpments

Landform position (two-dimensional): Shoulder

Landform position (three-dimensional): Tread

Custom Soil Resource Report

Down-slope shape: Linear

Across-slope shape: Linear

Parent material: Lacustrine deposits derived from basic igneous rock

Typical profile

H1 - 0 to 13 inches: cobbly loam

H2 - 13 to 18 inches: clay

H3 - 18 to 60 inches: weathered bedrock

Properties and qualities

Slope: 0 to 9 percent

Depth to restrictive feature: 8 to 20 inches to paralithic bedrock

Natural drainage class: Well drained

Runoff class: Very high

Capacity of the most limiting layer to transmit water (Ksat): Very low (0.00 in/hr)

Depth to water table: More than 80 inches

Frequency of flooding: None

Frequency of ponding: None

Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)

Available water storage in profile: Very low (about 2.0 inches)

Interpretive groups

Land capability classification (irrigated): 6e

Land capability classification (nonirrigated): 6e

Hydrologic Soil Group: D

Ecological site: SHALLOW LOAMY (R021XE133CA)

Hydric soil rating: No

Minor Components

Daphnedale

Percent of map unit: 8 percent

Hydric soil rating: No

Ager

Percent of map unit: 3 percent

Hydric soil rating: No

Barnard

Percent of map unit: 2 percent

Hydric soil rating: No

Casuse

Percent of map unit: 2 percent

Hydric soil rating: No

150—Ladd sandy loam, 0 to 2 percent slopes

Map Unit Setting

National map unit symbol: jb54

Elevation: 4,300 to 5,050 feet

Custom Soil Resource Report

Mean annual precipitation: 12 to 16 inches
Mean annual air temperature: 45 to 46 degrees F
Frost-free period: 80 to 100 days
Farmland classification: Prime farmland if irrigated

Map Unit Composition

Ladd and similar soils: 85 percent
Minor components: 15 percent
Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Ladd

Setting

Landform: Alluvial fans
Landform position (two-dimensional): Footslope
Landform position (three-dimensional): Rise
Down-slope shape: Linear
Across-slope shape: Linear
Parent material: Alluvium derived from basic igneous rock

Typical profile

H1 - 0 to 12 inches: sandy loam
H2 - 12 to 40 inches: clay loam
H3 - 40 to 60 inches: sandy loam

Properties and qualities

Slope: 0 to 2 percent
Depth to restrictive feature: More than 80 inches
Natural drainage class: Well drained
Runoff class: Medium
Capacity of the most limiting layer to transmit water (Ksat): Moderately high (0.20 to 0.57 in/hr)
Depth to water table: More than 80 inches
Frequency of flooding: None
Frequency of ponding: None
Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)
Available water storage in profile: Moderate (about 8.7 inches)

Interpretive groups

Land capability classification (irrigated): 3s
Land capability classification (nonirrigated): 4s
Hydrologic Soil Group: C
Ecological site: LOAMY (R021XE131CA)
Hydric soil rating: No

Minor Components

Unnamed, calcareous subsoil

Percent of map unit: 6 percent
Hydric soil rating: No

Buntingville

Percent of map unit: 3 percent
Hydric soil rating: No

Ladd, sandy loam, 2 to 9% slopes

Percent of map unit: 2 percent

Hydric soil rating: No

Calimus

Percent of map unit: 2 percent

Hydric soil rating: No

Unnamed, loam surface

Percent of map unit: 2 percent

Hydric soil rating: No

151—Ladd sandy loam, 2 to 9 percent slopes

Map Unit Setting

National map unit symbol: jb55

Elevation: 4,300 to 5,050 feet

Mean annual precipitation: 12 to 16 inches

Mean annual air temperature: 45 to 46 degrees F

Frost-free period: 80 to 100 days

Farmland classification: Prime farmland if irrigated

Map Unit Composition

Ladd and similar soils: 85 percent

Minor components: 15 percent

Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Ladd

Setting

Landform: Alluvial fans

Landform position (two-dimensional): Footslope

Landform position (three-dimensional): Tread

Down-slope shape: Linear

Across-slope shape: Linear

Parent material: Alluvium derived from basic igneous rock

Typical profile

H1 - 0 to 12 inches: sandy loam

H2 - 12 to 40 inches: clay loam

H3 - 40 to 60 inches: sandy loam

Properties and qualities

Slope: 2 to 9 percent

Depth to restrictive feature: More than 80 inches

Natural drainage class: Well drained

Runoff class: High

Capacity of the most limiting layer to transmit water (Ksat): Moderately high (0.20 to 0.57 in/hr)

Depth to water table: More than 80 inches

Frequency of flooding: None

Frequency of ponding: None

Custom Soil Resource Report

Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)

Available water storage in profile: Moderate (about 8.7 inches)

Interpretive groups

Land capability classification (irrigated): 3e

Land capability classification (nonirrigated): 4e

Hydrologic Soil Group: C

Ecological site: LOAMY (R021XE131CA)

Hydric soil rating: No

Minor Components

Unnamed, calcareous subsoil

Percent of map unit: 5 percent

Hydric soil rating: No

Buntingville

Percent of map unit: 3 percent

Hydric soil rating: No

Calimus

Percent of map unit: 3 percent

Hydric soil rating: No

Lakeview, 0 to 2% slopes

Percent of map unit: 2 percent

Hydric soil rating: No

Modoc

Percent of map unit: 2 percent

Hydric soil rating: No

193—Tuff outcrop-Casuse, eroded complex, 2 to 15 percent slopes

Map Unit Setting

National map unit symbol: j b6j

Elevation: 4,000 to 4,700 feet

Mean annual precipitation: 8 to 14 inches

Mean annual air temperature: 46 to 50 degrees F

Frost-free period: 80 to 90 days

Farmland classification: Not prime farmland

Map Unit Composition

Tuff outcrop: 55 percent

Casuse and similar soils: 35 percent

Minor components: 10 percent

Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Tuff Outcrop

Setting

Landform position (two-dimensional): Shoulder
Landform position (three-dimensional): Free face
Down-slope shape: Linear
Across-slope shape: Linear
Parent material: Tuff

Typical profile

H1 - 0 to 10 inches: unweathered bedrock

Properties and qualities

Slope: 2 to 15 percent
Depth to restrictive feature: About 0 inches to lithic bedrock
Runoff class: Very high
Available water storage in profile: Very low (about 0.0 inches)

Interpretive groups

Land capability classification (irrigated): None specified
Land capability classification (nonirrigated): 8s
Hydric soil rating: No

Description of Casuse

Setting

Landform: Terraces
Landform position (two-dimensional): Shoulder
Landform position (three-dimensional): Side slope
Down-slope shape: Linear
Across-slope shape: Linear
Parent material: Residuum weathered from tuff

Typical profile

H1 - 0 to 2 inches: sandy loam
H2 - 2 to 12 inches: clay loam
H3 - 12 to 60 inches: weathered bedrock

Properties and qualities

Slope: 2 to 15 percent
Depth to restrictive feature: 8 to 20 inches to paralithic bedrock
Natural drainage class: Well drained
Runoff class: Medium
Capacity of the most limiting layer to transmit water (Ksat): Very low (0.00 to 0.00 in/hr)
Depth to water table: More than 80 inches
Frequency of flooding: None
Frequency of ponding: None
Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)
Available water storage in profile: Very low (about 2.0 inches)

Interpretive groups

Land capability classification (irrigated): None specified
Land capability classification (nonirrigated): 6e
Hydrologic Soil Group: D
Ecological site: SHALLOW LOAMY (R021XE133CA)

Hydric soil rating: No

Minor Components

Daphnedale, cobbly loam, 0 to 9% slopes

Percent of map unit: 7 percent

Hydric soil rating: No

Ladd

Percent of map unit: 3 percent

Hydric soil rating: No

194—Tuff outcrop-Casuse, eroded complex, 30 to 50 percent slopes

Map Unit Setting

National map unit symbol: jb6k

Elevation: 4,000 to 4,700 feet

Mean annual precipitation: 8 to 14 inches

Mean annual air temperature: 46 to 50 degrees F

Frost-free period: 80 to 90 days

Farmland classification: Not prime farmland

Map Unit Composition

Tuff outcrop: 60 percent

Casuse and similar soils: 30 percent

Minor components: 10 percent

Estimates are based on observations, descriptions, and transects of the mapunit.

Description of Tuff Outcrop

Setting

Landform position (two-dimensional): Shoulder

Landform position (three-dimensional): Free face

Down-slope shape: Concave

Across-slope shape: Convex

Parent material: Tuff

Typical profile

H1 - 0 to 60 inches: bedrock

Properties and qualities

Slope: 30 to 50 percent

Depth to restrictive feature: About 0 inches to paralithic bedrock

Runoff class: Very high

Available water storage in profile: Very low (about 0.0 inches)

Interpretive groups

Land capability classification (irrigated): None specified

Land capability classification (nonirrigated): 8s

Hydric soil rating: No

Description of Casuse

Setting

Landform: Escarpments

Landform position (two-dimensional): Toeslope, shoulder

Landform position (three-dimensional): Free face, side slope

Down-slope shape: Linear, concave

Across-slope shape: Linear, convex

Parent material: Residuum weathered from tuff

Typical profile

H1 - 0 to 2 inches: cobbly sandy loam

H2 - 2 to 12 inches: clay loam

H3 - 12 to 60 inches: weathered bedrock

Properties and qualities

Slope: 30 to 50 percent

Depth to restrictive feature: 8 to 20 inches to paralithic bedrock

Natural drainage class: Well drained

Runoff class: High

Capacity of the most limiting layer to transmit water (Ksat): Very low (0.00 to 0.00 in/hr)

Depth to water table: More than 80 inches

Frequency of flooding: None

Frequency of ponding: None

Salinity, maximum in profile: Nonsaline to very slightly saline (0.0 to 2.0 mmhos/cm)

Available water storage in profile: Very low (about 2.0 inches)

Interpretive groups

Land capability classification (irrigated): None specified

Land capability classification (nonirrigated): 6e

Hydrologic Soil Group: D

Ecological site: SHALLOW LOAMY (R021XE133CA)

Hydric soil rating: No

Minor Components

Casuse, sandy loam, 2 to 9% slopes

Percent of map unit: 6 percent

Hydric soil rating: No

Daphnedale

Percent of map unit: 4 percent

Hydric soil rating: No

201—Water

Map Unit Composition

Water: 100 percent

Custom Soil Resource Report

Estimates are based on observations, descriptions, and transects of the mapunit.

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Custom Soil Resource Report

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Budgetary Cost Estimates

4

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Opinion of Probable Project Cost
City of Alturas WWTP
Alternative 1 Cost Summary

Item	Description	Units	Quantity	Unit Cost	Total Cost
<u>Demolition</u>					
1	Decommission Plant Components ²	LS	1	\$ 25,000	\$ 25,000
2	Removing Existing Trickling Filter	LS	1	\$ 41,000	\$ 41,000
<u>New Unit Processes</u>					
3	Spiralift 1/4" Screen and Housing	LS	1	\$ 309,000	\$ 309,000
4	New Trickling Filters (x 4, with flow control and recirculation box, pumps, etc.)	LS	1	\$ 1,744,000	\$ 1,744,000
5	Nitrification/Denitrification	LS	1	\$ 900,000	\$ 900,000
6	New Aerobic Digester	EA	2	\$ 386,000	\$ 772,000
7	Metals Removal	LS	1	\$ 1,500,000	\$ 1,500,000
<u>Rehabilitated/Upgraded Facilities</u>					
8	Rehabilitate Existing Influent Pump Stations - New VFD Pumps and Piping	LS	1	\$ 200,000	\$ 200,000
9	Convert disinfection system to hypochlorite	LS	1	\$ 50,000	\$ 50,000
10	Rehabilitate Secondary Clarifier (No. 1)	LS	1	\$ 300,000	\$ 300,000
11	Replace Back-up Generator and Electrical	LS	1	\$ 57,000	\$ 57,000
				Mobilization (12%):	\$ 708,000
				Subtotal :	\$ 6,606,000
				Contingency (30%):	\$ 1,982,000
				Subtotal Construction:	\$ 8,588,000
				Engineering/Construction Management (18%):	\$ 1,546,000
				Environmental/Permitting (5%):	\$ 430,000
				Administration/Legal (4%):	\$ 344,000
				Total Project:	\$ 12,890,000

Notes:

1. Refer to subsequent pages for additional information on each line item.
2. Decommissioning components of the existing plant involve termination of electrical connections throughout the plant and remove elevated mechanical infrastructure from the site. Components include Grit Chamber, Grinder, Primary Clarifier, and Digester. Concrete Substructures to remain.
3. Values rounded up to nearest \$1,000



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JOB	518004	ALT 1.
SHEET NO.	2	OF 7
CALC'ED BY	PCM	DATE 3/12/2020
CHECKED BY	AHR	DATE 11/6/2020

Item No.: 1 Decommission Plant Components

Terminate Electrical Connections to WWTF treatment components and Removal of Existing Mechanical Infrastructure including the secondary clarifier distribution arm/baffling, Grit Chamber baffles and Mixer, Digester Roof and Mixer, and Primary Clarifier Scum baffles, drive housing, decking, etc.

\$ 25,000.00 Allowance

Item No.: 2 Removing Existing Trickling Filter

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Remove Existing Media Bed	1	LS	\$ 2,500.00	\$ 2,500.00	3
Removal of Existing Piping/Emitters	1	LS	\$ 1,750.00	\$ 1,750.00	3
Cleaning piping, media, and emitters	1	LS	\$ 3,000.00	\$ 3,000.00	3
Drain and Cap Existing Piping	1	LS	\$ 2,250.00	\$ 2,250.00	3
Clean out Trickling Filter Basin	1	LS	\$ 3,000.00	\$ 3,000.00	1
Demolish CMU block retaining wall	3000	SF	\$ 3.63	\$ 10,903.20	4
Demolish Original Concrete Wall	11940	CF	\$ 0.68	\$ 8,171.74	4
Earthwork back to Existing Grade	757	CY	\$ 12.39	\$ 9,383.36	4
Subtotal :				\$ 41,000.00	

Notes:

1. CMU Wall 8" wide CMU Blocks 2 walls L = 142' H = 7' ; 2 walls L = 72' H = 7'
2. Original Concrete Structure 2 walls L = 142' H = 5' W = 1'; 2 walls L = 72' H = 5' W = 1'; Concrete Floor L = 140' H = 1' W = 70'
3. Earthwork to Existing Grade Dimensions Assumed; L = 142' W = 72' H = 2' and Fill could be allocated from location onsite.

Sources:

1. Per Steve B
2. Per Anders R.
3. Per Phil M.
4. DCD Simple Estimator 2018



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JOB	518004	ALT 1.
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Item No.: 3 Spiralift 1/4" Screen and Housing

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Franklin Miller Spiralift	1	LS	\$ 87,000.00	\$ 87,000.00	1
Installation of Spiralift	1	LS	\$ 174,000.00	\$ 174,000.00	2
Building for New Headworks	240	SF	\$ 200.00	\$ 48,000.00	2
Subtotal:				\$ 309,000.00	

Note:

Installation Cost Assumed to be 2 x the Unit Budgetary Cost

Building Approximately 12' x 20' around Spiralift to protect it from the Environment

Sources:

1. Franklin Miller Budgetary Cost Estimate
2. Per Anders R.



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JOB	518004	ALT 1.
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Item No.: 4 New Trickling Filters (x 4, with flow control and recirculation box, pumps, etc.)

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Clearing and Grubbing Area	0.34	Acre	\$ 6,127.76	\$ 2,083.44	4
Rough Grading	1097	CY	\$ 12.39	\$ 13,592.66	3
Ditch Excavation	587	CY	\$ 14.48	\$ 8,499.76	4
General Excavation	1047	CY	\$ 4.99	\$ 5,226.99	3
Class 2 Agg Base (2' compacted)	94	Ton	\$ 24.08	\$ 2,269.49	4
Concrete Tanks (Forms and Material Cost and Installation	188	CY	\$ 954.00	\$ 179,352.00	2
Cost for 27' Distribution Arms	1	LS	\$ 606,000.00	\$ 606,000.00	5
Material and Installation Cost for Drive System and Control Panel	1	LS	\$ 140,000.00	\$ 140,000.00	5
Material and Installation Cost for Trickling Filter Media/Supports	4	EA	\$ 132,000.00	\$ 528,000.00	6
Piping	1057	LF	\$ 150.00	\$ 158,550.00	1
Recirculation Pumps, Control Boxes, Valving	1	LS	\$ 100,000.00	\$ 100,000.00	1

Subtotal: \$ 1,744,000.00

Notes:

1. Trickling Filter Beds Demensions; Diameter = 27', Height = 18'
2. Total Construction Cost (2 x the Unit Budgetary Cost)

Source:

1. Per Anders
2. Per Steve
3. DCD Simple Estimator
4. Odot 2018 Weighted Averages Awarded Price
5. WesTech Budgetary Quote
6. Brentwood Budgetary Quote Please note the distribution system for the TF don't include a steel bridge that supports the whole drive unit.



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JOB	518004	ALT 1.
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Item No.: 5 Nitrification/Denitrification

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Budgetary Quote from Triplepoint Environmental, NitrOx+D	1	LS	\$ 600,000.00	\$ 600,000.00	-
Site Work Installation	1	LS	\$ 3,000,000.00	\$ 300,000.00	-
Subtotal Per NitrOx+D Unit				\$ 900,000.00	

Item No.: 6 New Aerobic Digester

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Excavation and Grading	1	LS	\$ 7,500.00	\$ 7,500.00	1
Rebar and Foundation	1	LS	\$ 10,000.00	\$ 10,000.00	1
Piping including Earthwork	1	LS	\$ 81,000.00	\$ 81,000.00	1
Mixer	1	LS	\$ 97,000.00	\$ 97,000.00	1
Installation of Mixer	1	LS	\$ 145,500.00	\$ 145,500.00	4
Digester with Reinforcement	1	LS	\$ 45,000.00	\$ 45,000.00	1
Subtotal Per Digester:				\$ 386,000.00	

Notes:

1. 20% added to vendor costs to account for contractor markup
2. Aerobic Digester Demensions; H = 18' Dia 30'
3. Mixer cost provided by WesTech
4. Installation of Mixer assumed to be 1.5 x Unit Budgetary Cost

Sources:

1. 2007 National Construction Estimator



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JOB	518004	ALT 1.
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CHECKED BY	AHR	DATE 11/6/2020

Item No.: 7

Metals Removal

Item	Quantity	Unit	Unit Cost	Total	Source
Metals Removal	1	LS	\$ 1,500,000.00	\$ 1,500,000.00	-

Subtotal: \$ 1,500,000.00

Notes:

Sources:

1. Jim Coskey, JBI Water

Process:

Precipitation using contact clarifier

Need pH above 10.3

Will require pH reduction after process

See report text for additional description



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JOB	518004	ALT 1.
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CALC'ED BY	PCM	DATE 3/12/2020
CHECKED BY	AHR	DATE 11/6/2020

Item No.: 8 Rehabilitate Existing Influent Pump Stations - New VFD Pumps and Piping

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
750 HP Variable Drive Pump	2	EA	\$ 30,000.00	\$ 60,000.00	1
350 HP Variable Drive Pump	1	EA	\$ 15,000.00	\$ 15,000.00	1
Piping, Valving, and Support	1	LS	\$ 40,000.00	\$ 40,000.00	2
Crane and Access Point	1	LS	\$ 75,000.00	\$ 75,000.00	1
Improve Existing Electrical	1	LS	\$ 10,000.00	\$ 10,000.00	1

Subtotal : \$ 200,000.00

Notes:

1. 20% added from Vendors Price for Shipping:
2. (2) 750 Hp Variable Drive Pumps
3. (1) 350 Hp Variable Drive Pump
4. Currently No Crane/ Access to remove Pumps Safely

Sources:

1. Per Anders R.
2. DCD Simple Estimator 2018

Item No.: 9 Convert disinfection system to hypochlorite

Additional detail for capital construction line items

Assume conversion from gaseous to liquid disinfection and dechlorination in existing building.

Allowance: \$50,000



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JOB	518004	ALT 1.
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Item No.: 10 Rehabilitate Secondary Clarifier (No. 1)

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Material and Installation Cost for	1	LS	\$ 260,000.000	\$ 260,000.000	1
Material and Installation Cost for	1	LS	\$ 16,000.00	\$ 16,000.00	1
Replace Existing Valving	1	LS	\$ 15,000.00	\$ 15,000.00	2
Miscellaneous Repair of	1	LS	\$ 5,000.00	\$ 5,000.00	3
Sludge Pump	1	LS	\$ 4,000.00	\$ 4,000.00	3

Subtotal: \$ 300,000.000

Notes:

1. (1) 12" Ductile Iron Gate Valve
2. Total Construction Cost Assumed to be 2 x the Unit Budgetary Cost

Sources:

1. WesTech Budgetary Quote (Clarifier has steel bridge associated with the distributor arm that holds the weight of
2. DCD Simple Estimator 2018
3. Per Anders R.

Item No.: 11 Replace Back-up Generator and Electrical

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Generac Commercial 150KW	1	LS	\$ 34,000.00	\$ 34,000.00	1
Earthwork and Concrete Pad for	1	LS	\$ 3,000.00	\$ 3,000.00	2
Electrical Connection	1	LS	\$ 20,000.00	\$ 20,000.00	3

Subtotal: \$ 57,000.00

Sources:

1. Generac Website
2. DCD Simple Estimator 2018
3. Per Anders R.

**Opinion of Probable Project Cost
City of Alturas WWTP
Alternative 2 Cost Summary**

Item	Description	Units	Quantity	Unit Cost	Total Cost
Demolition					
1	Decommission Plant Components ²	LS	1	\$ 25,000	\$ 25,000
2	Remove Existing Trickling Filter	LS	1	\$ 41,000	\$ 41,000
New Unit Processes					
3	Spiralift 1/4" Screen and Housing	LS	1	\$ 309,000	\$ 309,000
4	New Trickling Filters (x 4, with flow control and recirculation box, pumps, etc.)	LS	1	\$ 1,744,000	\$ 1,744,000
5	New Aerobic Digester	EA	2	\$ 386,000	\$ 772,000
6	Effluent Pump Station	LS	1	\$ 900,000	\$ 900,000
7	Effluent Force Main	LS	1	\$ 582,000	\$ 582,000
8	Infiltration Ponds	LS	1	\$ 157,000	\$ 157,000
Rehabilitated/Upgraded Facilities					
9	Replace Back-up Generator and Electrical	LS	1	\$ 56,000	\$ 56,000
10	Rehabilitate Existing Influent Pump Stations - New VFD Pumps and Piping	LS	1	\$ 200,000	\$ 200,000
11	Rehabilitate Secondary Clarifier (No. 1)	LS	1	\$ 300,000	\$ 300,000
12	Effluent Flow Meter	LS	1	\$ 15,000	\$ 15,000
Other					
13	Groundwater Monitoring Wells	LS	1	\$ 60,000	\$ 60,000
				Mobilization (12%):	\$ 620,000
				Subtotal :	\$ 5,781,000
				Contingency (30%):	\$ 1,735,000
				Subtotal Construction:	\$ 7,516,000
				Land Acquisition:	\$ 361,000
				Engineering/Construction Management (18%):	\$ 1,353,000
				Environmental/Permitting (5%):	\$ 376,000
				Administration/Legal (4%):	\$ 301,000
				Total Project:	\$ 11,642,000

Notes:

1. Refer to subsequent pages for additional information on each line item.
2. Decommissioning components of the existing plant involve termination of electrical connections throughout the plant and remove elevated mechanical infrastructure from the site. Components include Grit Chamber, Grinder, Primary Clarifier, and Digester. Concrete Substructures to remain.
3. Values rounded up to nearest \$1,000

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JOB	518004	ALT 2.
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Item No.: 1**Decommission Plant Components**

Terminate Electrical Connections to WWTF treatment components and Removal of Existing Mechanical Infrastructure including the secondary clarifier distribution arm/baffling, Grit Chamber baffles and Mixer, Digester Roof and Mixer, and existing Grinder, and Primary Clarifier Scum baffles, drive housing, decking, etc.

\$ 25,000.00 Allowance

Item No.: 2**Remove Existing Trickling Filter**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Remove Existing Media Bed	1	LS	\$ 2,500.00	\$ 2,500.00	3
Removal of Existing Piping/Emitters	1	LS	\$ 1,750.00	\$ 1,750.00	3
Cleaning piping, media, and emitters	1	LS	\$3,000	\$ 3,000.00	3
Drain and Cap Existing Piping	1	LS	\$2,250	\$ 2,250.00	3
Clean out Trickling Filter Basin	1	LS	\$3,000	\$ 3,000.00	1
Demolish CMU block retaining wall	3000	SF	\$ 3.63	\$ 10,903.20	4
Demolish Original Concrete Wall	11940	CF	\$ 0.68	\$ 8,171.74	4
Earthwork back to Existing Grade	757	CY	\$ 12.39	\$ 9,379.23	4
Subtotal :				\$ 41,000.00	

Notes:

1. CMU Wall 8" wide CMU Blocks 2 walls L = 142' H = 7' ; 2 walls L = 72' H = 7'
2. Original Concrete Structure 2 walls L = 142' H = 5' W = 1'; 2 walls L = 72' H = 5' W = 1'; Concrete Floor L = 140' H = 1' W = 70'
3. Earthwork to Existing Grade Dimensions Assumed; L = 142' W = 72' H = 2' and Fill could be allocated from location onsite.

Sources:

1. Per Steve B
2. Per Anders R.
3. Per Phil M.
4. DCD Simple Estimator 2018

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Item No.: 3**Spiralift 1/4" Screen and Housing**

Additional detail for capital construction line items

1.2% added from Vendors Price for Shipping:

Item	Quantity	Unit	Unit Cost	Total	Source
Franklin Miller Spiralift Screen	1	LS	\$ 87,000.00	\$ 87,000.00	1
Installation fo Spiralift	1	LS	\$ 174,000.00	\$ 174,000.00	2
Building for New Headworks	240	SF	\$ 200.00	\$ 48,000.00	2
Subtotal:				\$ 309,000.00	

Note:

Installation Cost Assumed to be 2 x the Unit Budgetary Cost.

Building 12' x 20' around Helicoil Sieve to protect it from freezing.

Sources

1. Franklin Miller Budgetary Cost Estimate
2. Per Anders R.



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JOB	518004	ALT 2.
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Item No.: 4 New Trickling Filters (x 4, with flow control and recirculation box, pumps, etc.)

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Clearing and Grubbing Area	0.34	Acre	\$ 6,127.76	\$ 2,083.44	4
Rough Grading	1097	CY	\$ 12.39	\$ 13,592.66	3
Ditch Excavation	587	CY	\$ 14.48	\$ 8,499.76	4
General Excavation	1047	CY	\$ 4.99	\$ 5,226.99	3
Class 2 Agg Base (2' compacted)	94	Ton	\$ 24.08	\$ 2,263.52	4
Concrete Tanks Forms and	188	CY	\$ 954.00	\$ 179,352.00	2
Material and Installation Cost for 27' Distribution Arms (x4)	1	LS	\$ 606,000.00	\$ 606,000.00	5
Material and Installation Cost for Drive System and Control Panel (x4)	1	LS	\$ 140,000.00	\$ 140,000.00	5
Material and Installation Cost for Trickling Filter Media/Supports	4	EA	\$ 132,000.00	\$ 528,000.00	6
Piping	1057	LF	\$ 150.00	\$ 158,550.00	3
Recirculation Pumps, Control Boxes, Va	1	LS	\$ 100,000.00	\$ 100,000.00	1
Subtotal:				\$ 1,744,000.00	

Notes:

1. Trickling Filter Beds Demensions; Diameter = 27', Height = 18'
2. Total Construction Cost (2 x the Unit Budgetary Cost)

Source:

1. Per Anders
2. 2007 National Construction Estimator
3. DCD Simple Estimator
4. Odot 2018 Weighted Averages Awarded Price
5. WesTech Budgetary Quote
6. Brentwood Budgetary Quote

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Item No.: 5**New Aerobic Digester**

Additional detail for capital construction line items

Notes:

Item	Quantity	Unit	Unit Cost	Total	Source
Excavation and Grading	1	LS	\$ 7,500.00	\$ 7,500.00	1 \$ 7,500.00
Rebar and Foundation	1	LS	\$ 10,000.00	\$ 10,000.00	1 \$ 10,000.00
Piping including Earthwork	1	LS	\$ 81,000.00	\$ 81,000.00	1 \$ 75,000.00
Mixer	1	LS	\$ 97,000.00	\$ 97,000.00	1 \$ 97,000.00
Installation of Mixer	1	LS	\$ 145,500.00	\$ 145,500.00	4 \$ 145,500.00
Digester with Reinforcement	1	LS	\$ 45,000	\$ 45,000.00	1 \$ 45,000.00

Subtotal Per Digester: \$ 386,000.00

Notes:

1. 20% added to vendor costs to account for contractor markup
2. Digester Dimensions H = 18' Dia 30'
3. Mixer cost provided by WesTech
4. Installation of Mixer assumed to be 1.5 x Unit Budgetary Cost

Sources:

1. 2007 National Construction Estimator

Item No.: 6 Effluent Pump Station

Additional detail for capital construction line items

Notes:

This would include the installation of a dry and wet well with capacity to pump from the post treated plant to the new infiltration beds. This would include (3) pumps. This amount would include connection to electric.

\$ 900,000.00 Allowance

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Item No.: 7**Effluent Force Main**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Install 8" Force Main From WWTF to Infiltration Ponds	11000	LF	\$ 50.00	\$ 550,000.00	1
Install 8" Force Main Ductile Iron Bridge Crossing with Supports	500	LF	\$ 60.00	\$ 30,000.00	1
Bridge Supports	1	LS	\$ 1,300.00	\$ 1,300.00	2
Subtotal:				\$ 582,000.00	

Note:

1. Cal Trans Guidance for Zone 1 & 2
2. DCD Simple Estimator

Item No.: 8**Infiltration Ponds**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Earthwork	26482	CY	\$ 4.94	\$ 130,932.30	1
Valving, Piping, and Restraints	1	LS	\$ 17,070.00	\$ 17,070.00	1
Erosion Control	1	LS	\$ 8,000.00	\$ 8,000.00	2
Subtotal:				\$ 157,000.00	

Notes:

1. 20% added to vendor costs to account for contractor markup

Source:

1. 2018 DCD Simple Estimator
2. Per Anders

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Item No.: 9**Replace Back-up Generator and Electrical**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
150 KW Generator	1	EA	\$ 34,000.00	\$ 34,000.00	1
Earthwork and Concrete Pad for	1	EA	\$ 2,000.00	\$ 2,000.00	2
Electrical Connection	1	EA	\$ 20,000.00	\$ 20,000.00	3
Subtotal:				\$ 56,000.00	

Notes:

1. One generator to replace existing plant back-up generator, second generator to be placed at new pump station.

Source:

1. Generac Website
2. 2018 DCD Simple Estimator
3. Per Anders R.

Item No.: 10**Rehabilitate Existing Influent Pump Stations - New VFD Pumps and Piping**

Additional detail for capital construction line items

1.2% added from Vendors Price for Shipping:

Item	Quantity	Unit	Unit Cost	Total	Source
750 HP Variable Drive Pump	2	EA	\$ 30,000.00	\$ 60,000.00	1
350 HP Variable Drive Pump	1	EA	\$ 15,000.00	\$ 15,000.00	1
Piping, Valving, and Support	1	LS	\$ 40,000.00	\$ 40,000.00	2
Crane and Access Point	1	LS	\$ 75,000.00	\$ 75,000.00	1
Improve Existing Electrical System in	1	LS	\$ 10,000.00	\$ 10,000.00	1
Subtotal :				\$ 200,000.00	

Notes:

1. 20% added from Vendors Price for Shipping:
2. (2) 750 Hp Variable Drive Pumps
3. (1) 350 Hp Variable Drive Pump
4. Currently No Crane/ Access to remove Pumps Safely

Sources:

1. Per Anders
2. DCD Simple Estimator

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Item No.: 11**Rehabilitate Secondary Clarifier (No. 1)**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
(1) 28' Rake Arm, Surface Skimmer, and Walkway	1	LS	\$ 260,000.000	\$ 260,000.000	1
FRP Effluent Weir & Baffles	1	LS	\$ 16,000.00	\$ 16,000.00	1
Replace Existing Valving	1	LS	\$ 15,000.00	\$ 15,000.00	2
Miscellaneous Repair of Concrete	1	LS	\$ 5,000.00	\$ 5,000.00	3
Sludge Pump	1	LS	\$ 4,000.00	\$ 4,000.00	
Subtotal:				\$ 300,000.000	

Notes:

1. Construction Cost Assumed to be 2 x the Unit Budgetary Cost.
2. 12" Ductile Iron Gate Valve

Sources:

1. WesTech Budgetary Quote
2. 2018 DCD for Gate Valve

Item No.: 12**Effluent Flow Meter**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Flow meter (mag meter)	1	EA	\$ 15,000.00	\$ 15,000.00	1
Subtotal:				\$ 15,000.00	

Notes:

Sources:

1. Per Anders

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Item No.: 13**Groundwater Monitoring Wells**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Groundwater Monitoring Wells	3	EA	\$ 20,000.00	\$ 60,000.000	1
			Subtotal:	\$ 60,000.000	

Notes:

1. Actual cost of the monitoring wells dependent on depth to groundwater.

Sources:

1. Per Anders

Cost of Purchasing Property for Ponds

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Parcel Property	277	AC	\$ 1,300.00	\$ 360,100.000	1
			Subtotal:	\$ 361,000.000	

Notes:

1. Parcel Cost was provided by City of Alturas. Cost of parcel is \$1000/ac. Estimated cost (\$1300/ac) includes additional transaction fees (closing costs), legal costs, survey, etc.

Sources:

1. Per Anders

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Opinion of Probable Project Cost
City of Alturas WWTP
Alternative 3 Cost Summary

Item	Description	Units	Quantity	Unit Cost	Total Cost
Demolition					
1	Decommission Existing Plant ²	LS	1	\$ 25,000	\$ 25,000
New Unit Processes					
2	Back-up Generators ³	LS	1	\$ 124,000	\$ 124,000
3	Flow monitoring	LS	1	\$ 15,000	\$ 15,000
4	Spiralift 1/4" Screen w/ Construction and Housing	EA	1	\$ 309,000	\$ 309,000
5	Influent Pump Station	LS	1	\$ 900,000	\$ 900,000
6	Influent Force Main	LS	1	\$ 582,000	\$ 582,000
7	Aeration Basin	LS	1	\$ 1,037,000	\$ 1,037,000
8	Infiltration Ponds	LS	1	\$ 156,000	\$ 156,000
9	Groundwater Monitoring Wells	LS	1	\$ 60,000	\$ 60,000
10	Blower Building	SF	400	\$ 200	\$ 80,000
				Mobilization (12%):	\$ 395,000
				Subtotal :	\$ 3,263,000
				Contingency (30%):	\$ 979,000
				Subtotal Construction:	\$ 4,242,000
				Land Acquisition:	\$ 361,000
				Engineering/Construction Management (18%):	\$ 764,000
				Environmental/Permitting (5%):	\$ 213,000
				Administration/Legal (4%):	\$ 170,000
				Total Project:	\$ 5,750,000

Notes:

1. Refer to subsequent pages for additional information on each line item.
2. Decommissioning of the most existing plant components including termination of electrical connections throughout the plant and removal of elevated mechanical infrastructure from the site. Concrete substructures to remain.
3. According to the Operator of Record for the City of Alturas WWTF, the current generator is not a reliable back-up power source. With new infrastructure being proposed, it should also have a source of backup power, therefore two (2) generators have been recommended for this upgrade. One (1) generator to provide backup power for the influent pump station and one (1) generator to provide backup power the aeration ponds.
4. Values rounded up to the nearest \$1,000

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Item No.: 1**Decommission Existing Plant**

Terminate Electrical Connections to WWTF treatment components and Removal of Existing Mechanical Infrastructure.

\$ 25,000.00 Allowance

Item No.: 2**Back-up Generators**

Additional detail for capital construction line items

Notes:

Item	Quantity	Unit	Unit Cost	Total	Source
150 KW Generator ¹	2	EA	\$ 34,000.00	\$ 68,000.00	2
Electrical connection, automatic transfer switch	2	EA	\$ 25,000.00	\$ 50,000.00	
Earthwork and Concrete Pad for Generator	2	LS	\$ 3,000.00	\$ 6,000.00	1

Subtotal: \$ 124,000.00

Note:

1. Two Generator have been specified for this Alternative to provide back up power at both the Pump Station and the Aeration Ponds. The generator at the pump station will replace the existing generator located at the existing WWTP.

Source:

1. DCD Simple Estimator
2. Generac Website

Item No.: 3**Flow monitoring**

Additional detail for capital construction line items

Allowance for influent flow monitoring: \$ 15,000
Magmeter at influent pump station discharge

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Item No.: 4 Spiralift 1/4" Screen w/ Construction and Housing

Additional detail for capital construction line items

1.2% added from Vendors Price for Shipping:

Item	Quantity	Unit	Unit Cost	Total	Source
Franklin Miller Spiralift Screen	1	LS	\$ 87,000.00	\$ 87,000.00	1
Installation of Spiralift	1	LS	\$ 174,000.00	\$ 174,000.00	2
Building for New Headworks	240	SF	\$ 200.00	\$ 48,000.00	2
Subtotal:				\$ 309,000.00	

Note:

Installation Cost Assumed to be 3 x the Unit Cost

Building 12' x 20' around Helicoil Sieve

Sources:

1. Franklin Miller Budgetary Cost Estimate

2. Per Anders R.

Item No.: 5 Influent Pump Station

Additional detail for capital construction line items

Installed \$ 900,000 As per Anders and previous projects experience

Subtotal: \$ 900,000

Item No.: 6 Influent Force Main

Additional detail for capital construction line items

Notes: 1. 20% added to vendor costs to account for contractor markup

Item	Quantity	Unit	Unit Cost	Total	Source
Install 8" Force Main From WWTF	11000	LF	\$ 50.00	\$ 550,000.00	1
Install 8" Ductile Iron From WWTF	500	LF	\$ 60.00	\$ 30,000.00	1
8" Bridge Support	1	LS	\$ 1,300.00	\$ 1,300.00	

Source:

1. Per Anders

Subtotal: \$ 582,000.00

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Item No.: 7**Aeration Basin**

Additional detail for capital construction line items

1.2% added from Vendors Price for Shipping:

Item	Quantity	Unit	Unit Cost	Total	Source
Excavation	55543	CY	\$ 4.12	\$ 228,737.18	2
Rough Grading	21450	CY	\$ 4.94	\$ 106,053.09	2
Electrical Connection	1	LS	\$ 40,000.00	\$ 40,000.00	1
Valving and Piping	1	LS	\$ 20,195.00	\$ 20,195.00	2
60 mm Pond Liner	145368	SF	\$ 0.80	\$ 116,294.40	1
Triple Lagoon System	1	LS	\$ 505,000.00	\$ 505,000.00	3
Pontoon Boat with Jib for	1	LS	\$ 20,000.00	\$ 20,000.00	1

Subtotal: \$ 1,037,000.00

Note:

1. Total Construction cost (2 x Budgetary Unit value)
2. Backup Generator accounted for under Item 1.

Sources:

1. Per Anders R.
2. DCD Simple Estimator 2018
3. Triple Point Lagoons

Item No.: 8**Infiltration Ponds**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Earthwork	26482	CY	\$ 4.94	\$ 130,932.30	3
Valving, Piping, and Restraints	1	LS	\$ 20,000.00	\$ 20,000.00	3
Erosion Control	1	LS	\$ 5,000.00	\$ 5,000.00	1

Subtotal: \$ 156,000.00

Source:

1. Per Anders
2. 2007 National Construction Editor
3. DCD Simple Estimator

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Item No.: 9**Groundwater Monitoring Wells**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Groundwater Monitoring Wells	3	EA	\$ 20,000.00	\$ 60,000.00	1
Subtotal:				\$ 60,000.00	

Sources:

1. Per Anders

Item No.: 10**Blower Building**

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Blower Building	400	SF	\$ 200.00	\$ 80,000.00	1
Subtotal:				\$ 80,000.00	

Notes:

1. 20' x 20' shed for Blower infrasturcture and storage of equipment.

Sources:

1. Per Anders

Cost of Purchasing Property for Ponds

Additional detail for capital construction line items

Item	Quantity	Unit	Unit Cost	Total	Source
Property	277	AC	\$ 1,300.00	\$ 361,000.00	1
Subtotal:				\$ 361,000.00	

Source:

1. Per City of Alturas

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**Opinion of Annual Wastewater Operational Costs
City of Alturas Wastewater Planning Study**

Item	Description	Alternative 1 Rehab WWTP w/ River Discharge	Alternative 2 Rehab WWTP w/ Land Discharge	Alternative 3 New WWTP w/ Land Discharge
1	Labor	\$ 85,000	\$ 85,000	\$ 85,000
2	Chemicals	\$ 95,000	\$ 39,500	\$ -
3	Testing/Reporting	\$ 35,000	\$ 24,500	\$ 24,500
4	Sludge/solids hauling	\$ 2,000	\$ 2,000	\$ 1,000
5	Electrical Costs	\$ 28,500	\$ 30,900	\$ 48,800
6	Permitting	\$ 5,000	\$ 5,000	\$ 5,000
Annual O&M costs:		\$ 250,500	\$ 186,900	\$ 164,300

Notes:

1. Other costs not listed are considered relatively equal between the three alternatives.



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Operations Cost Details for all alternatives

Item No.: 1 Labor

For all three alternatives, assume the following for labor including benefits for one full time person.
Per Jason Diven, City of Alturas

Annual Labor: \$ 85,000 per year

Item No.: 2 Chemicals

For FY 2018/2019, the City spent \$98,647 on chemicals, which include:
pH adjustment, coagulation, chlorination, dechlorination
No further breakdown of costs was available.

Assume the following:

1. For Alternative 1, assume minor reduction in coagulation, but increase for lime and pH adjustment.
2. For Alternative 2, assume reduction in coagulation, chlorination, and dechlorination, use 40% of recent costs.
3. For Alternative 3, assume full elimination of chemicals, use \$0.

	Alt 1	Alt 2	Alt 3
Chem costs:	\$ 95,000	\$ 39,500	\$ -

Item No.: 3 Testing/Reporting

Current budget for testing fees is \$35,000.
Assume same for Alt 1 with river discharge.
For Alts 2 and 3, reduced effluent testing but groundwater monitoring; assume 70% of current budget.

	Alt 1	Alt 2	Alt 3
Testing:	\$ 35,000	\$ 24,500	\$ 24,500

Item No.: 4 Sludge/solids hauling

Sludge handling in most recent year was \$973.
Alt 1 includes solids removal at headworks plus sludge handling.
Alt 2 includes solids removal at headworks plus sludge handling.
Alt 3 includes only solids removal at headworks.
Assume following:

	Alt 1	Alt 2	Alt 3
Testing:	\$ 2,000	\$ 2,000	\$ 1,000

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Operations Cost Details for all alternatives**Item No.: 5 Electrical Costs**

Notes: 1. Assumed rate of: \$0.17 per kWh Pacific Power
\$0.07 per kWh Surprise Valley Electric
Rates cover meter charges for SVE

Notes:

2. 1 hp = 0.746 kW

3. Electrical budget for current FY is \$28,500

Alt 1 Rehab WWTP w/ River Discharge

Assume same electrical costs as currently: \$ 28,500 per year

Alt 2 Rehab WWTP w/ Land Discharge

Assume same electrical costs as currently: \$ 28,500

Plus effluent pumping of approximately: \$ 2,400

Total: \$ 30,900 per year

Effluent pumping based on 108 MG pumped per year using 14,000 kwh

All power in Pacific Power service area.

Alt 3 New WWTP w/ Land Discharge

Headworks plus pumping in Pacific Power service area

Blowers at new WWTP in SVE service area

Headworks screen (allowance): \$ 600

Influent pumping (same as Alt 2): \$ 2,400

Blowers (2x50hp=746 kW), running 24/7 = 653,496 kwh per year

Electrical cost for blower: \$ 45,800

Total: \$ 48,800

Item No.: 6 Permitting

Notes: Annual permit estimates were derived from the following fee equation, provided in the California Code of Regulations (Fee Schedule) by the California State Water Resources Control Board:

"Fee equals \$2,572 plus 4,548 multiplied by the permitted flow, in mgd"

flow = 0.5 mgd

Annual Permit Fee = \$ 4,846 *

* The annual permit fee is rounded to the nearest thousands for the planning stage.

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Present Value Analysis
City of Alturas Wastewater Planning Study

Data mostly from other spreadsheets

Discount rates from OMB Circular A-94 Appendix C November 2019

Duration 20 years

Nominal rate 2.3%

Real rate 0.3%

Project Component Alternatives	Capital Cost	Annual O&M	Present Value O&M	Salvage Value	Present Value Salvage	Net Present Value
Alternative 1	\$ 12,890,000	\$ 250,500	\$4,855,598	\$ 4,253,700	\$ 4,006,344	\$ 13,739,254
Alternative 2	\$ 11,642,000	\$ 186,900	\$3,622,799	\$ 3,841,860	\$ 3,618,452	\$ 11,646,347
Alternative 3	\$ 5,750,000	\$ 164,300	\$3,184,729	\$ 1,897,500	\$ 1,787,159	\$ 7,147,571

For salvage value, assume overall average 30-year life w/ straightline depreciation, so after 20 years, salvage is 33%.

Net Present Value = Capital Cost + Present Worth O&M - Present Worth Salvage.

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Aerator Manufacturer Design Data 5

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Basis of Design

Alturas, CA

16-Oct-19

Aeration Design Calculations**SUMMARY - General Design Parameters**

v3.6.3	Design Scenario Name			ADF
1	Influent Flowrate	MGD		0.500
2	Influent Concentration	mg/L		240.0
3	Effluent Concentration (summer)	mg/L		10.7
4	Effluent Concentration (winter)	mg/L		29.0
5	Actual Oxygen Supplied	lb/day		1522.8
6	Has air been provided for nitrification?			No
7	Number of Aerators			27
8	Estimated Tubing Length	ft		4000
9	Airflow	scfm		955
10	Design Pressure (includes cushion)	psig		7.51
11	Projected Brake Horsepower	bhp		35.55
12	Min. Design Horsepower	hp		53

SUMMARY - Aerators

		Air Supplied Via:	Manifolds
Cell Name	Aerator Type		
Cell 1	750T		21
Cell 2	750T		6
Cell 3	750T		0
Cell 4	750T		0
Cell 5	750T		0
Cell 6	750T		0

SUMMARY - Biological Treatment Calculations

Item	Description	Units	ADF
1	Number of Treatment Cells		2
2	Flow Regime		Series
3	Site Elevation - HWL	MSL - ft	4370

Cell 1

	4	Wastewater Flowrate	MGD	0.500
	5	Treatment Volume	M-Gal	6.7
	6	Treatment Time	days	13.3
	7	Treatment Type	-	Partial Mix
	8	Standard Reaction Rate, k_{20}	days ⁻¹	0.28
	9	Design Water Temp	°C	20
	10	Design Reaction Rate, k_T	days ⁻¹	0.122
	11	Biological Treatment Efficiency	%	78.9%
Summer	12	Influent BOD Loading	lb/day	1,000
	13	Influent BOD Concentration	mg/L	240.0
	14	BOD Removed	lb/day	788
	15	Effluent BOD Loading	lb/day	211
	16	Effluent BOD Concentration	mg/L	50.7
	17	Design Water Temp	°C	0.5
Winter	18	Biological Treatment Efficiency	%	65.2%
	19	BOD Removed	lb/day	651.8
	20	Effluent BOD Concentration	mg/L	83.5
	N1	Influent NBOD Loading	lb/day	104
	N2	Influent NBOD Concentration	mg/L	25.0
	N3	NBOD Removed* (Assumed)	lb/day	-
	N4	Effluent NBOD Loading*	lb/day	104
	N5	Effluent NBOD Concentration*	mg/L	25

Cell 2

	17	Wastewater Flowrate	MGD	0.500
	18	Treatment Volume	M-Gal	6.7
	19	Treatment Time	days	13.3
	21	Treatment Type	-	Partial Mix
	22	Standard Reaction Rate, k_{20}	days ⁻¹	0.28
Summer	20	Design Water Temp	°C	20
	23	Design Reaction Rate, k_T	days ⁻¹	0.122
	24	Biological Treatment Efficiency	%	78.9%
	25	Influent BOD Loading	lb/day	211
	26	Influent BOD Concentration	mg/L	50.7
	27	BOD Removed	lb/day	166
	28	Effluent BOD Loading	lb/day	45
	29	Effluent BOD Concentration	mg/L	10.7
Winter	17	Design Water Temp	°C	0.5
	18	Biological Treatment Efficiency	%	65.2%
	19	BOD Removed	lb/day	226.8
	20	Effluent BOD Concentration	mg/L	29.0
	N6	Influent NBOD Loading	lb/day	104
	N7	Influent NBOD Concentration	mg/L	25.0
	N8	NBOD Removed* (Assumed)	lb/day	-
	N9	Effluent NBOD Loading*	lb/day	104
	N10	Effluent NBOD Concentration*	mg/L	25

Cell 3

Cell 4

Cell 5

Cell 6

SUMMARY - Aeration Calculations

	Item	Description	Units	ADF
	1	Site Elevation	ft	4370
	N1	O ₂ Loading Factor (BOD ₅)	lb-O ₂ /lb-BOD	1.5
	N2	O ₂ Loading Factor (NBOD ₅)	lb-O ₂ /lb-NBC	4.6
	2	Alpha-value, α		0.60
	3	Beta-value, β		0.95
	4	Theta-value, θ		1.02

Cell 1

	5	Lagoon Side Water Depth	ft	12
	6	Air Release Depth	ft	11.25
	N3	AOR - BOD	lb/day	1183
	N4	AOR - NBOD	lb/day	0
	7	AOR - Total	lb/day	1183
	8	SOTE/ft	%/ft	1.89%
	9	SOTE	%	21.27%
	10	Design DO Concentration	mg/L	2.0
	11	FTE		6.36%
	12	Air requirement	scfm	742
	13	Airflow per aeration unit	scfm/unit	35.3
	14	Aerator Type		750T
	15	Number of aeration units	units	21
	16	Water Pressure	psi	4.87
	17	Aerator Pressure Loss	psi	0.75
	18	Header/Feeder Pressure Allowance	psi	0.89
	19	Total Operating Pressure	psig	6.51
	20	Design Motor Pressure	psig	7.51

Cell 2

	21	Lagoon Side Water Depth	ft	12
	22	Air Release Depth	ft	11.25
	N5	AOR - BOD	lb/day	340
	N6	AOR - NBOD	lb/day	0
	23	AOR	lb/day	340

24	SOTE/ft	%/ft	1.89%
25	SOTE	%	21.27%
26	Design DO Concentration	mg/L	2.0
27	FTE		6.36%
28	Air requirement	cfm	213
29	Airflow per aeration unit	cfm	35.6
30	Aerator Type		750T
31	Number of aeration units	units	6
32	Water Pressure	psi	4.87
33	Aerator Pressure Loss	psi	0.75
34	Header/Feeder Pressure Allowance	psi	0.89
35	Total Operating Pressure	psig	6.51
36	Design Motor Pressure	psig	7.51



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BUDGETARY ESTIMATE – LAGOON AERATION

PROJECT NO.: 3339

PROJECT NAME: Alturas WWTP

PROJECT LOCATION: Alturas, CA

DATE: October 22, 2019

PREPARED FOR

Anders Rasmussen, P.E.
SHN Engineers & Geologists

c/o
Kyle Menath
JBI Water & Wastewater

PREPARED BY

Triplepoint Environmental, LLC
Tom Daugherty, Western Region Manager
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Fax: (312) 957-4712
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Basis of Design

Site desires to evaluate lagoon based treatment using diffused aeration. A summary basis of design is in the table below. A detailed basis of design is attached herewith.

Parameter	Influent Average	Effluent Design Criteria
Design Flow	0.50 MGD	0.50 MGD ¹
BOD	240 mg/L	25 mg/L
TSS	300 mg/L	TBD
NH3-N	25 mg/L	TBD

1. It is understood daily peaks may reach 1.0 MGD. If the max month/average day is > 500,000 gpd TPE will recalculate and provide an updated scope of supply.

Discussion

Layout: Triplepoint designed a two cell lagoon system. Both cells measure 400' x 240' at the surface with 3:1 slope and 12' water depth. Each cell has a volume of 6.67 MG featuring 13.3 treatment days. Cell 1 will receive 21 aerators and Cell 2 will receive six aerators based on the design inputs.

Blower and Control Panel: Three Blowers are proposed at 30 HP each at the specified design depth. Two blowers are duty and one is standby. The NEMA 3R control panel features dual VFDs and switching to allow the third blower to be operated by an existing VFD.

Installation: Triplepoint will provide a turnkey installation quote upon request.

Triplepoint has a robust solution for retrofitting a lagoon or servicing individual aerators without dewatering.



The aerators are placed into position from a floating vessel. Buoys are attached to the aerators by stainless cables facilitating individual retrieval for maintenance. No electrical or moving parts are in the water.

Scope of Supply

Equipment	Quantity	Unit
Ares 750T Tube Aerators with Coarse and Fine Bubble	27	ea
Buoys with SS tether cables, SS quick disconnects	27	ea
High-flow Flexible Weighted Airline	4200	ft
SS Full-Port Ball valves for individual aerator control and barb set	27	ea
6-port Custom Welded 304 SS air distribution manifold (2 caps)	4	ea
4-port Custom Welded 304 SS air distribution manifold	2	ea
316 Stainless Steel hose clamps	60	ea
30 HP 460/60/3 Blower with Sound and Weather package (two duty one standby)	3	ea
NEMA 3R Control Panel with dual VFDs and A/B switch	1	ea
Blower spares, filter, belts, oil, spec grease	1	lot
Detailed Installation and layout plan (Shop Drawings)	1	ea
Person Days Triplepoint Installation Supervision, Start-up, and Training	4	ea
Optional: DO Real timer Blower Control (\$9,000-\$18,000 depending on options)	0	ea
Optional: Additional Year of Warranty (\$5,616)	0	ea
Installation quote upon request	0	ea
Freight FOB factory (TBD)	0	lot
TOTAL PRICE:	\$252,479	

Conditions of Sale

Price and Payment

The quote in this proposal is in US Dollars and does not include applicable federal or state taxes, fees, or tariffs. It remains valid for a period of 45 days. Fifty (50) percent of the quote price is due upon contract acceptance, forty (40) percent upon shipment and ten (10) percent upon startup.

Installation

Triplepoint Environmental will provide installation supervision as part of this proposal along with certification of proper installation once complete. All installation labor is the responsibility of the customer. A separate proposal can be supplied for aeration installation. A floating vessel is required for installation unless cells are drained.

Supplied by Others

Air headers are not included in this scope of supply. Installation and blower connection to air distribution header and integration to SCADA or other plant specific data recording schema are not included. Site specific preferred embodiments of installation such as exterior conduit runs, cable ties, and the like are not included.

Delivery

The MARS Aeration diffusers and tubing will be delivered within a period of 10-14 weeks after submittal approval or receipt of purchase order. All packing and shipping costs are FOB origination unless otherwise quoted herein. Customer is responsible for paying all taxes and fees associated with shipping.

Blowers

All blowers to provide the recommended airflow for each MARS option at the recommended pressure found in the basis of design document attached herewith.

Warranty

Triplepoint Environmental offers the most competitive warranty in the industry, ensuring that your MARS/Ares products are free from defects in material or workmanship for a period of one (1) year from the date of installation completion.

Limits of Liability

Triplepoint Environmental shall not be liable for any loss of profits, business, goodwill, interruption of business, nor for incidental or consequential merchantability or fitness of purpose, damages related to this quote.

CONFIDENTIALITY NOTICE

The MARS/Ares Aeration system is the subject of one or more confidential patents filed in the United States Patent Office. The Client, Engineer, and any other parties contracted recognize the importance of maintaining the continued confidentiality of the design of the MARS/Ares Aeration system. The Client, Engineer and any other parties contracted agree that they shall not sell, transfer or disclose any such confidential information relating to the design of the MARS/Ares Aeration system to any other person, organization, or corporation without the express written authorization of Triplepoint Environmental LLC and pursuant to an enforceable agreement of confidentiality, except as required by law or as necessary in connection with the use, operation, maintenance, repair, or replacement of the system. Additionally, The Client, Engineer and any other parties contracted all agree to preserve the confidentiality of this proposal and all materials attached and not to distribute or copy such materials for any other parties not previously authorized by Triplepoint Environmental LLC.



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